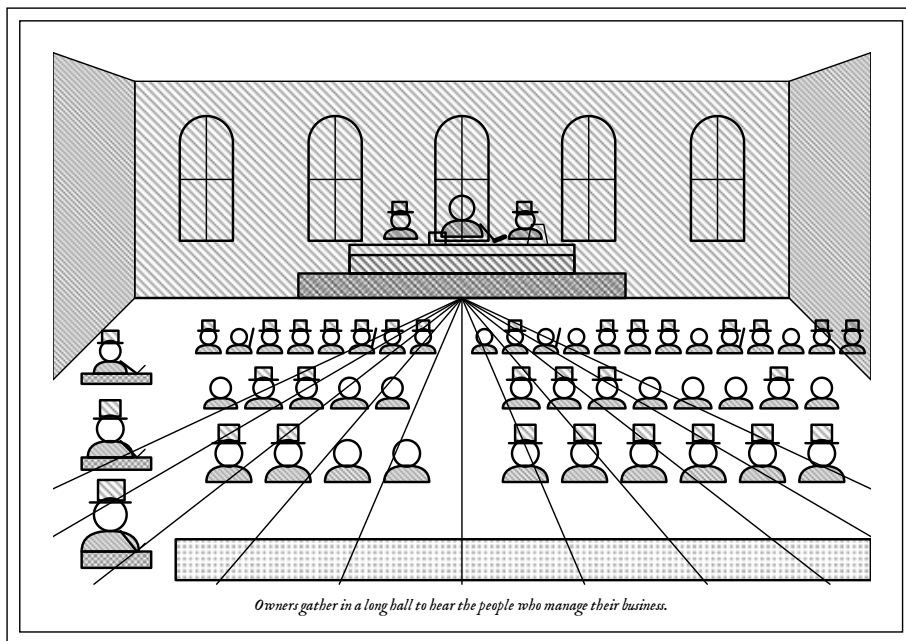




CHAPTER III

The Goal of the Firm and Agency

What should a business aim for, and who makes sure it does?



A business needs one clear aim. This chapter explains why the aim is the wealth of the owners, not only this year's profit, and why hired managers may follow a different aim. It names the clash that follows and the cost it brings.

A TREATISE ON THE PRINCIPLES OF FINANCE



*Being a Plain Account of Money, Capital, Risk and the
Management of Funds*

M. MEHU, G. MOMO

2.7 A small manufacturer, in one year, buys a new machine, pays for it partly with a loan, and pays its owners a share of profit. List the three decisions and explain how the dividend decision affects the financing decision.

2.8 Tailor A spends Tk 70 for each Tk 100 of sales and Tailor B spends Tk 80. Which has the greater operating efficiency, and why does efficiency differ from profitability?

Question. Which choice is the investment decision, which the financing decision and which the dividend decision? How much profit was kept? What does the new loan cost each year, and how large is the bank loan afterwards?

Reasoning. Begin with the profit. The family withdrew Tk 120,000, which is the dividend, so the retained earnings are Tk 480,000 less Tk 120,000, which is Tk 360,000. The choice of how much to pay out and how much to keep is the dividend decision.

Next, the corner. Opening it commits Tk 350,000 to a long-lasting asset, so this is the investment decision. Paying for it is a separate choice. The owners supply Tk 150,000 of retained earnings and a lender supplies Tk 200,000, and Tk 150,000 plus Tk 200,000 is Tk 350,000, which matches the cost. Choosing this mixture of owners and lender is the financing decision.

Then count what remains. Tk 360,000 less Tk 150,000 leaves Tk 210,000 of retained earnings, which buys textbook stock. The bank loan was Tk 600,000, and the new loan adds Tk 200,000, so it becomes Tk 800,000. The yearly charge on the new loan is 10% of Tk 200,000, which is 10 divided by 100 times Tk 200,000, or Tk 20,000.

Conclusion. The dividend decision kept Tk 360,000. The investment decision chose the corner at Tk 350,000. The financing decision paid for it with Tk 150,000 of retained earnings and a Tk 200,000 loan. The loan costs Tk 20,000 a year, and the bank loan after the new borrowing is Tk 800,000.

Check. The Tk 150,000 for the corner and the Tk 210,000 for stock add to Tk 360,000, so all retained earnings are used and none is counted twice. The two sources of the corner add to its cost. The loan grew by one third of its old size, from Tk 600,000 to Tk 800,000, which is a sensible rise for a purchase partly paid by borrowing.

* * *

SUMMARY

The financial manager plans, raises and controls the money of a business, and this duty exists in every business whatever the title. Three decisions make up the work. The investment decision chooses what to buy, the financing decision chooses how to pay, and the dividend decision chooses how much profit to pay out and how much to keep as retained earnings. The success of the business each day shows in its operating efficiency, liquidity and profitability.

THINK IT THROUGH

2.T1 A bakery has Tk 150,000 of spare profit and must weigh three options: a new oven, a payout to the owners, or repaying a loan. Name the decision type in each option.

2.T2 In the bakery, say what changes in its sources of funds if it repays the loan, and what changes if it pays the owners a dividend.

The questions ask for each decision, each amount and the cost of the loan.

The first step is the dividend decision, which fixes how much profit stays.

The next steps separate what is bought from how it is paid for.

The last steps count the remaining profit, the loan and its yearly charge.

The conclusion names each decision and the resulting figures.

The check confirms that the money is fully accounted for.

The chapter in brief.

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First edition

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Retained earnings connect the third decision to the second. Profit that stays in the business still belongs to the owners, so it is a source of funds from the owners, which Chapter 1 called equity. It is also the easiest source to use, because nobody new must be asked. This is why the decisions cannot be fully separated: the more profit is paid out, the less is left to fund purchases, and the more the business may need to borrow. At Meghna and Sons, Tk 150,000 of the Tk 360,000 kept went into the corner. This is the kept profit that the financing paragraph above described.

PAUSE AND CHECK 2.P2 A tailor earns a profit of Tk 90,000 and pays Tk 25,000 of it to her family, who own the shop. How much is the dividend, and how much are the retained earnings?

The table below sets the three decisions side by side. Read each row from left to right: the decision, the question it asks, and the choice made at Meghna and Sons.

THE THREE CORE DECISIONS COMPARED

DECISION	QUESTION IT ASKS	AT MEGHNA AND SONS
Investment	Which long-lasting assets should we buy?	A stationery corner costing Tk 350,000
Financing	How should we raise the money?	Tk 150,000 of kept profit and a Tk 200,000 bank loan
Dividend	How much profit do we pay out, and how much do we keep?	Tk 120,000 paid out; Tk 360,000 kept

The figure below draws the same decisions as three arrows around a box of funds. Read it arrow by arrow. The arrow into the box shows money arriving from owners and lenders. The arrow to the right shows money leaving to buy assets. The dashed arrow downward shows money leaving as dividends. Each arrow carries the name of its decision in plain words.

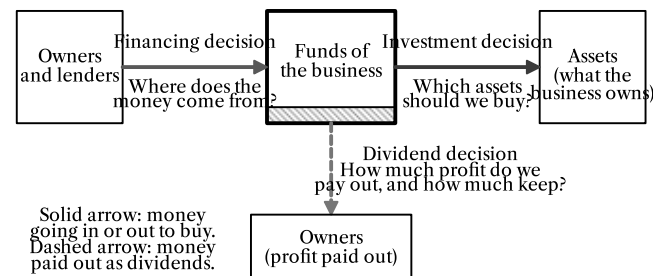


FIG. 2.1. The three decisions as three arrows around the funds of a business

Before moving on. The investment decision asks what to buy, the

Retained earnings link the dividend decision to the financing decision.

The table compares the three decisions side by side.

The figure shows the three decisions as three arrows around the funds of a business.

The section ends with a short

Financial control means checking that money is spent and recorded as planned, and acting when it is not. Suppose the tailor planned Tk 30,000 for cloth but the bill is Tk 36,000. Control notices the difference of Tk 6,000 and asks why. Perhaps cloth became more expensive, or perhaps she bought cloth she does not need. Then comes the act: change a price, change a supplier or change the plan. Control that finds a difference and does nothing has no value.

The viewpoint changes with the person. Owners rely on the financial manager to use their money wisely and to report on it honestly. A lender relies on the manager to repay on the agreed day. Employees rely on the manager to make sure money is available for wages. The financial manager therefore answers to several groups at once, which is part of what makes the work demanding.

The role has limits. The financial manager is not the accountant of Chapter 1, section 1.2. The accountant records what has happened, while the financial manager uses those records to decide what should happen next. The financial manager also does not choose which books the shop stocks; Imran does. The role supplies money and judges its use, but it does not run the other functions. In a very small business one person may hold both roles, though the two kinds of work remain different.

Before moving on. A financial manager is the person who plans, raises and controls the money of a business, whatever the title. Planning decides in advance how much money is needed and when. Control checks what happened against the plan and acts on the difference. *You should now be able to say* what each of the three verbs means and give a small example of each.

* * *

2.2 THE THREE CORE DECISIONS

Begin at home. A household has saved Tk 60,000 and has some spare income each month. It faces three questions. First, what should we buy with our money? Perhaps a fridge. Second, how should we pay for it: from our savings, or by borrowing from a relative? Third, how much of our spare income should we give to the grandparents, and how much should we keep? These questions sound alike, because each one concerns the same money. Yet each asks something different, and a household that mixes them up chooses badly.

A business asks the same three questions, and each question has a name. We take them one at a time, because the names are similar and the questions are not.

The first is the investment decision. The **investment decision** is choosing which long-lasting assets or projects a business should buy. A project is a planned piece of work that needs money, such as opening a second branch. A bakery that must choose between a new oven and a delivery van makes an investment decision, because either would serve the bakery for many years.

Case file. illustrative. On 10 February 2024 the owners of

Financial control.

Checking that money is spent and recorded as planned, and acting when it is not.

Financial control compares what happened with the plan and acts on any difference.

The owner, the lender and the employee each depend on the financial manager in a different way.

Recall: finance beside accounting, Chapter 1, section 1.2, 'Finance Among the Business Functions'.

The role has limits: it is not accounting, and it does not replace the other managers.

The section ends with a short recap.

GO SLOWLY HERE.

Even a household faces three separate questions about the same money.

A business asks the same three questions, and each has a name.

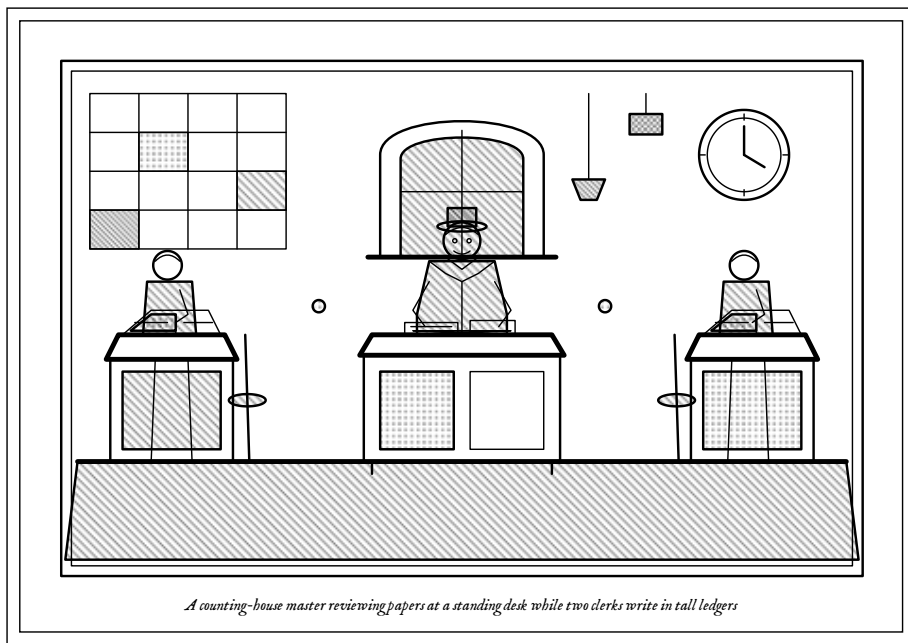
Investment decision.

Choosing which long-lasting assets or projects a business should buy.

The investment decision asks what to buy.

The case file shows the

11.3 Degree of Operating Leverage	108
11.4 Operating Leverage and Business Risk	109
XII. FINANCIAL AND TOTAL LEVERAGE	113
12.1 Borrowing and Fixed Financing Charges	114
12.2 Degree of Financial Leverage	116
12.3 Degree of Total Leverage	117
12.4 Leverage and the Risk of a Firm	118
XIII. VALUING BONDS	123
13.1 What a Bond Is	124
13.2 The Idea of Value	126
13.3 The Value of a Bond	126
13.4 Discount, Par and Premium Bonds	128
XIV. VALUING COMMON AND PREFERRED STOCK	133
14.1 Rights of Shareholders	134
14.2 Valuing Preferred Stock and Level Dividends	135
14.3 Valuing Stock with Growing Dividends	136
14.4 Comparing Bonds and Stock	137
XV. WORKING CAPITAL AND THE CASH CYCLE	141
15.1 What Working Capital Is	142
15.2 The Cash Conversion Cycle	144
15.3 Float	146
15.4 Estimating Working Capital Needs	147
XVI. SHORT-TERM FINANCING AND MATCHING	151
16.1 Financing by Time Span	152
16.2 Trade Credit	153
16.3 Other Short-Term Sources	155
16.4 The Matching Principle	155
XVII. LONG-TERM FINANCING AND THE COST OF CAPITAL	161
17.1 Term Loans and Debt Instruments	162
17.2 Owners' Funds as Long-Term Finance	163
17.3 Raising Funds in the Capital Market	164
17.4 Long-Term Lenders and Investors in Bangladesh	165
17.5 The Cost of Capital	166
XVIII. LEASING	171
18.1 What a Lease Is	172
18.2 Kinds of Lease	173
18.3 The Lease or Buy Decision	174
GLOSSARY	179
ANSWERS TO QUESTIONS	191
FURTHER READING	201
INDEX	203



CHAPTER II

The Financial Manager's Work

Who decides where the money goes, and how do they decide well?



This chapter introduces the financial manager and the three decisions this person makes about money: what to buy, how to pay for it, and how much profit to pay out. It ends by showing how these decisions affect the daily success of a business. A bookshop in Dhaka supplies the figures, and a London bank supplies a warning from history.

A merchant of that time faced a problem: carrying coin from one city to another was slow and dangerous. The bank offered a solution, the bill of exchange. A merchant paid money to the bank in one city and received a written order. The order told a branch or agent in another city to pay an agreed sum there. No coin travelled; only the paper did. The bank earned its charge from the difference in the rates at which currencies were exchanged.

Seen with this chapter's words, the bank received money from depositors, which was a source of funds, and used it to make loans and to finance trade, which were uses. It also moved funds between cities, so that merchants did not have to. The historian Raymond de Roover told the story in *The Rise and Decline of the Medici Bank, 1397-1494* (1963). The bank ended in 1494, when the Medici were expelled from Florence.

REVIEW QUESTIONS

- 1.1 A stall owner has Tk 1,200 in her purse and a table full of vegetables. In your own words, explain what finance means for her. Name one thing she must find and one thing she must decide.
- 1.2 A tailor records the money received each month and also decides whether to borrow to buy a second sewing machine. Which task is accounting and which is finance? Explain the difference.
- 1.3 A new bakery begins with the owner's savings and a loan. Name its two sources of funds, say which is equity and which is debt, and give two uses of the funds.
- 1.4 A customer takes goods on 28 June and pays on 28 July. Explain why the sale and the cash flow fall in different months.
- 1.5 On Monday customers pay the stall owner Tk 3,000, and she pays the wholesaler Tk 2,000. State the cash flow in, the cash flow out and the net movement.
- 1.6 The same stall owner now takes a loan of Tk 800 from her cousin. Is this a source or a use of funds? Is it equity or debt? What does she owe afterwards?
- 1.7 A school canteen is run by a committee, using money from a school fund. Name one source of funds and one use of funds for the canteen.
- 1.8 A small clinic is funded by its owners' savings and a bank loan. Name its sources of funds and two assets it might hold. Where does finance sit among the clinic's business functions, and what does it do for the others?

The bill of exchange solved the problem of carrying coin from city to city.

In this chapter's words, the bank gathered sources of funds and put them to use.

A stall owner has a full table of goods but too little money to buy more.

Her choices raise three questions, and finance exists to answer them.

You need nothing beforehand except your own experience of spending and borrowing.

Each idea is built in the same order, and small signs help you follow it.

The chapter takes three steps, from the idea of finance to the flow of funds.

Every household already practises finance on a small scale.

GO SLOWLY HERE.

Finance. The work of finding money and deciding how to use it, so a business or person can reach chosen aims.

Finance is finding money and

Picture a woman who sells vegetables at a market stall. On Monday evening her table still holds tomatoes, onions and gourds, and at the prices on her sign they are worth Tk 5,000.¹ Yet at five o'clock on Tuesday morning, at the wholesale market, her purse holds only Tk 1,200, and the onions she needs cost Tk 2,000. The stall is full, and she still cannot buy what she needs. Her trouble is older than any bank: the goods a person owns and the money a person holds are not the same thing.

She has choices. She may buy fewer onions. She may ask a cousin for a loan until Friday. She may sell the tomatoes cheaply this evening to receive money at once. Each choice raises the same questions. Where will the money come from? What will it be spent on? What will she owe, and to whom, afterwards? These questions are the beginning of finance. This chapter turns them into plain ideas and gives each idea a name.

You will need: nothing beyond your own experience of spending, saving and borrowing. Every other word is explained when it first appears.

This book builds each idea in the same order: a plain statement, an example, a rule, and then the limits of the rule. A short note in the side column sums up each paragraph in one line. When a new term appears, it is printed in bold, defined at once and repeated in the side column; a fuller meaning is in the glossary. Questions marked Pause let you test yourself before reading on, and their answers are given at the back. This first chapter goes slowly on purpose, because every term in it is new.

The chapter takes three steps. Section 1.1 explains what finance is. Section 1.2 shows where finance sits among the kinds of work that a business performs. Section 1.3 follows money from the people who supply it to the things the business buys with it. A running case, the bookshop Meghna and Sons in Dhaka, appears from section 1.2 onward.

* * *

1.1 WHAT FINANCE IS

Begin with your own week. Money reaches you from somewhere: an allowance, a wage, a gift from a relative. Money leaves you for a purpose: bus fares, food, a phone, fees. Sometimes you keep part of it for later. Sometimes you borrow from a friend and promise to repay. In each case, someone decides where money should come from and where it should go. That someone is practising finance, on a small scale.

Finance is the work of finding money and deciding how to use it, so that a business or a person can reach chosen aims. The stall owner who asks her cousin for Tk 800 and decides to spend it on onions is doing finance twice: once to find the money, and once to choose its use.

¹ Tk is the written short form of taka, the money of Bangladesh. The book writes amounts as Tk 5,000, with a comma after each group of three digits.

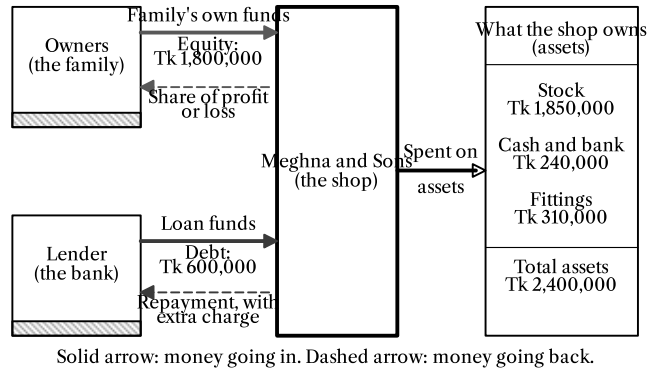


FIG. 1.1. Funds flowing between the owners, the lender, the shop and its assets, Meghna and Sons, 1 January 2024

Before moving on. A source of funds is where money comes from, a use of funds is what it buys or keeps, and assets are the things the business owns. Equity is the owners' money and debt is borrowed money, and on any one date the sources equal the assets. *You should now be able to say* which part of a business's funds is equity, which is debt, and what each one bought.

* * *

WORKED CASE: SOURCES AND USES AT MEGHNA AND SONS

Case file, illustrative. *Situation.* At 1 January 2024, Meghna and Sons records stock of Tk 1,850,000, a cash and bank balance of Tk 240,000 and fittings of Tk 310,000. It owes a bank loan of Tk 600,000 at 10% a year, and the family's own funds are Tk 1,800,000. Sales in 2023 were Tk 7,200,000.

Question. First, which figures are sources of funds and which are uses? Second, do the totals agree? Third, what is the family's wealth in the shop, and does the sales figure belong in the sorting?

Reasoning. Begin by sorting. The bank loan and the family's own funds answer the question "where did the money come from?", so they are sources: debt of Tk 600,000 and equity of Tk 1,800,000. Stock, the cash and bank balance and fittings answer the question "what did the money buy or keep?", so they are uses, and each one is an asset.

Next, compare the totals. Sources: Tk 600,000 plus Tk 1,800,000 is Tk 2,400,000. Assets: Tk 1,850,000 plus Tk 240,000 plus Tk 310,000 is Tk 2,400,000. Then find wealth. Assets of Tk 2,400,000 less the liability of Tk 600,000 leave Tk 1,800,000.

Last, consider sales. The figure of Tk 7,200,000 describes the whole of 2023, which is a period. The sorting describes one date, 1 January 2024. Sales therefore do not belong in the list of sources or uses. They belong with the ideas of period and cash flow, and sales

The section ends with a short recap.

The worked case sorts each figure into its place.

Three tasks are set: sort, compare, and find wealth.

The first step asks of each figure: did the money come from here, or go to here?

The second step adds each side and compares the totals.

The sales figure describes a year, while the sorting describes one date.

* * *

1.2 FINANCE AMONG THE BUSINESS FUNCTIONS

A bookshop needs more than one kind of work, and someone must look after each.

Business function. One main kind of work in a business, such as selling, making or managing money.

A business function is one main kind of work in a business.

Account. Word in use. In daily life, a record of money in a bank. In accounting, a named record of one kind of item, such as Cash or Sales.

The running case shows two functions shared between two brothers.

Managers decide; finance supplies the money to carry out each decision and the facts to judge it.

Finance depends on the other functions as much as they depend on it.

Finance is close to accounting and to economics, but its main question differs.

Think of a bookshop on a busy street. Someone chooses and buys the books. Someone serves the readers. Someone records what was sold. Someone makes sure the shop can pay its suppliers. These are different kinds of work, and no one person can do all of them well at all times. Where does the work of finance sit among them?

A **business function** is one main kind of work in a business, such as selling, making or managing money. In a bookshop, buying books from publishers is one function, selling to readers is another, and looking after money is a third. Larger businesses give each function its own team; a small business may give several functions to one person.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhaka, founded on 12 January 1998 by Abdur Rahim Chowdhury. Since 2019 it has been run by his sons Imran and Tanvir. Imran is responsible for the shop and buying. Tanvir is responsible for accounts and money.

Managers decide what a business will do. They choose goods, prices, staff and suppliers. Every such decision has a money side. Suppose a bookshop wishes to add a shelf of stationery. Someone must find the money for the new goods and the shelf. Someone must judge whether the new goods will bring in enough to repay that money. Finance serves the other functions in this way, by supplying funds and by judging the money side of their decisions. It informs the decision; it does not replace it.

The dependence runs both ways. A business with perfect money records but no customers will fail, because finance cannot create sales. In a larger business, the person who leads this function is called the financial manager; Chapter 2, The Financial Manager's Work, describes the role. In the running case, one brother holds that duty beside his work on accounts.

Two nearby subjects are often confused with finance. Accounting is the work of recording and reporting what has happened to the money of a business. Economics studies how people and societies choose when money and resources are limited. Finance draws on both, but its main question looks forward: what should we do with money, and where do we find it? The table below sets the three side by side, using a tailor's shop. Read each row from left to right: the subject, the question it asks, and an example.

FINANCE, ACCOUNTING AND ECONOMICS COMPARED

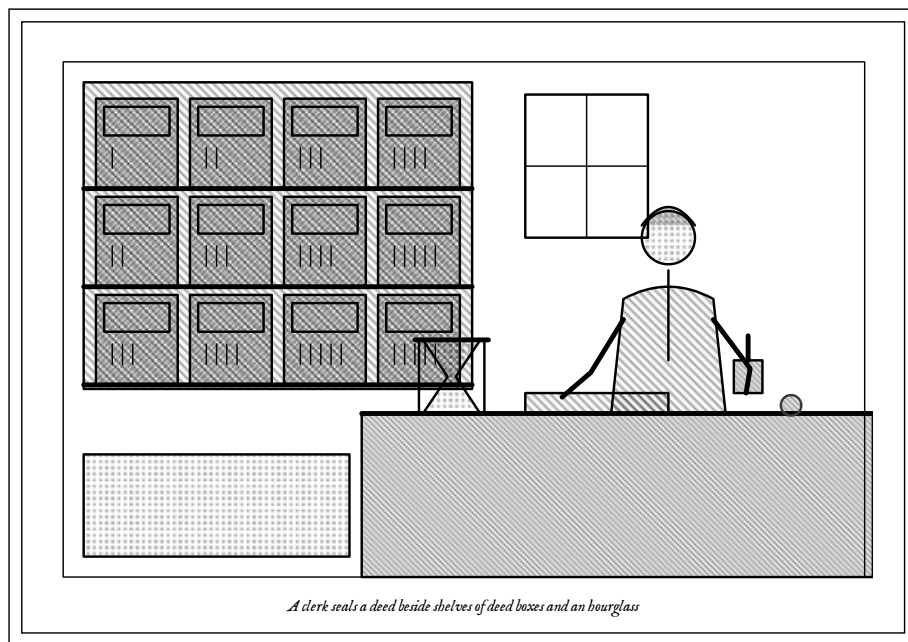
SUBJECT	MAIN QUESTION	EXAMPLE AT A TAILOR'S SHOP
Accounting	What has happened to our	Recording that Tk 6,000 was

What you should already know before you begin.

A puzzle: which of two shops was run better?

A second puzzle: what happens when the owner is not the manager?

The chapter answers two questions, one about aims and one about people.



A clerk seals a deed beside shelves of deed boxes and an hourglass

CHAPTER VI

Uneven Cash Flows

Can two plans that pay the same total be worth different amounts?



This chapter extends present value and future value to payments that change in size from year to year. It shows how to value such a series, how to split it into a level part and an odd part, and how to compare two series at a common rate.

GO SLOWLY HERE.

A market stall shows how a bigger profit today can hurt the stall tomorrow.

Profit maximisation. The aim of earning the largest profit possible in a period, usually one year.

The first and oldest answer to the question of aims is profit maximisation.

A business that never earns a profit cannot survive, so the

You will need two ideas from earlier chapters. The first is *wealth*, first defined as a stock of value (Chapter 1, section 1.1).¹ The second is *profitability* (Chapter 2, section 2.3).² Terms such as dividend, retained earnings and financial manager are used here as they were taught earlier.

Two friends each run a small shop that sells school supplies. At the end of the year the first friend reports a larger profit than the second. Most people answer at once that the first friend ran the better business. Then we learn more. The first friend stopped buying new stock in the last months, and the shelves are nearly bare. The second friend kept the shelves full, and a large order from a school is already waiting for the new year. The answer is now less certain.

Now change the story. The owner of one shop moves to another city and hires a manager to run it. The owner wants a business that stays strong for many years. The manager is paid a bonus that depends on this year's results. Will the manager choose what the owner would choose? The manager may be honest and hard-working and still face a different set of rewards.

These two stories lead to two questions. What should a business aim for? And what happens when the people who own a business are not the people who run it? The first question is answered by the *goal of the business*. The second is answered by a body of ideas called agency theory. We take the first question in sections 3.1 and 3.2 and the second in section 3.3. A worked case then uses the facts of Meghna and Sons to test every idea.

* * *

3.1 PROFIT MAXIMISATION AND ITS LIMITS

Begin with a stall in a market. The stall holder sells tomatoes and could earn more this week by mixing in spoiled fruit and charging the usual price. This week the profit is larger. Next week the buyers who were cheated stay away, and the profit falls below its old level. The choice that gave the best result today gave a poor result over a longer time. Keep this picture in mind, because the whole of section 3.1 repeats it in business form.

The oldest answer to the question "what should a business aim for?" has a name. **Profit maximisation** is the aim of earning the largest profit possible in a period, usually one year. A bookshop that chooses between two plans for its stock, and picks the plan with the larger profit for the year, follows this aim. The aim has real strengths. It is easy to state, it is easy to measure from the records, and it is linked to profitability, the idea you met in Chapter 2.

Profit matters for every person connected with a business. For

¹ Chapter 1, 'Money, Finance and the Firm', section 1.1 'What Finance Is': wealth was first defined there as a stock of value.

² Chapter 2, 'The Financial Manager's Work', section 2.3: profitability was taught there. Section 2.2 'The Three Core Decisions' shows what owners give up or gain through dividend and financing decisions.

AMORTISATION OF A LOAN OF TK 100,000 AT 10% (TK)

YEAR	OPENING BALANCE	INTEREST	PART OF LOAN REPAID	CLOSING BALANCE
1	100,000.00	10,000.00	30,211.48	69,788.52
2	69,788.52	6,978.85	33,232.63	36,555.89
3	36,555.89	3,655.59	36,555.89	0.00

Before moving on. To find an unknown payment, divide the known sum by the right factor: the target by the future value factor for a sinking fund, the loan by the present value factor for amortisation. *You should now be able to say* which factor belongs to saving and which to borrowing.

* * *

WORKED EXAMPLE: SAVING FOR A VAN

Case file, illustrative. Meghna and Sons, 15 August 2024. A van replacement needs Tk 500,000 in 5 years. The interest rate is 8% a year, and the deposits are equal and made at each year end. The equal year-end deposit is Tk 85,228.23.

Given. Target Tk 500,000 at the end of year 5; equal ordinary deposits; interest rate 8% a year; $n = 5$. *Find.* The equal year-end deposit P . *Formula.* Deposit = target $\div$ future value factor, so $P = FVA \div [(1 + i)^n - 1] \div i$. *Steps.* The factor for 8% and 5 payments is 5.866601 (section 5.2). Then $P = Tk\ 500,000 \div 5.866601 = Tk\ 85,228.23$. *Answer.* Meghna and Sons must deposit Tk 85,228.23 at the end of each of 5 years. *Check.* Reverse the step: $Tk\ 85,228.23 \times 5.866601 = Tk\ 500,000.02$. The two poisha of difference arise because the deposit was rounded to the nearest poisha.

Compare the parts. Five deposits total $5 \times Tk\ 85,228.23 = Tk\ 426,141.15$. Interest therefore supplies about Tk 73,859 of the Tk 500,000, which is about 15%. A business that saves over a longer period needs smaller deposits, because interest has more time to work.

* * *

SUMMARY

An annuity is a series of equal payments at equal intervals for a set number of periods. It is ordinary when payments fall at period ends and due when they fall at period starts; a perpetuity never ends. The future value of an ordinary annuity is $P \times [(1 + i)^n - 1] \div i$, and its present value is $P \times [1 - (1 + i)^{-n}] \div i$. For an annuity due, multiply either result by $1 + i$. A perpetuity is worth $P \div i$. A target sum divided by the future value factor gives the deposit of a sinking fund, and a loan divided by the present value factor gives the instalment under amortisation.

THINK IT THROUGH

Recap: a target sum divided by the future value factor gives the deposit; a loan divided by the present value factor gives the instalment.

The van problem asks for an unknown payment and then checks it by future value.

Worked example 6 divides to find the deposit and then reverses the step as a check.

Deposits supply most of the target and interest supplies the rest.

An annuity is a series of equal payments; two factors value it, and division finds an unknown payment.

Value. Word in use. In daily life, how much we like or need something. In finance, an amount of money that a market would pay for something.

The owners' share in a business also has a market value.

Wealth maximisation. The aim of making the market value of the owners' stake as large as possible over time.

Wealth maximisation is the improved answer to the question of aims.

The owners of a business hold a stake in it, meaning a share of what the business owns after its debts are paid. That stake also has a market value: the price at which it could be sold today. For a large business whose common stock is traded in a public market, the price can be read each day. For a small family business, no daily price exists, so owners estimate what a buyer would pay.

We can now state the improved answer. **Wealth maximisation** is the aim of making the market value of the owners' stake as large as possible over time. Remember that wealth, from Chapter 1, is a stock of value. Wealth maximisation says that the owners' stock of value should grow, and that the growth is measured by market value, not by the profit of one year. A bookshop that gives up a small profit this year to win a large loyal customer group has chosen this aim if buyers would pay more for the business afterwards.

IN SHORT

Wealth maximisation means making the market value of the owners' stake as large as possible over time.

PAUSE AND CHECK 3.P1 A shop owner can raise this year's profit by cutting the quality of goods, but customers will leave later. Explain in two sentences why wealth maximisation would reject the cut.

Why does this aim solve the two problems from section 3.1? A buyer who wants to purchase a business thinks about all the profits that the business is likely to earn, not the profit of the last year only. A buyer pays more for profit that comes sooner, because it can be used sooner. A buyer pays less for profit that is uncertain, because it may not arrive. So market value already contains an opinion about timing and about risk. When we aim at market value, we aim at what the owners could gain by selling their stake. Return to the restocking choice of 15 May 2024. A buyer would see that the cut lowers the next year's profit by more than it raises this year's, and would pay less for the business. The cut lowers the owners' wealth.

The table that follows compares the two aims. Read each row from left to right: the first column names a matter, and the next two columns say how each aim treats it.

THE TWO AIMS COMPARED

MATTER	PROFIT MAXIMISATION	WEALTH MAXIMISATION
What is measured	Profit of one period	Market value of the owners' stake
Time	Looks at one year	Looks at all future years
Risk	Not considered	Included in the price a buyer will pay
Main weakness	Can favour a short-term gain	Needs a market price, or a careful estimate

Stakeholder. An investor or

A business affects many people besides its owners. A stakeholder

Given. Tk 120,000 paid at the end of each of 4 years; discount rate 8% a year. *Find.* The present value of the four payments. *Formula.* $PVA = P \times [1 - (1 + i)^{-n}] \div i$, with $P = \text{Tk } 120,000$, $i = 0.08$ and $n = 4$. *Steps.* First the power: $1.08^{-4} = 0.73502985$. Next the factor: $(1 - 0.73502985) \div 0.08 = 0.26497015 \div 0.08 = 3.312127$. Then the value: $\text{Tk } 120,000 \times 3.312127 = \text{Tk } 397,455.24$. The case file gives Tk 397,455.22, because the factor shown is rounded to six places; the unrounded factor gives the case figure. *Answer.* The four payments are worth Tk 397,455.22 today, although they total Tk 480,000. *Check.* The table below discounts each payment alone, as in worked example 3.

Read each row from left to right: the payment, multiplied by the discount multiplier for that year, gives its present value. The multipliers are shown to six places, and the present values are computed from unrounded multipliers.

PRESENT VALUE OF EACH RENT PAYMENT AT 8% (TK)

YEAR	PAYMENT	DISCOUNT MULTIPLIER	PRESENT VALUE
1	120,000	0.925926	111,111.11
2	120,000	0.857339	102,880.66
3	120,000	0.793832	95,259.87
4	120,000	0.735030	88,203.58
Total	480,000		397,455.22

Before calculators, people did not calculate powers each time. They read the multiplier from a **factor table**. A factor table is a printed table of ready-made multipliers for given rates and numbers of periods. A reader finds the row for the number of periods, moves across to the column for the factor needed, and multiplies that factor by the payment. The table below lists the two annuity factors for 8% a year.

ANNUITY FACTORS FOR AN INTEREST RATE OF 8% A YEAR

NUMBER OF PAYMENTS	FUTURE VALUE FACTOR	PRESENT VALUE FACTOR
1	1.000000	0.925926
2	2.080000	1.783265
3	3.246400	2.577097
4	4.506112	3.312127
5	5.866601	3.992710

Row 5 of the first column holds 5.866601, the factor used for the savings plan in section 5.2. Row 4 of the second column holds 3.312127, the factor used for the rent offer. The same method serves for any rate, provided that the table lists it.

The present value of an annuity due follows the same reasoning as before. Each payment comes one period sooner, so it is discounted one period less and is worth $(1 + i)$ times as much. Therefore the present value of an annuity due equals the present value of the ordinary annuity multiplied by $(1 + i)$. For three payments of Tk 1,000 at 10%: $\text{Tk } 2,486.85 \times 1.10 = \text{Tk } 2,735.54$. Check by payment: the first payment, made today, is worth Tk 1,000, and

Worked example 4 values the landlord's offer in the standard order.

The table discounts each rent payment separately; the present values total the answer.

Printed tables store factors, so that no powers need to be calculated.

Factor table. A printed table of ready-made multipliers for given rates and numbers of periods.

Two cells of the table are the factors used in the case file.

For an annuity due, each payment is one period closer, so the present value is the ordinary value times $1 + i$.

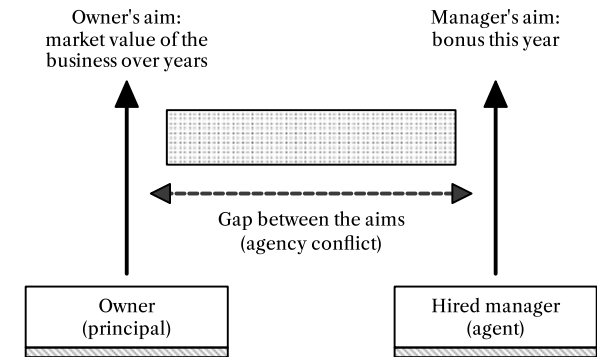


FIG. 3.1. Owner and hired manager with the gap between their aims

The following case file shows an agency conflict with exact figures.

Case file, illustrative. Date: 1 April 2024. Business: Meghna and Sons opens a second branch and hires Mr Shafiq to run it. His pay is a yearly bonus of 5% of branch sales. Plan X (steady pricing): sales Tk 1,500,000; profit before bonus Tk 150,000; bonus Tk 75,000; owners keep Tk 75,000. Plan Y (deep discounts and credit to schools): sales Tk 2,000,000; profit before bonus Tk 70,000; bonus Tk 100,000; owners keep minus Tk 30,000.

The next table puts the figures in one place. Read down each plan's column: the sales come first, then the profit before the bonus, then the bonus, and last the amount the owners keep.

PLAN X AND PLAN Y FOR THE SECOND BRANCH (TK)

ITEM	PLAN X	PLAN Y
Sales	1,500,000	2,000,000
Profit before bonus	150,000	70,000
Bonus (5% of sales)	75,000	100,000
Owners keep	75,000	-30,000

The manager gains Tk 25,000 more under Plan Y, because Tk 100,000 less Tk 75,000 is Tk 25,000. The owners are worse off by Tk 105,000, because Tk 75,000 less minus Tk 30,000 is Tk 105,000. The manager's bonus rises with sales, while the owners' share depends on profit. Deep discounts raise sales and lower profit, so the bonus rule rewards the plan that the owners would reject. The manager prefers Plan Y and the owners prefer Plan X. **PAUSE AND CHECK 3.P2** In the case file, which figure in the manager's pay rule causes the conflict, and why?

The viewpoints differ. The owner sees a bonus rule that pays for sales whatever the result for profit. The manager sees a rule to follow, and a manager who follows it has not broken any promise. A lender, who is a person that gives money to the business and

The case file shows two plans and an agent who gains while the owners lose.

A manager paid a share of sales is offered two plans.

The table shows the plans side by side.

Each side prefers a different plan.

Viewpoints: the same rule looks different to owner, manager and lender.

Given. Tk 60,000 deposited at the end of each of 5 years; interest rate 8% a year. *Find.* The sum held at the end of year 5. *Formula.* $FVA = P \times [(1 + i)^n - 1] \div i$, with $P = \text{Tk } 60,000$, $i = 0.08$ and $n = 5$. *Steps.* First the power: $1.08^5 = 1.46932808$. Next the factor: $(1.46932808 - 1) \div 0.08 = 0.46932808 \div 0.08 = 5.866601$. Then the sum: $\text{Tk } 60,000 \times 5.866601 = \text{Tk } 351,996.06$. *Answer.* The deposits reach Tk 351,996.06 at the end of year 5. *Check.* The deposits total $5 \times \text{Tk } 60,000 = \text{Tk } 300,000$. The answer exceeds this by Tk 51,996.06, which is the interest earned. The table below confirms the figure year by year.

To see the factor at work, follow the balance through the five years. Each year the opening balance earns 8% interest, and then the deposit of Tk 60,000 arrives at the year end. Read each row from left to right: opening balance plus interest plus deposit gives the closing balance.

BALANCE OF THE DEPOSITS, YEAR BY YEAR (TK)

YEAR	OPENING BALANCE	INTEREST AT 8%	DEPOSIT AT YEAR END	CLOSING BALANCE
1	0.00	0.00	60,000.00	60,000.00
2	60,000.00	4,800.00	60,000.00	124,800.00
3	124,800.00	9,984.00	60,000.00	194,784.00
4	194,784.00	15,582.72	60,000.00	270,366.72
5	270,366.72	21,629.34	60,000.00	351,996.06

The last closing balance equals the answer from the formula, so the shortcut and the long method agree. The next figure draws the closing balances as bars. Each bar stands for the balance at the end of one year, and its height shows money in taka. The lower, line-hatched part of a bar is the total deposited so far; the upper, cross-hatched part is interest earned. Notice that the interest part grows faster each year, because interest is earned on earlier interest.

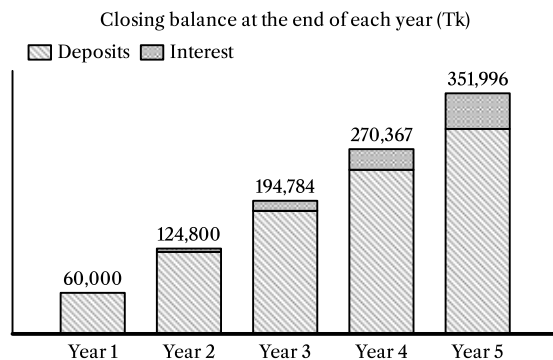


FIG. 5.2. Closing balance of the savings plan at the end of each year, split into deposits and interest

Now consider the annuity due. If each deposit is made one year

Worked example 2 repeats the method with the case figures, in the standard order from Given to Check.

The table follows the balance year by year: opening balance, interest, deposit, closing balance.

The last closing balance matches the formula, and the figure splits each balance into deposits and interest.

An annuity due earns one

The question asks which aim each side follows and what the conflict costs.

Reasoning works through the plans and then through the restocking choice.

The conclusion names the preferences and the aim that favours the cut.

The check confirms the arithmetic and the sense of the result.

A ladder of situations shows the idea in new settings.

Familiar case: a tailor paid on monthly profit.

Reshaped case: a driver paid per trip.

New setting: a school principal and a bursar.

Combined case: three aims at once.

Summary of the chapter.

would raise this year's profit by Tk 80,000 and lower next year's profit by Tk 150,000.

Question. Which plan does each side prefer, and what is the loss in value from the conflict? Which aim, profit or wealth, favours the restocking cut?

Reasoning. First we find the bonus under each plan: 5% of Tk 1,500,000 is Tk 75,000, and 5% of Tk 2,000,000 is Tk 100,000. Second we find what the owners keep: Tk 150,000 less Tk 75,000 is Tk 75,000 under Plan X, and Tk 70,000 less Tk 100,000 is minus Tk 30,000 under Plan Y. Third we compare: the manager gains Tk 25,000 more under Plan Y and the owners lose Tk 105,000. Fourth we turn to the restocking choice. This year's profit rises by Tk 80,000, and next year's falls by Tk 150,000, so the two years together show a fall of Tk 70,000.

Conclusion. The manager prefers Plan Y and the owners prefer Plan X. The loss in value from the conflict is Tk 105,000, and the cost of preventing it, such as checking the records, is added to that loss to find the agency cost. A rule of one-year profit favours the restocking cut, while wealth maximisation rejects it, because the market would pay less for the owners' stake.

Check. Tk 75,000 plus Tk 30,000 is Tk 105,000, which agrees with the loss. Tk 100,000 less Tk 75,000 is Tk 25,000, which agrees with the manager's gain. The result is sensible, because a bonus on sales rewards sales and does not reward profit.

THE SAME IDEA IN FOUR SITUATIONS

The idea is useful only if you can recognise it elsewhere. Each situation below changes the setting while keeping the idea.

A familiar case. A tailor manages a shop for its owner and is paid on the monthly profit. The tailor skips the buying of fabric to show a larger profit this month. The owner loses the sales of later months. Here the tailor is the agent and the owner is the principal.

A reshaped case. A taxi owner pays a hired driver for each trip. The driver takes many short trips and drives hard, and the car wears out faster. The driver's reward is the number of trips, and the owner's reward is the condition and profit of the car.

A new setting. The principal of a school wants the best use of limited fees. The bursar, who keeps the school accounts, is judged on spending less than the budget. The bursar may delay repairs that the principal wants. The setting has no profit, yet the pattern is the same.

A combined case. A hired manager wants a larger bonus, the owners want a larger market value, and a lender wants safe repayment with interest. All three may prefer different plans, and the manager's choice affects the other two.

SUMMARY

The goal of the business is wealth maximisation, which is the aim of making the market value of the owners' stake as large as possible over time. Profit maximisation, the aim of the largest profit in a period, ignores timing and risk. Stakeholders are affected by the

The timing of the first payment needs care. Does it fall at the end of the first period or at its start? The answer matters, because a payment made one period earlier has one more period in which to earn interest. Finance therefore separates two kinds of annuity.

An **ordinary annuity** is an annuity whose payments fall at the end of each period. An **annuity due** is an annuity whose payments fall at the start of each period. Take three payments of Tk 1,000 over three years. In an ordinary annuity they are made at the end of year 1, year 2 and year 3. In an annuity due they are made at the start of year 1, year 2 and year 3. The start of year 1 is today, and the start of year 2 is the end of year 1. Loan instalments and bank interest are usually ordinary annuities, because they are paid after the period has run. Rent is often an annuity due, because landlords ask for it in advance. When no kind is stated, take the annuity to be ordinary.

The figure below shows both kinds on a timeline. Read each line from left to right. The ticks on a line are moments in time, numbered 0, 1, 2 and 3, where 0 is today and each later number is the end of a year. Each block is one payment of Tk 1,000. On the upper line the blocks sit at the ends of the years, at 1, 2 and 3. On the lower line they sit at the starts, at 0, 1 and 2. The lower blocks are shifted one step to the left, and that shift is the whole difference between the two kinds.

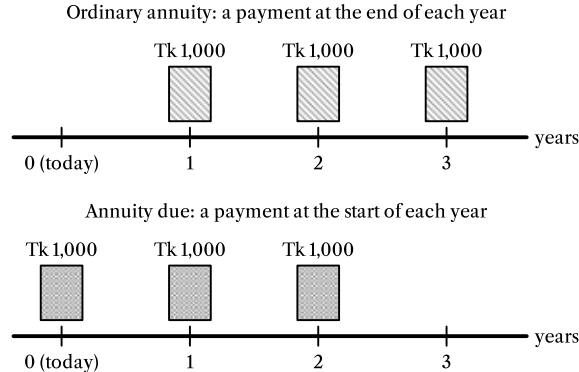


FIG. 5.1. Two timelines of three yearly payments: at the end of each year (ordinary annuity) and at the start of each year (annuity due)

Four viewpoints help here. The saver sees an annuity as a way to grow a sum by regular deposits. The borrower sees it as a series of instalments that must be met on time. A landlord, as owner of the property, sees it as income. The financial manager sees it as a pattern whose worth must be measured, so that rent, loans and savings can be compared fairly.

One case is unusual but useful. Suppose that the payments never stop. A series of equal payments with no final date is a **per-**

Whether a payment falls at the start or at the end of a period changes its value.

An ordinary annuity pays at period ends; an annuity due pays at period starts.

Ordinary annuity. An annuity whose payments fall at the end of each period.

Annuity due. An annuity whose payments fall at the start of each period.

Due. Word in use. In daily life, owed or expected on a date, as in "the rent is due on Friday". In finance, the word "due" in annuity due means payment at the start of each period.

The figure reads from left to right: ticks on the line are moments in time, and blocks are payments.

To the saver an annuity grows a sum; to the borrower it is a burden; to the financial manager it is a pattern to value.

A perpetuity is an annuity with no final payment.

Perpetuity. An annuity that

it defined agency cost and applied it to the way a business is owned and financed.

REVIEW QUESTIONS

- 3.1 Explain why profit maximisation can mislead when returns come at different times or carry different risk.
- 3.2 Explain wealth maximisation in plain words to a village shopkeeper who has never heard the term.
- 3.3 Describe an agency conflict between a taxi owner and a driver who is paid for each trip.
- 3.4 Name three ways in which owners can bring the interests of a manager closer to their own, and say what each costs.

What to bring from Chapter 1.

A friend offers the same amount today or a year from today.

Raising the later amount shows where people change their minds.

The chapter turns this feeling into arithmetic.

GO SLOWLY HERE.

A farmer's seed shows why money in hand early can become more later.

Time value of money. The idea that money received sooner is worth more than the same amount received later.

The farmer's lesson has a name: the time value of money.

Three reasons support the idea, and this chapter uses the first.

You will need two ideas from Chapter 1, section 1.1: *funds* and *cash flow*.¹ Debt, equity and wealth are used here as they were taught earlier.

A friend offers you a choice. You may take Tk 100,000 today, or you may take Tk 100,000 one year from today. The friend is honest, and you are certain you will be paid on the agreed day. Which do you choose? Nearly everyone chooses today. Yet the two amounts are equal. The choice seems to say that the same number of taka is not the same thing at two different times.

Now suppose the friend raises the later offer to Tk 105,000, then to Tk 110,000, then to Tk 120,000. At some amount you will change your answer and choose to wait. The amount at which you change measures how much you value time. It differs between people, but it is never zero. The habit is not foolish. Money in hand can be used, kept safe or put to work, and money promised for later can do none of these things until the day arrives.

This chapter turns that feeling into arithmetic. Section 4.1 explains the idea and the words around it. Section 4.2 moves an amount forward in time to find what it will grow to. Section 4.3 moves an amount back in time to find what it is worth today. Section 4.4 allows growth to happen more than once a year. A set of worked examples closes the chapter, and every one of them can be done by hand. We study one amount at a time here; payments that repeat are the subject of Chapter 5, 'Annuities'.

* * *

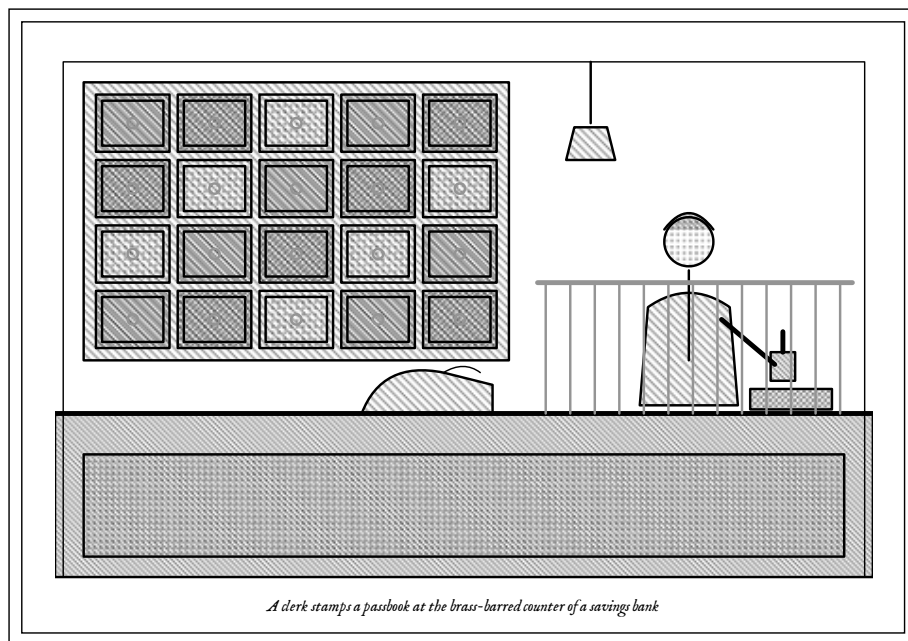
4.1 WHY TIME CHANGES THE VALUE OF MONEY

Begin with a farmer who has 10 kilograms of rice seed. The farmer can eat the seed today, or sow it and wait for the harvest. If the farmer sows it, the harvest in six months may be 40 kilograms. Ten kilograms today is therefore worth more than ten kilograms after the harvest, because the seed of today can become forty. Keep this picture in mind. Seed that is sown grows, and seed that is eaten is gone. Money in hand can be sown in the same way, as a deposit in a bank or as stock for a shop.

We can now name the idea. The **time value of money** is the idea that money received sooner is worth more than the same amount received later. Tk 100,000 received today is worth more than Tk 100,000 received in a year, because today's money can be put to work for that year. The idea does not depend on prices rising. It holds even when every price stays the same, because the reason is the earning power of funds, not the cost of goods.

Three reasons support the idea. First, money in hand can earn: it can be lent, deposited or used to buy stock that will be sold at a profit. Second, prices usually rise over time, so a taka buys a little less each year. Third, a promise for later may fail, while money in hand is certain. This chapter works with the first reason and treats every future amount as certain. Chapter 9, 'Return and Risk', deals

¹ Chapter 1, 'Money, Finance and the Firm', section 1.1 'What Finance Is': funds and wealth define what is being timed.



A clerk stamps a passbook at the brass-barred counter of a savings bank

CHAPTER V

Annuities

How does one value a promise that arrives in equal instalments?



This chapter teaches equal payments made at equal intervals. It shows how to find their future value and their present value, how to read a factor table, and how to find an unknown payment for a savings target or a loan.

COMMON MISTAKES AND CORRECTIONS

MISTAKE	CORRECTION
Tk 100,000 today and Tk 100,000 in a year are worth the same because the numbers match	Money today can be put to work, so the later amount has a lower present value
Simple and compound interest are equal	They agree only in the first period; after that compound interest is larger
Present value is the amount you will receive	Present value is today's worth of that amount; the amount received later is the future value
The discount rate and the interest rate are unrelated	They can be the same number, used in opposite directions
A yearly rate can be used for half-years without change	Divide the rate by the periods in a year and multiply the years by the same number

IN HISTORY

In 1683 the Swiss mathematician Jacob Bernoulli (1654–1705) studied a question about compound interest. Suppose Tk 1 is lent at 100% a year. Added once, the interest gives Tk 2 at the end of the year. Added twice, at 50% each half-year, it gives Tk 2.25. Bernoulli asked what happens if interest is added more and more often. The amount keeps rising, but each step adds less than the one before.

Bernoulli showed that the amount never reaches Tk 3, and that it stays above Tk 2. The limit is the number now written *e*, about 2.71828; the letter was introduced later by Leonhard Euler, who first used it in a letter in 1731. In this chapter the same idea appears in a smaller form. At 12% a year, the effective annual rate rises from 12% to 12.36% with half-yearly compounding and to 12.55% with quarterly compounding, but it can never rise above about 12.75%, however often interest is added.

SOURCES

Jacob Bernoulli's study of compound interest in 1683 is the first known step towards the number *e*. It matters here because it showed that adding interest more often raises the total, but only up to a limit.

Leonhard Euler, *Introductio in analysin infinitorum* (1748), gave the full treatment of *e* and showed that it is the limit of Bernoulli's expression. It matters because it fixed the number and its notation for later use in finance and science.

REVIEW QUESTIONS

- Find the future value of Tk 15,000 at 9% a year for 6 years.
- Find the present value of Tk 80,000 due in 4 years at 11% a year.

In 1683 Jacob Bernoulli asked what happens when interest is added more and more often.

*Bernoulli showed the amount stays between 2 and 3, and the limit became the number *e*.*

*Bernoulli's 1683 problem is the early source of the limit that became the number *e*.*

Euler's book of 1748 gave the number its full treatment.

Single sum. One payment or receipt made at one point in time.

A single sum is one payment at one moment.

Simple interest. Interest worked out on the original amount only.

The first answer: interest on the original amount only.

Compound interest.

Interest worked out on the original amount plus interest already earned.

The second answer: interest on the amount plus earlier interest.

Simple interest ignores earlier interest; compound interest includes it.

Future value. What an amount today will grow to after a stated time at a stated rate.

Future value is the amount an account will hold later, found by multiplying once for each year.

The future value factor is a number that can be used for any amount.

First worked example: given, find and formula.

First worked example: steps, answer and check.

The business case uses the same method on a larger deposit.

We first need a name for the Tk 10,000. A **single sum** is one payment or receipt made at one point in time. The deposit of Tk 10,000 today is a single sum. Payments that repeat each year are not single sums; Chapter 5 handles them.

The first answer is **simple interest**, which is interest worked out on the original amount only. Under simple interest the second year's interest is again 10% of Tk 10,000, which is Tk 1,000, so the account holds Tk 12,000 after two years. Each year adds the same amount. In words: interest = original amount × interest rate × number of years.

The second answer is **compound interest**, which is interest worked out on the original amount plus interest already earned. In the second year the bank pays 10% of Tk 11,000, which is Tk 1,100, so the account holds Tk 12,100. The extra Tk 100 is interest earned on the first year's interest. Interest on interest is the source of the extra growth. Repeating this addition year after year is called compounding.

PAUSE AND CHECK 4.P1 Using the deposit of Tk 10,000 at 10% a year, find the amount in the account after three years under compound interest. Work year by year.

Before moving on to the shortcut. Simple interest uses the original amount every year. Compound interest uses the original amount plus the interest already earned, so it grows faster.

Year-by-year working becomes slow when the time is long, so a shortcut is useful. Each year the amount is multiplied by (1 + *i*). In the example it is multiplied by 1.10, and after two years it has been multiplied twice. The **future value** is what an amount today will grow to after a stated time at a stated rate. In words: future value = single sum × (1 + interest rate) to the power of the number of years. In symbols: FV = S × (1 + *i*) to the power of *n*, where FV is the future value, S is the single sum today, *i* is the interest rate for one period as a decimal, and *n* is the number of periods.

The number (1 + *i*) to the power of *n* is called the future value factor. It does not depend on the size of the single sum, so it can be worked out once and used for any amount. For 10% and 2 years it is 1.10 × 1.10 = 1.21.

Given. S = Tk 10,000; *i* = 10% a year = 0.10; *n* = 2 years. *Find.* The future value, FV. *Formula.* In words: future value equals the single sum times one plus the interest rate, to the power of the number of years. In symbols: FV = S × (1 + *i*) to the power of *n*.

Steps. (1) 1 + *i* = 1 + 0.10 = 1.10. (2) 1.10 to the power of 2 = 1.10 × 1.10 = 1.21. (3) FV = Tk 10,000 × 1.21 = Tk 12,100. *Answer.* The Tk 10,000 will grow to Tk 12,100 in two years. *Check.* Year by year, Tk 10,000 became Tk 11,000 and then Tk 12,100, so the two methods agree. The result is Tk 100 above the simple interest total of Tk 12,000, as expected.

Case file, illustrative. Date: 1 July 2024. Meghna and Sons deposits Tk 200,000 at 8% a year for 5 years. Simple interest gives Tk 200,000 × (1 + 0.08 × 5) = Tk 280,000. Compound interest gives Tk 200,000 × 1.08 to the power of 5 = Tk 293,865.62, using a future value factor of 1.469328.

Case file, illustrative. Date: 1 July 2024. A deposit of Tk 200,000 at 8% a year is compounded half-yearly at 4% for 10 periods and grows to Tk 296,048.86. The effective annual rate is 8.16%.

The working is $Tk\ 200,000 \times 1.04$ to the power of 10 = Tk 200,000 $\times 1.480244$ = Tk 296,048.86. The effective annual rate is $1.04 \times 1.04 - 1 = 0.0816$, which is 8.16%. Yearly compounding gave Tk 293,865.62 in section 4.2, so compounding twice a year earns Tk 2,183.24 more over the five years. A common slip is to use 8% for each half-year, or to use 5 periods with the half-year rate. Always change both the rate and the number of periods.

Before moving on. The compounding period is the time between additions of interest. The rate per period is the yearly rate divided by the periods in a year, and the number of periods is the years times the periods in a year. The effective annual rate is the yearly rate that gives the same growth. You should now be able to say why 12% compounded half-yearly is the same as 12.36% compounded once a year.

WORKED EXAMPLES

These examples are a little harder than those in the sections. Each follows the same order: Given, Find, Formula, Steps, Answer and Check.

A REALISTIC FUTURE VALUE

Given. S = Tk 36,500; i = 7.5% a year = 0.075; n = 6 years. *Find.* The future value, FV. *Formula.* In words: future value equals the single sum times one plus the interest rate, to the power of the number of years. In symbols: $FV = S \times (1 + i)$ to the power of n.

Steps. (1) $1 + i = 1.075$. (2) Multiply by 1.075 six times, keeping six decimal places: 1.075, 1.155625, 1.242297, 1.335469, 1.435629, 1.543302. So 1.075 to the power of 6 = 1.543302. (3) $FV = Tk\ 36,500 \times 1.543302 = Tk\ 56,330.52$. A calculator that keeps every digit gives Tk 56,330.51; the one paisa gap comes from rounding the factor. *Answer.* Tk 36,500 grows to about Tk 56,331 in six years. *Check.* Simple interest would give $Tk\ 36,500 \times (1 + 0.075 \times 6) = Tk\ 36,500 \times 1.45 = Tk\ 52,925$. The compound result is higher, as it should be.

FINDING A MISSING RATE

Given. S = Tk 25,000; FV = Tk 33,275; n = 3 years. *Find.* The interest rate, i. *Formula.* Future value factor = $FV \div S$. Then find that factor in the row for n years. *Steps.* (1) $33,275 \div 25,000 = 1.331$. (2) In the row for 3 years of the future value table, the factors are 1.191016 at 6%, 1.259712 at 8% and 1.331000 at 10%. *Answer.* The interest rate is 10% a year. *Check.* $Tk\ 25,000 \times 1.331 = Tk\ 33,275$, which agrees with the given amount.

FINDING A MISSING TIME

Given. S = Tk 40,000; FV = Tk 58,773.12; i = 8% a year. *Find.* The number of years, n. *Steps.* (1) $58,773.12 \div 40,000 = 1.469328$. (2) In the 8% column of the future value table, the factor 1.469328 appears in the row for 5 years. *Answer.* The time is 5 years. *Check.*

The business case compounds half-yearly.

Working of the case file and a comparison with yearly compounding.

A short summary of section 4.4 with a test sentence.

The examples here are a little harder than those in the sections.

Third worked example: given, find and formula.

Third worked example: Steps, answer and check.

Fourth worked example: a rate to be found from the table.

Fifth worked example: a time to be found from the table.

4.3 PRESENT VALUE OF A SINGLE SUM

GO SLOWLY HERE.

A target amount shows the need to work backwards.

Present value. What an amount to be received later is worth today at a stated rate.

Present. Word in use. In daily life, the time now, or a gift. In finance, the worth today of an amount due later.

Present value is the worth today of a later amount.

Discounting. Finding a present value by working backwards from a future amount at a rate.

Discounting moves an amount back in time by dividing once for each year.

Discount rate. The rate used to turn a future amount into a present value.

The discount rate is the rate used when working backwards.

The formula mirrors the one in section 4.2.

Second worked example: given, find and formula.

Second worked example: Steps, answer and check.

Tables of factors save repeated multiplication.

Now reverse the question. You know that you will need Tk 12,100 in two years, and a bank pays 10% a year with yearly compounding. How much must you deposit today? Section 4.2 gave a way from today to a later date. This section gives the way back. Think of one road: in section 4.2 we travelled forward along it, and here we travel the same road in the opposite direction.

The answer to the question has a name. The **present value** is what an amount to be received later is worth today at a stated rate. If Tk 12,100 is due in two years and the rate is 10%, its present value is the deposit that grows to Tk 12,100 in two years. Section 4.2 showed that Tk 10,000 grows to Tk 12,100 in two years at 10%. So the present value of Tk 12,100 is Tk 10,000.

The working has a name too. **Discounting** is finding a present value by working backwards from a future amount at a rate. Compounding multiplies by $(1 + i)$ for each year, and discounting divides by $(1 + i)$ for each year. Year by year: $Tk\ 12,100 \div 1.10 = Tk\ 11,000$, and $Tk\ 11,000 \div 1.10 = Tk\ 10,000$. Each step back removes one year of interest.

PAUSE AND CHECK 4.P2 Using the same rate of 10% a year, find the present value of Tk 13,310 due in three years. Divide by 1.10 once for each year.

The rate used in discounting has its own name. The **discount rate** is the rate used to turn a future amount into a present value. In our example it is 10%, the same number as the interest rate, because 10% is what the money could earn. One number serves in two directions: it is the interest rate when we move forward and the discount rate when we move back. Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', and the chapter after it use the discount rate to value projects.

The shortcut mirrors the one in section 4.2. In words: present value = future amount $\div (1 + \text{interest rate})$ to the power of the number of years. In symbols: $PV = FV \div (1 + i)$ to the power of n. Dividing by a factor is the same as multiplying by its reciprocal, $1 \div (1 + i)$ to the power of n, which is called the present value factor. For 10% and 2 years it is $1 \div 1.21 = 0.826446$.

Given. FV = Tk 12,100; i = 10% a year = 0.10; n = 2 years. *Find.* The present value, PV. *Formula.* In words: present value equals the future amount divided by one plus the interest rate, to the power of the number of years. In symbols: $PV = FV \div (1 + i)$ to the power of n.

Steps. (1) $1 + i = 1.10$. (2) 1.10 to the power of 2 = 1.21. (3) $PV = Tk\ 12,100 \div 1.21 = Tk\ 10,000$. *Answer.* The present value of Tk 12,100 due in two years is Tk 10,000. *Check.* Grow the answer forward: $Tk\ 10,000 \times 1.21 = Tk\ 12,100$. The present value is below the future amount, as it must be when the rate is above zero.

Tables of factors save repeated arithmetic. To read a table, find the row for the number of years, move across to the column for the rate, and read the factor. The first table gives future value

100,000 = Tk 4,000. The return is $6,000 \div 4,000 = \text{Tk } 10,000$. The rate is $10,000 \div 100,000 \times 100\% = 10\%$. *Answer.* The bond gave a return of Tk 10,000, a rate of return of 10% for the year. *Check.* Ten percent of Tk 100,000 is Tk 10,000, so the two answers agree.

Three limits deserve care. First, a return can be negative: if the bond could be sold only at Tk 96,000, the change in value would be -Tk 4,000 and the return would be Tk 2,000. Second, a rate means little unless the period is stated, so we say "10% a year". Third, an unsold gain is a gain on paper only, because the price may change before the sale. Different people also read return differently: the owner sees a reward for waiting, the financial manager sees a figure to set against the required return, and a lender sees only the interest.

Before moving on. Return is the total gain or loss over a period, income plus any change in value. The rate of return divides it by the amount first invested, so that investments of different size can be compared.

* * *

9.2 WHAT RISK IS

You can walk to a shop by two routes. On average each takes 25 minutes. Route A takes between 20 and 30 minutes, whatever the day. Route B takes 10 minutes on a quiet day and 40 minutes when the traffic is heavy. If you must catch a bus at a fixed hour, which route do you choose? Most people choose route A, although the averages are equal. The difference between the routes is not the average. It is the uncertainty of the arrival time.

Finance gives the name **risk** to this uncertainty. Risk means the chance that actual results will differ from those we expected, especially for the worse. A savings bond that pays a fixed amount has very little risk. A shop that hopes to sell Tk 7,200,000 of books in a year, but may sell less, has more. The word does not mean certain loss. A certain loss has no uncertainty, so it carries no risk.

How do we recognise more risk? Look at the possible results. If every possible result lies close to the expected one, the risk is small. If some results lie far above and others far below, the risk is large. Route B has more risk than route A for exactly this reason. The rest of this chapter turns that idea into numbers.

Test. To judge the risk of a choice, ask how far the actual result could fall from the expected result, and how likely that is. A choice whose results all lie near the expected one has low risk.

The puzzle at the start of the chapter shows why risk matters to a business. Owners and lenders can place their funds in safe or in uncertain uses. They accept the uncertain use only if they expect a higher return. This link between a higher expected return and

Return can be negative, depends on the period and may exist only on paper.

The section's two ideas are return and rate of return.

GO SLOWLY HERE.

Two routes with the same average time still feel different.

Risk. the chance that actual results differ from those expected, especially for the worse

Risk. Word in use. In daily life, the danger of harm, such as the risk of a fall. In finance, the chance that results differ from what was expected.

Risk is the chance that results differ from what was expected, especially for the worse.

The more widely the possible results are scattered, the greater the risk.

The reader gains a plain test for the amount of risk.

People bear risk only when they expect something in return.

YOU WILL NEED

Chapter 5, section 5.3: the present value of an annuity. Chapter 4, section 4.3: the present value of a single sum.

Two plans that pay the same total may still differ in value, because the order of the payments differs.

The shortcut for equal payments cannot be used here, because these payments are not equal.

The remedy is to return to single sums, and to learn a tidy way of working with them.

Wages that rise, fees that change and takings that follow the seasons all have payments of different sizes.

A mixed stream is a series of payments or receipts that differ in size from one period to the next.

Mixed stream. A series of payments or receipts that differ in size from one period to the next.

Stream. Word in use. In daily life, a small river or a flow of water. In finance, a series of payments or receipts arriving over time.

A single sum, an annuity and a mixed stream differ in how

Two stalls at a book fair each promise you Tk 6,000 over three years, paid at the end of each year. The first stall pays Tk 1,000, then Tk 2,000, then Tk 3,000. The second pays the same amounts in the opposite order: Tk 3,000, then Tk 2,000, then Tk 1,000. Both plans total Tk 6,000. Are they worth the same to you? Most people feel that the second plan is better, because the largest payment arrives soonest. The feeling is correct, but a feeling is not a number. How much better is it?

Chapter 5 gave a fast way to value equal payments. That way cannot be used directly now, because the payments differ from year to year. Yet real receipts seldom stay equal. A shop's sales grow as customers learn of it. A machine earns less as it ages. A loan may be repaid in large amounts first and small amounts later. If we could value only equal payments, we could value very little of business life.

The remedy lies in Chapter 4. Each payment, taken alone, is a single sum received on one date, and a single sum can be discounted. Once every payment has been discounted, the results can be added. This chapter shows the method, shows where the shortcut of Chapter 5 still helps, and shows how to compare two series fairly.¹

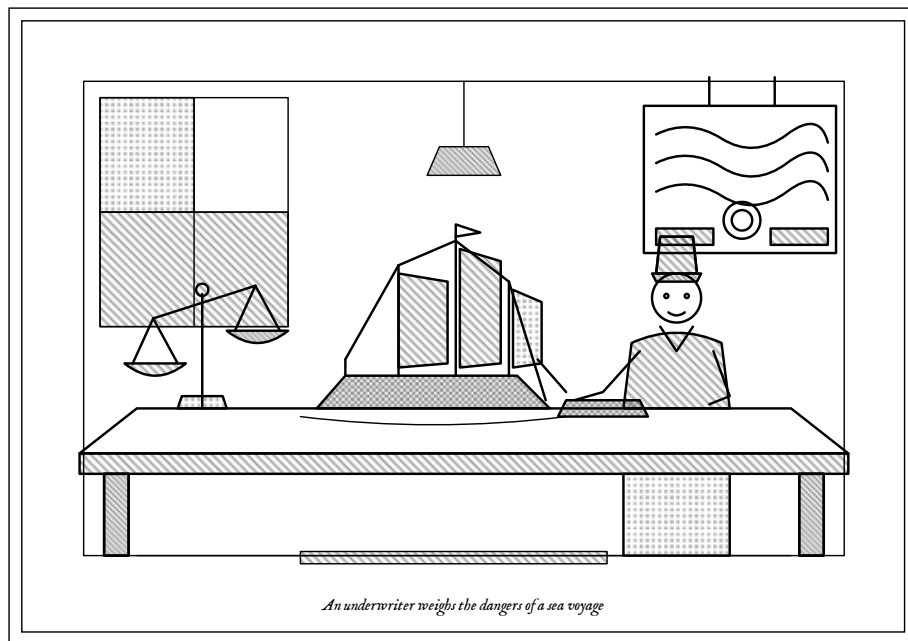
* * *

6.1 WHAT A MIXED STREAM IS

Begin with everyday life. A worker's wage may rise by a fixed amount each year. School fees change when a child moves to a higher class. The takings of a shop are larger before a festival and smaller after it. In each case a series of money amounts arrives over time, and the amounts are not equal. Finance has a name for this shape.

That name is **mixed stream**. A mixed stream is a series of payments or receipts that differ in size from one period to the next. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of three successive years form a mixed stream. A receipt is money coming in; a payment is money going out. The valuing method is the same for both, and in this chapter we use receipts. The word *mixed* tells us that the sizes are mixed, not equal.

Three shapes of cash flow now stand side by side. A single sum is one payment on one date. An annuity is several equal payments at ¹ Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'; Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'.



CHAPTER IX

Return and Risk

Which is the better bet: a small sure gain, or the chance of a large one?



This chapter shows how to state what an investment pays and how uncertain that payment is. You will describe a gain as a percentage; list the outcomes that might happen, find their average and measure how widely they scatter. Meghna and Sons uses these tools to compare two product lines.

The future value of a mixed stream grows each receipt to the last date and totals the results.

The case file gives two stall plans, and their figures are valued at 12% in the tables.

The first table discounts every receipt of both plans at 12%; the totals are the present values.

The entries of B, rounded to the poisha, total two poisha less than the case figure, because of rounding.

The second table grows each receipt to year 5 at 12% and totals the results.

continues for as many years as the stream lasts. Each term is a single sum from Chapter 4, and the formula only totals them.

The future value of a mixed stream is the sum that all receipts reach on the date of the last receipt, when each has grown at compound interest. A receipt in year 1 grows for two years, one in year 2 grows for one year, and one in year 3 does not grow. In the clean example: Tk 1,000 $\times$ 1.21 = Tk 1,210; Tk 2,000 $\times$ 1.10 = Tk 2,200; and Tk 3,000 stays Tk 3,000. The future value is Tk 1,210 + Tk 2,200 + Tk 3,000 = Tk 6,410. Check: Tk 4,815.93 $\times$ 1.331 = Tk 6,410, because the future value equals the present value grown for three years.

Case file, illustrative. Meghna and Sons, 10 October 2024. Two book-fair stall plans, with receipts at year end for 5 years. Stream A: Tk 40,000; Tk 55,000; Tk 60,000; Tk 70,000; Tk 80,000. Stream B: Tk 80,000; Tk 70,000; Tk 60,000; Tk 55,000; Tk 40,000. Both total Tk 305,000. At 12% a year, the present value of A is Tk 212,147.18 and of B Tk 227,589.53. The future value at year 5 of A is Tk 373,875.81 and of B Tk 401,090.51.

Worked example 2 repeats the method with realistic figures. Read each row of the first table from left to right: the receipt, multiplied by the discount multiplier for its year, gives its present value. The multipliers are shown to six places, and the present values are calculated from unrounded multipliers.

PRESENT VALUE OF EACH RECEIPT AT 12% (TK)

YEAR	DISCOUNT MULTIPLIER	RECEIPT A	PRESENT VALUE A	RECEIPT B	PRESENT VALUE B
1	0.892857	40,000	35,714.29	80,000	71,428.57
2	0.797194	55,000	43,845.66	70,000	55,803.57
3	0.711780	60,000	42,706.81	60,000	42,706.81
4	0.635518	70,000	44,486.27	55,000	34,953.49
5	0.567427	80,000	45,394.15	40,000	22,697.07
Total		305,000	212,147.18	305,000	227,589.53

The present value of A is Tk 212,147.18, and of B Tk 227,589.53, as in the case file. The five rounded entries of B total Tk 227,589.51, two poisha less, because each entry was rounded separately; the unrounded sum is the case figure. A poisha is one hundredth of a taka.

For the future value, each receipt is grown to the end of year 5. A receipt of year 1 grows for four years, so its growth multiplier is $1.12^4 = 1.573519$. The multiplier for year 2 is $1.12^3 = 1.404928$, for year 3 it is $1.12^2 = 1.2544$, for year 4 it is 1.12, and year 5 has none.

FUTURE VALUE AT YEAR 5 OF EACH RECEIPT, AT 12% (TK)

YEAR	GROWTH MULTIPLIER	FUTURE VALUE A	FUTURE VALUE B
1	1.573519	62,940.77	125,881.55
2	1.404928	77,271.04	98,344.96
3	1.254400	75,264.00	75,264.00
4	1.120000	78,400.00	61,600.00
5	1.000000	80,000.00	40,000.00

Joel Dean (1906–1979) was an American economist who wrote *Capital Budgeting* (Columbia University Press, 1951). Later writers often credit this book, and his consulting work, with moving discounted cash flow methods into company practice. The tools of this chapter descend from their work.

SOURCES

Irving Fisher, *The Theory of Interest* (1930). It matters because it explains why a sum due later is worth less today and shows how to value a stream of future receipts.

Joel Dean, *Capital Budgeting* (1951). It matters because it is an early book that treats the choice of investments as a management method for companies, and it helped to spread discounting.

REVIEW QUESTIONS

- 8.1 Find the payback period of a project with an initial investment of Tk 400,000 and inflows of 120,000, 150,000, 180,000 and 200,000 in years 1 to 4.
- 8.2 Find the net present value of the same project at a required return of 10%. Should it be accepted?
- 8.3 State the accept-or-reject rule for payback period, net present value, internal rate of return and the profitability index.
- 8.4 For the same project, calculate the net present value at 20% and at 25%, then estimate the internal rate of return by interpolation.
- 8.5 Project M has inflows with a present value of Tk 330,000 and an initial investment of Tk 300,000. Project N has a present value of Tk 540,000 and an initial investment of Tk 500,000. Find the profitability index and net present value of each. Which ranks first by each measure?
- 8.6 In one sentence each, say what the payback period, net present value, internal rate of return and profitability index measure.
- 8.7 When two mutually exclusive projects rank differently by payback and by net present value, which should decide, and why?

Dean carried discounting into company practice.

Fisher gave the theory.

Dean gave the practice.

total Tk 305,000, but the bars of B are taller at the left, where money is worth more.

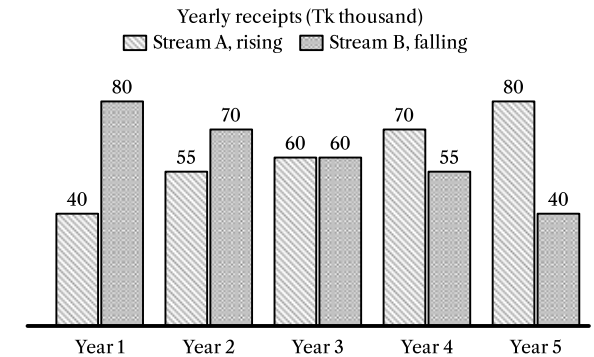


FIG. 6.1. Yearly receipts of two five-year streams: A rising, B falling, with the same total

The case file shows that B is worth more at 12%, although both streams total the same.

The gap of Tk 15,442.35 in present value is the price of receiving the large amounts later.

The ranking of two streams may depend on the discount rate, so it is wise to test more than one rate.

The ranking reverses because a higher rate punishes the distant payment of F more than the early payments of E.

To test a rate, repeat the whole comparison at a lower and a higher rate and watch whether the ranking changes.

Case file, illustrative. Meghna and Sons, 10 October 2024. At 12% a year, B is worth more than A because larger receipts come sooner. The present value of A is Tk 212,147.18 and of B Tk 227,589.53. The future value at year 5 of A is Tk 373,875.81 and of B Tk 401,090.51.

The difference in present value is Tk 227,589.53 – Tk 212,147.18 = Tk 15,442.35. The difference in future value is Tk 401,090.51 – Tk 373,875.81 = Tk 27,214.70. Neither gap comes from the totals, which are equal. Both come from timing alone. Stream B delivers Tk 80,000 in year 1, and that money can start earning interest at once.

A discount rate is rarely known with certainty, so a careful reader tests the ranking at more than one rate. Sometimes the ranking holds at every rate, as with A and B. Sometimes it reverses. Take stream E, which pays Tk 30,000 at the end of each of 5 years, and stream F, which pays Tk 200,000 once, at the end of year 5. At 5% a year, E is worth Tk 30,000 × 4.329477 = Tk 129,884, using the annuity factor of Chapter 5, and F is worth Tk 200,000 × 0.783526 = Tk 156,705. At 15% a year, E is worth Tk 30,000 × 3.352155 = Tk 100,565 and F is worth Tk 200,000 × 0.497177 = Tk 99,435. At 5% F wins; at 15% E wins.

The reason is plain. A higher discount rate shrinks distant money faster than near money. The single large payment of F lies five years away, so it loses most. The yearly payments of E are nearer on average, so they lose less. The lesson is not to fear the reversal but to expect it: when two streams have different totals and different timing, test the rate. When they have the same total, as A and B do, the stream with earlier receipts wins at every positive rate.

The test has three steps. First choose a low rate and a high rate that bracket the likely one, for example 5% and 15%. Second, value each stream at both rates, using the same method each time. Third, compare the rankings. If the same stream wins at both rates, the decision is secure. If the winner changes, there is a rate between

8.4 WHEN THE METHODS DISAGREE

Table 8.2 collects the results for Press and Fit-out. Read each row as one tool and each column as one project. The last column names the project that tool prefers.

RESULTS FOR THE TWO PROJECTS

TOOL	PRESS	FIT-OUT	PREFERS
Payback period	3.26 years	2.80 years	Fit-out
Net present value at 12%	Tk 197,055	Tk 86,464	Press
Internal rate of return	19.9%	16.6%	Press
Profitability index	1.22	1.10	Press

Payback prefers Fit-out; the other three prefer Press. The reason is the shape of the cash flows. Fit-out is front-loaded, so it recovers its money first. Press brings large cash in year 5, which payback ignores and the discounted tools count.

When rankings conflict for mutually exclusive projects, let net present value decide. It uses all cash flows, respects timing and states the result as a gain in taka. Payback still has a role as a second check, for example when the business is short of cash. Even net present value and the internal rate of return can disagree when projects differ in size or timing; again, net present value is the guide.

Situation matters as well. An owner with plenty of funds will weigh the extra Tk 110,591 of value heavily. An owner who must repay a loan soon may value the early cash of Fit-out. Even so, this is a reason to look at the cash plan, not to change the ranking by value. The ranking stays the same; only the owner's readiness to accept a lower value for speed changes.

* * *

WORKED EXAMPLE: CHOOSING BETWEEN PRESS AND FIT-OUT

We now combine the tools and justify a choice, in the standard order: Given, Find, Formula, Steps, Answer and Check.

Given. The case file of 2 February 2025: Press and Fit-out, each with an initial investment of Tk 900,000, a required return of 12% and the inflows stated earlier. *Find.* Which project Meghna and Sons should accept. *Formula.* Payback period as above; net present value equals present value of inflows minus initial investment; accept if the value is at least zero; choose the highest value among exclusive projects.

Steps. Press by year: $246,429 + 220,026 + 196,451 + 175,403 + 258,747 = 1,097,056$. This differs from the earlier 1,097,055 by Tk 1, because each term was rounded; the unrounded total is 1,097,055.07. Both agree that NPV is Tk 197,055 for Press, and Table 8.1 gives Tk 86,464 for Fit-out. The profitability indices are 1.22 and 1.10.

The same two projects now receive four different tests.

Payback disagrees because it ignores late cash.

Net present value decides because it measures worth in taka.

The choice also depends on the business's situation.

The four tools meet in one decision.

Given and Find: the data and the question.

Steps: check each project on the discounted tools.

* * *

SUMMARY

A mixed stream is a series of payments or receipts that differ in size from one period to the next. Discounted cash flow values it by turning each receipt into a present value and adding. The future value grows each receipt to the last date and adds. A level part of a stream may use the annuity factor, and the rest is valued as single sums. Streams are compared at one date and one discount rate; the ranking may change with the rate, so test more than one.

THINK IT THROUGH

6.T1 A supplier offers two payment plans for the same machine. Plan X pays Tk 90,000, Tk 60,000 and Tk 30,000 at the ends of years 1, 2 and 3. Plan Y pays Tk 30,000, Tk 60,000 and Tk 90,000 at the same dates. At 10% a year, which plan costs less in present value, and why?

6.T2 Explain in plain words why two streams with the same total can have different present values.

WHERE PEOPLE GO WRONG

Mistake: "The two plans total the same, so they are worth the same." The total adds money from different dates as if the dates did not matter. Timing changes value, and the present value shows it.

Mistake: using one multiplier for all years. The multiplier for year 3 is smaller than that for year 1. Using one multiplier for every year treats the stream as if every receipt came on the same date.

Mistake: using the annuity factor on unequal receipts. The factor assumes equal payments. Split the stream into a level part and an odd part first, or discount each receipt alone.

Mistake: ranking streams at one rate and then trusting the result. Where totals differ, the ranking can reverse. Test a low rate and a high rate before deciding.

IN HISTORY

Benjamin Franklin (1706–1790) died in Philadelphia on 17 April 1790. In a codicil to his will he left 1,000 pounds each to Boston, where he was born, and to Philadelphia, where he lived most of his adult life. The money was to be lent to young married craftsmen who had served an apprenticeship. Part of it could be used after 100 years, and the rest after 200 years, in 1990.

Franklin's plan is a mixed stream seen from the owner's side. No money left the fund for a century, one part was then withdrawn, and the remainder was withdrawn two centuries after his death. Compound interest worked on the fund during all that time. Accounts of the sums reached in 1990 differ: they are usually given

A mixed stream is valued by discounting each receipt; streams are compared at a common rate.

The total of a stream is not its value, because the total ignores timing.

Each receipt needs the multiplier for its own year, not the multiplier of another year.

The annuity factor applies only to the level part of a stream, never to the whole stream.

A ranking found at one rate may not hold at another.

Benjamin Franklin's bequest of 1790 grew for two centuries with withdrawals at set dates.

The bequest is a mixed stream: nothing for a century, then a withdrawal, then the remainder.

value, so a project that passes at 12% might fail at 18%. Because of this, a careful business tests a project at more than one rate. Many students find this uncomfortable, because the method looks exact. The arithmetic is exact; the inputs are estimates.

Now take Fit-out at 12%. Table 8.1 sets out the work. Read each row as one year: the inflow, the divisor (1.12 to the power of the year number) and the present value, which is the inflow divided by the divisor. The last row adds the present values.

PRESENT VALUE OF PROJECT FIT-OUT AT 12%, IN TAKA

YEAR	INFLOW	DIVISOR	PRESENT VALUE
1	400,000	1.120000	357,143
2	300,000	1.254400	239,158
3	250,000	1.404928	177,945
4	200,000	1.573519	127,104
5	150,000	1.762342	85,114
Total			986,464

NPV = 986,464 – 900,000 = Tk 86,464. The value is positive, so Fit-out is acceptable at 12%. *Check.* The inflows total 1,300,000, and the present value is much lower, as we expect from discounting.

For Press the inflows in years 1 to 4 are equal, so they form an annuity. The present value factor for 4 years at 12% is $(1 - 1.12^{-4}) \div 0.12 = 3.0373493$. Then $276,000 \times 3.0373493 = 838,308$. The year 5 inflow is $456,000 \div 1.762342 = 258,747$. The total is $838,308 + 258,747 = 1,097,055$, so NPV = $1,097,055 - 900,000 = \text{Tk } 197,055$. Press is also acceptable.

Press and Fit-out are mutually exclusive, so the shop can accept only one. When both pass the test, choose the larger net present value, because it measures the extra worth in taka today: Tk 197,055 against Tk 86,464. Press is the better project on this measure.

Before moving on. Net present value equals present value of inflows minus the initial investment. The required return is the discount rate and the hurdle. Accept when the value is zero or positive; among exclusive projects, choose the highest.

* * *

8.3 INTERNAL RATE OF RETURN AND PROFITABILITY INDEX

Net present value answers in taka. Many owners prefer an answer in percent, which is easy to compare with a loan rate. Two further tools give that kind of answer, and both build on net present value.

A realistic example uses five unequal inflows.

Fit-out has a positive net present value.

Press uses the annuity shortcut for years 1 to 4.

When only one project can be accepted, the larger net present value wins.

Recap: discount every inflow, subtract the initial investment, accept if the result is not negative.

Sometimes owners prefer one percentage to a sum of taka.

Fraction of year 4: $72,000 \div 276,000 = 0.26$. Payback period = $3 + 0.26 = 3.26$ years.

Fit-out. The running total of inflows is 400,000 after year 1 and 700,000 after year 2, still short of 900,000. In year 3 the inflow is 250,000. Unrecovered at the start of year 3: $900,000 - 700,000 = \text{Tk } 200,000$. Fraction of year 3: $200,000 \div 250,000 = 0.80$. Payback period = $2 + 0.80 = 2.80$ years. **Check.** After 3 years the inflows total 950,000, which is above 900,000, so recovery does fall in year 3.

Figure 8.1 shows the cumulative cash flow of Press. The horizontal axis shows years from the start; the vertical axis shows the running total in thousands of taka. Each point is the total at the end of a year. The line starts at -900 and rises with each inflow. The point where it crosses the zero line is the payback period, here at 3.26 years.

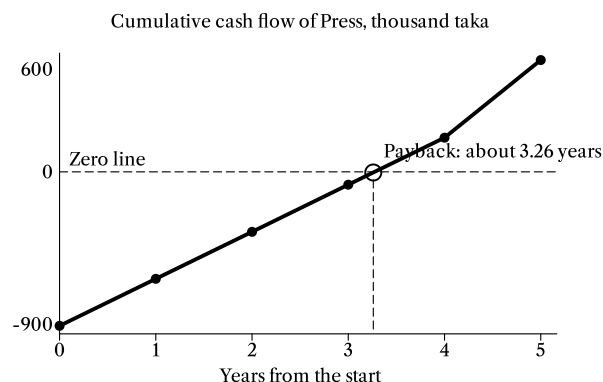


FIG. 8.1. Cumulative cash flow of Project Press, crossing zero at the payback point

To turn the tool into a decision rule, managers set a maximum acceptable payback period, for example 3 years. A project is accepted if its payback period is no longer than the limit. With a limit of 3 years, Fit-out passes at 2.80 years and Press fails at 3.26 years. The limit is a judgement, not a fact, and a different limit gives a different answer.

The tool has real limits. It ignores all cash flows after the payback point: Press earns Tk 456,000 in year 5, and payback never looks at it. It treats a taka in year 1 and a taka in year 3 as equal, so it ignores the time value of money. And the limit has no firm basis. Payback remains useful as a quick check of how long money is at risk, but it cannot be the only test.

The owner may like payback because it speaks of risk: the sooner the money returns, the less time the future has to spoil the plan. The financial manager values it as a quick screen, which removes weak proposals before slower work begins. A lender may care about it too, since a short payback suggests that loan instalments will be met. An employee or customer rarely sees the figure, but both feel its effects when a business cannot recover its money and must cut costs.

Fit-out recovers its money sooner.

The figure shows the running total rising through zero.

A decision rule needs a limit set by management.

Payback ignores later cash and the timing of money.

Different people use payback for different reasons.

A rickshaw owner faces a choice that is hard to undo.

You will need: Chapter 2, section 2.2, the investment decision.

The dealer's promise leaves many questions open.

Three questions turn hope into a method.

This chapter measures the cash; the next chapter judges it.

Imagine a man in Dhaka who hires out three rickshaws by the day. He has saved Tk 90,000, and a dealer offers him a fourth rickshaw for exactly that sum. The dealer promises that the new rickshaw will repay its price within a year. Should the owner agree?

The owner should hesitate, and the reasons are worth listing. The rickshaw will not earn money on every day. Some days bring rain, some weeks bring repairs, and a rival may begin work on the same road. The Tk 90,000 leaves his hands today, but the money it earns returns slowly, over years. And once the rickshaw is bought, he cannot easily recover his money, because a used rickshaw sells for much less than a new one.

So the owner needs more than hope. He must ask which payments will really happen, and when. He must ask which of them exist only because he buys the rickshaw. And he must ask whether the money coming in is worth more than the money going out, remembering that money received later is worth less than money received today. You met that idea in Chapters 4 to 6. Each question has a method, and the method has a name.

This chapter teaches the first half of that method. It explains what a decision of this size is called, why it deserves careful work, and which steps a business follows. Then it shows how to measure the cash that a project spends and receives. Chapter 8, 'Capital Budgeting: Evaluation Techniques', shows how to judge whether those cash flows are good enough. Our running example is a print-on-demand machine for Meghna and Sons, the family bookshop in Dhaka. The shop and its figures are illustrative.

* * *

7.1 WHAT CAPITAL BUDGETING IS AND WHY IT MATTERS

Begin with a choice from home. A family spends Tk 3,000 a week on food. If it buys too many onions, the harm is small, because the onions are eaten or wasted within days. Now suppose the same family decides to build a second room for Tk 400,000. The room cannot be returned to the builder, and it will serve the family for thirty years. Both choices are spending, but they are not the same kind of choice. The second is large, slow to repay and hard to reverse.

A business meets the same difference. In Chapter 2 you learned that the investment decision is the choice of which assets the business should hold, and that the financial manager shares in that choice. Most investment decisions are small and repeat every week, such as restocking a shelf. A few are large and rare. The large ones need a method of their own, and the name of that method is capital budgeting.

Capital budgeting means the process of planning, evaluating and choosing long-term investments. When Meghna and Sons asks whether to buy a print-on-demand machine that will work for five years, the shop is doing capital budgeting. The word *budgeting*

Spending on onions and spending on a new room are different kinds of choice.

Businesses face the same difference between small and large choices.

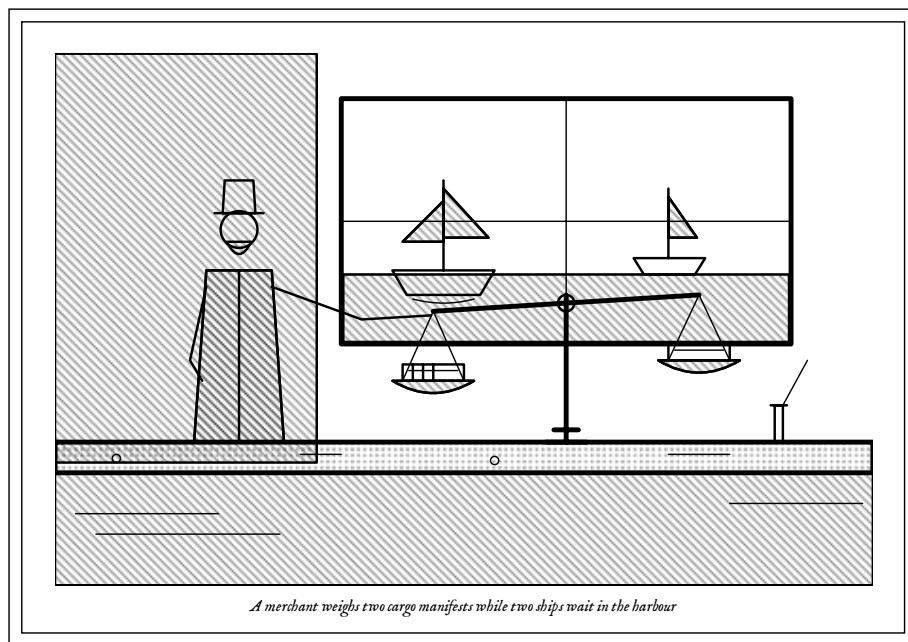
Recall: the investment decision, Chapter 2, 'The Financial Manager's Work', section 2.2 'The Three Core Decisions'.

Capital. Word in use. In daily life, money or wealth that a person owns. In finance, the long-term funds and assets that a business uses to earn profit.

Project. Word in use. In daily life, a piece of work such as a school project. In capital budgeting, one proposed investment, such as a machine or a branch, treated as a single choice.

The capital budgeting process has five steps.

The figure shows the five steps in order.



CHAPTER VIII

Capital Budgeting: Evaluation Techniques

Two ships sail tomorrow: which cargo deserves the merchant's money?



A business that has measured the cash flows of a project must still decide whether the project is worth its money.

This chapter teaches four tools for that judgement: payback period, net present value, internal rate of return and the profitability index. It also shows what to do when the tools give different answers.

The capital budgeting process has five steps, and the word *project* in them has the special meaning shown in the side note. Step one is to find proposals: ideas come from owners, staff, customers and changes in the market. Step two is to estimate the cash flows of each proposal, year by year. Step three is to evaluate the cash flows and choose. Step four is to implement the choice by buying, installing and training. Step five is to review the results by comparing real cash flows with the forecast.

Figure 7.1 shows the sequence. Read it from left to right. Each box is one step, and each solid arrow means that the next step begins when the last one ends. The dashed arrow below the boxes returns from the last step to the first: lessons from the review improve the next search for proposals. The labels under two boxes show where this book teaches that step.

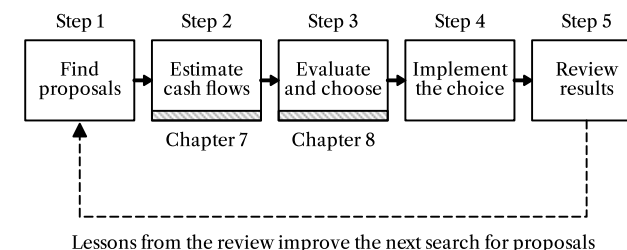


FIG. 7.1. *The five steps of the capital budgeting process*

Each step answers a question. Step one asks what we could do. Step two asks what money will move, and when. Step three asks whether the money is worth the cost of the funds; Chapter 8 teaches the tools. Step four asks how to do it well. Step five asks what we learn. The financial manager is most active in steps two, three and five, because they need figures and honest comparison.

The steps have exceptions. In a small business one person may perform several steps in a week, and the steps may overlap. A proposal may return to step two when new facts appear. Step two is usually the hardest, because it forecasts the future; an error there passes into every later step. This is why the rest of this chapter concentrates on it.

Case file, illustrative. 5 January 2025: Meghna and Sons holds two proposals. One is a print-on-demand machine with an initial investment of Tk 900,000. The other is a branch fit-out that needs Tk 900,000. The two cannot be run together. A feasibility survey for the machine, costing Tk 15,000, was paid in December 2024.

The case file shows why the rest of the process matters. Two proposals compete, and each needs a measured set of cash flows before any choice is made. Two new names describe how propos-

Each step has an owner and a question.

The steps are a guide, not a rigid rule.

Case: two proposals compete for the same funds.

The case shows why every proposal needs measured cash flows.

WHERE PEOPLE GO WRONG

A common belief is that profit and cash flow are the same. They are not. Depreciation reduces profit but no cash leaves; so operating cash flow is higher than EBIT less tax. The correction is to add depreciation again.

Another belief is that money already spent must count against a project. It cannot, because the money is gone either way. Only cash that changes with the decision is relevant.

A third mistake is to forget working capital, or to treat it as lost. It is paid at the start and returns at the end of the project, so it appears twice, with opposite signs.

A fourth mistake is to subtract loan interest in the yearly cash flow. In this method the cost of funds is allowed for in the discount rate, taught in Chapter 8, so interest is left out of the cash flows. This also keeps one cost from being counted twice.

IN HISTORY

The Padma Bridge in Bangladesh carries road traffic over the Padma River near Dhaka. It was inaugurated on 25 June 2022 and opened to traffic the next day. According to the Bangladesh Bridge Authority, as reported in the press, it cost Tk 307.70 billion, paid from the government's own funds. In 2012 the World Bank decided not to fund the project after it found signs of corruption involving a Canadian construction firm, and the government then decided to build with its own money.

The bridge shows all three marks of this chapter: it was large, long-lived and impossible to reverse. Its early estimate, near Tk 101.62 billion in 2007, rose after several revisions. The finance ministry gave the funds to the Bridge Authority as a loan, to be repaid over 35 years in 140 quarterly instalments from toll revenue and other income. This is not a full analysis, but it shows the central question of capital budgeting: will the cash and the wider benefit over a long life justify the money spent today?

REVIEW QUESTIONS

- 7.1 What is capital budgeting, and why does a business need it? Give the three marks of such a decision.
- 7.2 A bakery plans to buy a second oven. Apply the five steps of the capital budgeting process to this decision, with one sentence for each step.
- 7.3 Explain independent projects and mutually exclusive projects, with two examples of each.
- 7.4 A shop is considering a freezer. Which of these items are relevant cash flows, and why? (a) The freezer price, Tk 200,000. (b) Tk 5,000 paid last month for a report on freezers. (c) Tk 8,000 a year of extra electricity. (d) Tk 30,000 of extra stock needed at the start. (e) Shop rent of Tk 40,000 a year, paid with or without the freezer.
- 7.5 A project adds revenue of Tk 240,000 a year with cash costs of

Profit and cash flow are different things.

A sunk cost is not a reason to continue or to stop.

Working capital is both paid and returned.

Interest on loans is allowed for elsewhere.

A very large public project, judged on long-run cash and benefit.

The bridge shows all three marks and a loan repaid from tolls.

initial investment is paid at time zero, which means today, before year one begins.

Case file, illustrative. 5 January 2025: the print-on-demand machine for Meghna and Sons needs an initial investment of Tk 900,000. It is made up of the machine at Tk 780,000, installation at Tk 60,000 and working capital of Tk 60,000. The machine and installation together, Tk 840,000, are the cost of the asset.

OPERATING CASH FLOW

Once the machine works, it brings in cash each year. Take a tailor's machine that adds Tk 100,000 of revenue in a year, meaning extra money from customers, and costs Tk 40,000 in cash to run. Is the yearly cash flow Tk 60,000? Not quite, for two reasons: depreciation changes the tax, and the tax itself is a payment. We take them in turn, using small clean figures so that the method is the only thing to learn.

The first reason is **depreciation**. Depreciation means the part of an asset cost charged as an expense in each year of its use. A machine that costs Tk 100,000, works for five years and then has no resale value is charged at Tk 20,000 in each year. Charging the same amount each year is called the straight-line method. If a resale value is expected, subtract it first: the yearly charge is the cost, less the resale value, divided by the number of years. No cash leaves the business in those years, because the cash was paid at the start. But the charge lowers the profit on which tax is calculated, so it still changes the tax paid.

The second tool is **EBIT**, which stands for earnings before interest and tax. EBIT is the profit from operations before paying lenders or the tax office. For the tailor, EBIT equals revenue, less cash costs, less depreciation: $100,000 - 40,000 - 20,000 = \text{Tk } 40,000$. Some writers call EBIT operating profit, but this book uses EBIT from here on. Interest is left out on purpose. The cost of the funds is allowed for in the discount rate, which Chapter 8 teaches, so counting interest here as well would count the same cost twice.

The **operating cash flow** is the cash a project produces each year from running it, after tax. Calculate it in three moves. First find EBIT. Second subtract the tax on EBIT. Third add depreciation again, because depreciation was subtracted to reach EBIT but no cash was paid. For the tailor, with tax at 25%, the tax is $0.25 \times 40,000 = \text{Tk } 10,000$. Then $40,000 - 10,000 = 30,000$, and adding depreciation of 20,000 gives an operating cash flow of Tk 50,000.

IN SHORT

Operating cash flow = EBIT - tax on EBIT + depreciation. In symbols, $\text{OCF} = \text{EBIT} - (T \times \text{EBIT}) + D$, where OCF is operating cash flow, T is the tax rate and D is depreciation.

There is a second route to the same answer: revenue, less cash costs, less tax. For the tailor, $100,000 - 40,000 - 10,000 = \text{Tk } 50,000$. The two routes agree, and a good habit is to run both. If

Case: the initial investment adds three payments.

Operating cash flow is not the same as profit.

Cost. Word in use. In daily life, the price paid for something. Here, a cash cost is money paid out in a year to run the project; depreciation is also called a cost in accounts, but no cash leaves.

Depreciation. The part of an asset cost charged as an expense in each year of its use.

Depreciation spreads the cost of an asset over its years of use.

EBIT. Earnings before interest and tax: the profit from operations before paying lenders or the tax office.

EBIT is the profit from operations before interest and tax.

Operating cash flow. Cash a project produces each year from running it, after tax.

Operating cash flow is the cash a project makes each year after tax.

A second route gives the same answer, which is a useful check.

We can find the same number from the base figures alone. Interest comes out before tax, but the preferred dividend comes out after it, so we first convert the dividend to its before-tax equivalent by dividing by $(1 - T)$. In symbols: degree of financial leverage = $\text{EBIT} \div [\text{EBIT} - I - \text{Dp} \div (1 - T)]$. *Given.* EBIT Tk 80,000; I Tk 20,000; Dp Tk 12,000; T 25%. *Steps.* $12,000 \div 0.75 = \text{Tk } 16,000$. The bottom line is $80,000 - 20,000 - 16,000 = \text{Tk } 44,000$. Then $80,000 \div 44,000 = 1.8182$. *Answer.* The degree of financial leverage is 1.82. *Check.* It matches the $90.9\% \div 50\%$ found from the two EBIT levels, and $1.8182 \times 50 = 90.9$.

PAUSE AND CHECK 12.P1 A business has EBIT of Tk 50,000, interest of Tk 10,000 and no preferred stock. Find its degree of financial leverage.

Watch for these slips. Forgetting to divide the preferred dividend by $(1 - T)$ gives a wrong answer, because the dividend is paid after tax. Using after-tax interest is also wrong, because interest already stands before tax in the formula. The number holds near the EBIT at which it was found, and it changes at other levels of EBIT. With no fixed financing charges the degree is exactly 1.00, and as EBIT falls toward the charges the degree grows without limit.

Before moving on. Earnings per share is the profit left for common stock owners divided by the number of common shares. The degree of financial leverage shows how many percent earnings per share moves for each one percent move in EBIT. For the printing unit it is 1.82.

* * *

12.3 DEGREE OF TOTAL LEVERAGE

Think of two levers joined end to end. A push on the first moves the second, and the second lifts the weight. The printing unit has two such levers. Operating leverage turns a change in sales into a larger change in EBIT. Financial leverage turns a change in EBIT into a larger change in earnings per share. Chapter 11 measured the first lever with the degree of operating leverage.³ Section 12.2 measured the second. What does the whole chain do?

The table follows the rise from 8,000 to 9,000 packs. Read each row from left to right: the item, its value at 8,000 packs, its value at 9,000 packs, and the percentage change. Sales rise by 12.5%, EBIT by 50%, and earnings per share by 90.9%. Each change is larger than the one before.

ONE RISE IN SALES THROUGH BOTH LEVERS

ITEM	AT 8,000 PACKS	AT 9,000 PACKS	CHANGE
Sales	Tk 720,000	Tk 810,000	12.5%
EBIT	Tk 80,000	Tk 120,000	50.0%
Earnings per share	Tk 6.60	Tk 12.60	90.9%

³ Chapter 11, 'Operating Leverage and Break-even Analysis', section 11.3 'Degree of Operating Leverage', measured the first link in total leverage.

The same number can be found from one set of figures, without changing EBIT.

Common slips come from the preferred dividend and from using the number far from its base.

The section's ideas are earnings per share and the degree of financial leverage.

Two levers in a row multiply their effects, and each has been measured.

The table follows one rise in sales through both levers.

Business risk. the risk that operating profit varies because of sales, costs and trade conditions

Business risk comes from the trade itself.

Financial risk. the extra risk to owners that comes from funding part of a business by borrowing

Financial risk comes from borrowing.

The two kinds differ in cause and in who bears the extra risk.

To read the table, move along a row and compare the two columns.

The section ends with a recap and a test sentence.

GO SLOWLY HERE.

A list of possible outcomes needs a number for how likely each one is.

Probability. a number from 0 to 1 showing how likely an outcome is

Probability puts a chance on a scale from 0 (cannot happen) to 1 (certain).

Probability distribution. a list of possible outcomes with

Business risk is the risk that operating profit varies because of sales, costs and trade conditions. A bookshop shows the idea. Sales rise before school terms and fall in the holidays. A rival may open nearby, and the price of paper may rise. Every business in the same trade shares these dangers, whatever the way it is funded. Business risk grows when a business has many fixed operating costs, such as rent, which must be paid even when sales are poor. Chapter 11, 'Operating Leverage and Break-even Analysis', section 11.1 'Fixed and Variable Costs', teaches this link.

Financial risk is the extra risk to owners that comes from funding part of a business by borrowing. Suppose a bookshop borrows Tk 600,000 at 10% a year. The interest is Tk 60,000 every year, in a busy year and in a poor one. After the interest is paid, what remains for the owners rises and falls more sharply than operating profit does. Chapter 12, 'Financial and Total Leverage', section 12.1 'Borrowing and Fixed Financing Charges', teaches this link in full.

HOW THE TWO KINDS OF RISK DIFFER

POINT	BUSINESS RISK	FINANCIAL RISK
Source	Sales, costs and trade conditions	Borrowing, which brings fixed interest
Who meets it	Every business in the trade	Owners of a business that borrows
Can the business avoid it?	Not by changing its funding	Yes, by borrowing less

Read each row from left to right. In the first row the source differs: the trade creates one kind and borrowing creates the other. In the last row the difference is choice: the owners decide how much to borrow, but they cannot choose the conditions of the trade.

Before moving on. Risk is the chance that results differ from those expected, especially for the worse. Business risk comes from the trade, and financial risk comes from borrowing. *You should now be able to say* what risk means in finance, and why a certain loss is not risk.

* * *

9.3 DESCRIBING UNCERTAIN OUTCOMES

Return to the two routes. On any day the traffic is clear, usual or heavy, and we cannot know which in advance. We can say how often each one happens. Suppose that over 100 days the traffic is heavy on 25 of them. Then heavy traffic has a chance of 25 in 100. Finance writes such a chance as one number.

Probability is a number from 0 to 1 showing how likely an outcome is. Zero means the outcome cannot happen, and one means it is certain. A fair coin landing heads has a probability of 0.5. Heavy traffic on 25 days in 100 has a probability of $25 \div 100 = 0.25$.

A probability distribution is a list of possible outcomes with the probability of each adding up to one. The probabilities add up

and saves no tax. We will need this difference when we calculate. A business that cannot pay its interest risks action by its lender, which is why the charge is called fixed.

Common stock means ownership shares that receive what remains after all other claims, with no fixed dividend. Some countries call it ordinary shares or equity shares; this book uses common stock. The owners of common stock take the most risk and have the best chance of gain. If the business has Tk 33,000 left after all other claims, all of it belongs to them.

Preferred stock means ownership shares with a fixed dividend that is paid before any dividend on common stock. A business that issues Tk 100,000 of preferred stock at 12% promises Tk 12,000 each year to those owners, ahead of the common stock owners. Preferred stock owners do not usually vote, and they cannot ask for more than the fixed dividend. Terms differ between issues and between countries, and Chapter 14 teaches the rights of shareholders in full.¹

The table sets the three claims side by side. Read each row from left to right: who holds the claim, what the holder receives, whether the amount is fixed, and when it is paid.

THREE CLAIMS ON THE PROFIT OF A BUSINESS

CLAIM	WHAT IT RECEIVES	FIXED?	PAID
Lender	Interest	Yes	Before tax
Preferred stock	Preferred dividend	Yes	After tax, before common stock
Common stock	What remains	No	Last

Now we repeat the tea stall in business form, with small round numbers. *Given.* EBIT Tk 10,000; interest Tk 2,000; 1,000 common shares; ignore tax for now. *Find.* The profit for each share, at EBIT of Tk 10,000, Tk 12,000 and Tk 8,000. *Formula.* Profit for each share equals EBIT less interest, divided by the number of shares. The book gives this its proper name, earnings per share, in section 12.2. *Steps.* At Tk 10,000: $(10,000 - 2,000) \div 1,000 = \text{Tk } 8.00$. At Tk 12,000: $(12,000 - 2,000) \div 1,000 = \text{Tk } 10.00$. At Tk 8,000: $(8,000 - 2,000) \div 1,000 = \text{Tk } 6.00$. *Answer.* A 20% rise in EBIT lifts the profit for each share from Tk 8.00 to Tk 10.00, a rise of 25%; a 20% fall lowers it to Tk 6.00, a fall of 25%. *Check.* Without interest the profit for each share would be Tk 10.00, Tk 12.00 and Tk 8.00, and the change would be only 20% each way.

The extra swing has a name. **Financial leverage** is the effect of fixed financing charges in making earnings per share change by

¹ Chapter 14, 'Valuing Common and Preferred Stock', section 14.1 'Rights of Shareholders', teaches the rights of common and preferred shareholders.

Share. Word in use. In daily life, a part of something divided among people, such as a share of a meal. In finance, one unit of ownership of a business.

Common stock. ownership shares that receive what remains after all other claims, with no fixed dividend

Common stock owners receive whatever is left, so their gain rises and falls with profit.

Preferred stock. ownership shares with a fixed dividend that is paid before any dividend on common stock

Preferred stock owners are paid a fixed dividend before common stock owners receive anything.

The table compares the three claims on the profit of a business.

A clean example shows the lever with numbers small enough to follow.

Financial leverage. the effect of fixed financing charges in making earnings

Spread. Word in use. In daily life, to cover a surface thinly, such as butter on bread. In finance, how far outcomes lie from the expected value.

Standard deviation. a measure of how widely outcomes spread around the expected value

Standard deviation measures the typical distance of outcomes from the expected value.

The four steps turn distances into one number.

The table works through the four steps for both routes.

Each deviation is squared, weighted, added and then the root is taken.

The figure shows the spread as the width of a bell-shaped outline.

The **standard deviation** is a measure of how widely outcomes spread around the expected value. A small standard deviation means the outcomes huddle near the expected value. A large one means they lie far from it. We build it in four small steps, each in plain arithmetic.

Step one: find each outcome's deviation, which is the outcome less the expected value. *Step two:* square each deviation, because some are below the expected value and some above, and without squaring they would cancel to zero. *Step three:* multiply each squared deviation by its probability and add the results; this total is called the variance. *Step four:* take the square root, which returns the answer to the units of the outcomes. In symbols, standard deviation = $\sqrt{\sum p \times (x - \text{expected value})^2}$].

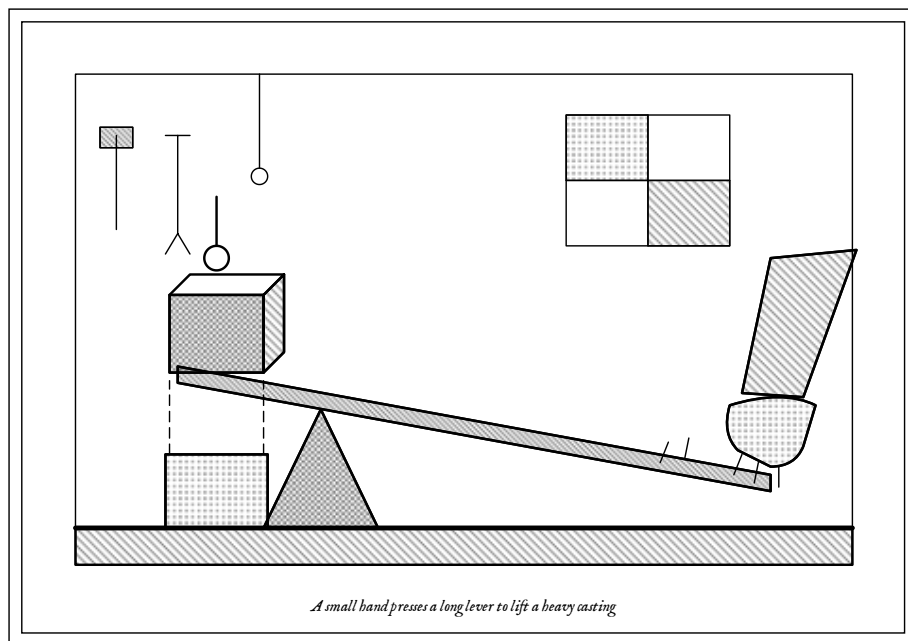
STANDARD DEVIATION OF TWO ROUTES (EXPECTED TIME 25 MINUTES)

TRAFFIC	PROBABILITY	A: DEVIATION	A: P × DEVIATION ²	B: DEVIATION	B: P × DEVIATIO
Clear	0.25	-5	6.25	-15	56.25
Usual	0.50	0	0.00	0	0.00
Heavy	0.25	+5	6.25	+15	56.25
Total (variance)	1.00		12.50		112.50

Given. The distributions and the expected time of 25 minutes. *Find.* The standard deviation of each route. *Steps.* Route A: $(-5)^2 = 25$, weighted $0.25 \times 25 = 6.25$, twice, so the variance is 12.50 and the square root is 3.54 minutes. Route B: $(-15)^2 = 225$, weighted $0.25 \times 225 = 56.25$, twice, so the variance is 112.50 and the square root is 10.61 minutes. *Answer.* Route A has a standard deviation of 3.54 minutes, route B 10.61 minutes. *Check.* Route B's deviations are three times larger than route A's, and 10.61 is about three times 3.54.

PAUSE AND CHECK 9.P2 Why does the calculation square each deviation before adding?

To read the next figure, look at the horizontal line: it shows the possible outcomes, from low to high. The height of an outline above any point shows how likely outcomes near that point are. Both outlines are centred on the same expected value, marked by the dashed vertical line. The narrow outline is tall because most outcomes lie near the centre. The wide outline is low because the outcomes are scattered far from it.



CHAPTER XII

Financial and Total Leverage

What happens to the owners when a business borrows to push its profit higher?



This chapter shows how borrowing and preferred stock make the profit for each owner's share change by more than the operating profit does. You will calculate that profit for each share, then two measures of how strongly a change in sales or in operating profit moves it. The printing unit data of Chapter 11 is used again, now with its financing added.

The first step weights each return by its probability.

The second step finds the spread of each line around its own expected value.

The third step divides each spread by its expected value and a judgement follows.

A reshaped case tests the same method on takings in taka.

A combined case ranks three assets by average outcome, spread and spread per unit of return.

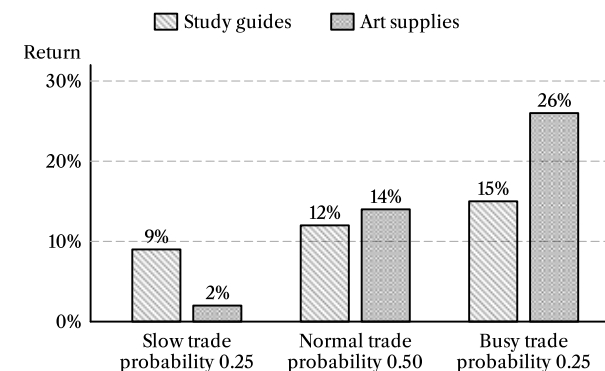


FIG. 9.2. Returns of two product lines in slow, normal and busy trade

Given. The distributions above. *Find.* The expected value, standard deviation and coefficient of variation of each line. *Steps for expected value.* Study guides: $0.25 \times 9 = 2.25$; $0.50 \times 12 = 6.00$; $0.25 \times 15 = 3.75$; the sum is 12%. Art supplies: $0.25 \times 2 = 0.50$; $0.50 \times 14 = 7.00$; $0.25 \times 26 = 6.50$; the sum is 14%.

Steps for standard deviation. Study guides: the deviations are -3, 0 and +3. The squares are 9, 0 and 9. Weighted: $2.25 + 0 + 2.25 = 4.50$, the variance. The square root is 2.12%. Art supplies: the deviations are -12, 0 and +12. The squares are 144, 0 and 144. Weighted: $36 + 0 + 36 = 72$, the variance. The square root is 8.49%.

Steps for the coefficient. Study guides: $2.12 \div 12 = 0.18$. Art supplies: $8.49 \div 14 = 0.61$. *Answer.* Art supplies have the higher expected return, 14% against 12%, and much more risk for each unit of return, 0.61 against 0.18. *Check.* Art supplies' deviations are four times as large as those of study guides, and 8.49 is about four times 2.12. Whether the extra return is worth the extra risk depends on the required return and on how much risk the owners will accept.

Reshaped case. Two market stalls have weekly takings with the same probabilities, 0.25, 0.50 and 0.25. Stall X takes Tk 8,000, Tk 10,000 or Tk 12,000. Stall Y takes Tk 6,000, Tk 12,000 or Tk 18,000. Stall X: expected value $2,000 + 5,000 + 3,000 = \text{Tk } 10,000$; deviations $\pm 2,000$; variance $0.25 \times 4,000,000 \times 2 = 2,000,000$; standard deviation Tk 1,414; coefficient 0.14. Stall Y: expected value $1,500 + 6,000 + 4,500 = \text{Tk } 12,000$; deviations $\pm 6,000$; variance $0.25 \times 36,000,000 \times 2 = 18,000,000$; standard deviation Tk 4,243; coefficient 0.35. Stall Y takes more and carries more risk for each taka.

Combined case. Three assets have expected values and standard deviations as shown. The table introduces them in a row each, with the coefficient calculated as standard deviation $\div$ expected value.

THREE ASSETS COMPARED

ASSET	EXPECTED VALUE	STANDARD DEVIATION	COEFFICIENT OF VARIATION
P	8%	2%	0.25

11.T2 At 9,000 packs, find the EBIT and the degree of operating leverage before and after the new machine. Which cost structure carries more business risk?

WHERE PEOPLE GO WRONG

Some students say that a fixed cost is the same for every unit. This is wrong. The total fixed cost stays the same, so the fixed cost per unit falls as more units are sold. The printing unit spreads Tk 240,000 over 6,000 packs at Tk 40 each, but over 8,000 packs at Tk 30 each.

Some students divide the fixed cost by the selling price. The price includes the variable cost, which must be paid for each unit. Only the contribution margin is left to cover the fixed cost, so it is the divisor.

Some students think a high degree of operating leverage is always an advantage. It multiplies gains when sales rise and multiplies losses when sales fall. Whether it helps depends on the likely path of sales.

Some students quote one degree of operating leverage for a whole business. It depends on the starting volume. The printing unit has 4.00 at 8,000 packs and 3.00 at 9,000 packs, and no value at break-even.

IN HISTORY

Henry Ford (1863–1947) was an American car maker who founded the Ford Motor Company in 1903 in Detroit, Michigan. The Model T went on sale in 1908. From 1913, at the Highland Park plant, on the edge of Detroit, Ford's factory used a moving assembly line, in which the car moved past the workers instead of the workers moving to the car. The line needed heavy spending on plant and machinery, which is fixed cost, and it cut the labour needed per car, which is variable cost.

The price of a Model T was about 850 dollars at its launch in 1908 and had fallen to 260 dollars for the Runabout model by 1925. Ford sold the car in far greater numbers than any maker before him. Large fixed costs are carried well only at high volume, which is the lesson of this chapter. A business that raises its fixed costs must be sure its sales will stay above its break-even point.

REVIEW QUESTIONS

- 11.1 Explain the difference between a fixed operating cost and a variable operating cost, and give one example of each from a bookshop.
- 11.2 A business sells at Tk 150 per unit, with a variable operating cost of Tk 95 per unit and a fixed operating cost of Tk 330,000 a year. Find the contribution margin and the operating break-even point.
- 11.3 For the business in Question 11.2, find EBIT at 5,000, 6,000

Fixed cost per unit is not fixed: only the total fixed cost stays the same.

The break-even point divides by the contribution margin, not by the price.

High leverage is not always good, because it magnifies losses as well as gains.

The degree of operating leverage is not one fixed number for a business: it changes with the volume.

Henry Ford's moving assembly line raised fixed cost, lowered variable cost per car, and led to a fall in the Model T's price.

The price of the Model T fell from about 850 dollars to 260 dollars, as high volume spread the fixed cost.

SOURCES

The landmark works behind the chapter are Knight, 1921, and Markowitz, 1952.

Frank Knight, in *Risk, Uncertainty and Profit* (1921), argued that profit is the reward for bearing uncertainty, and he separated risks that can be measured from those that cannot. Harry Markowitz, in "Portfolio Selection" (*Journal of Finance*, 1952), first treated the spread of returns as the measure of investment risk. The Lloyd's account draws on the history pages published by Lloyd's of London.

REVIEW QUESTIONS

- 9.1 A bond costs Tk 80,000, pays Tk 5,600 and could be sold for Tk 82,000. Find the return and the rate of return.
- 9.2 Explain the difference between business risk and financial risk, with one example of each.
- 9.3 An asset returns 6%, 10% or 14%, with probabilities 0.25, 0.50 and 0.25. Find the expected value and the standard deviation.
- 9.4 Asset M has an expected return of 10% and a standard deviation of 3%. Asset N has an expected return of 18% and a standard deviation of 9%. Find each coefficient of variation and say which carries more risk for each unit of return.

Figure 11.2 shows the result in pictures. Each pair of bars compares a value before and after the rise in sales. Both starting values are scaled to 100, so the height of the second bar shows the new value as a share of the first. The sales pair rises from 100 to 112.5, and the EBIT pair rises from 100 to 150. The second pair climbs four times as far as the first.

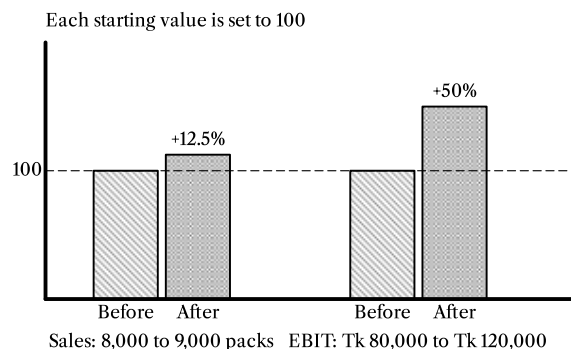


FIG. 11.2. Sales rise by 12.5% while EBIT rises by 50%, with each starting value set to 100

PAUSE AND CHECK 11.P2 A business has a contribution of Tk 60,000 and an EBIT of Tk 20,000. What is its degree of operating leverage, and by what percentage would EBIT change if sales fell by 10%?

The lever works downward as well. If sales fall from 8,000 to 7,000 packs, a fall of 12.5%, EBIT falls from Tk 80,000 to Tk 40,000. That is a fall of 50%, which is $12.5\% \times 4.00$. The case file lists this EBIT of Tk 40,000 for 7,000 packs. The same fixed cost that magnified the gain now magnifies the loss.

Four points limit the use of the number. First, it changes with volume: it falls from 4.00 to 3.00 as the unit moves from 8,000 to 9,000 packs. Second, it is highest close to break-even, where EBIT is small. Third, at break-even EBIT is zero, so the division cannot be performed and the value is undefined. Fourth, below break-even EBIT is negative and the value has no useful meaning. Common slips include dividing EBIT by contribution, which turns the answer upside down, and using the starting volume of a different rise.

Before moving on. Operating leverage is the effect of fixed cost in making EBIT move more than sales. The degree of operating leverage equals contribution divided by EBIT at the starting volume, which is 4.00 at 8,000 packs for the printing unit.

* * *

11.4 OPERATING LEVERAGE AND BUSINESS RISK

² Recall: business risk, Chapter 9, 'Return and Risk', section 9.2 'What Risk Is'.

Figure 11.2 shows the two changes side by side, scaled so that each starting value equals 100.

The lever works in both directions, so a fall in sales cuts EBIT by the same multiple.

The degree of operating leverage is highest near break-even, falls as volume grows, and cannot be used at break-even.

The degree of operating leverage is contribution divided by EBIT, and it is large when EBIT is small relative to contribution.

A certain gain and an uncertain gain raise one question: how much more must the second promise?

Different people give different answers, so we need a principle rather than a feeling.

The chapter gives a rule for the extra return, a way to separate removable danger from the rest, and a view of how markets use information.

The chapter relies on earlier ideas about risk, return and the benchmark for projects.

Lenders ask for more from a borrower who is less likely to repay.

Risk-free rate. the return on an investment with no chance of loss, such as a short government bill

The risk-free rate is the reward for waiting alone, with no uncertainty in it.

Premium. Word in use. In daily life, a payment for insurance, or an extra charge added to a price. In finance, the extra return demanded for bearing risk.

Risk premium. the extra

Suppose a relative offers you a choice. You may lend Tk 10,000 to the government for three months and receive 8% a year with certainty. Or you may lend the same sum to a stranger who plans to sell tea from a cart, and the stranger offers 9%. Would you accept the stranger's offer? Most people say no. One extra percentage point seems too little for the chance that the stranger repays nothing. What if the stranger offered 12%? Or 20%?

Two people may answer differently, and neither is foolish. Yet a business meets this question every day, with real money. It must decide whether a project that might earn 14% is worth the danger that it earns far less. If managers answered by feeling, a business would accept a good project in one month and refuse it in the next, for no reason that anyone could explain.

This chapter builds the principle in four steps. First: how much extra return does a given danger demand? Second: is all danger paid for, or can some of it be removed at no cost? Third: can we turn the answer into one formula that gives a required return? Fourth: how quickly do markets take in what is known? By the end you will calculate a required return by hand and use it to accept or reject an investment.

You will need Chapter 9, 'Return and Risk', sections 9.2 to 9.4, which define risk and show how to measure it with the expected value and the standard deviation. You will also need Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value', where the required return is the benchmark for a project.

* * *

10.1 RISK PREMIUM

Return to the tea seller and the government. Both want to borrow Tk 10,000 for three months. The government will almost certainly repay. The tea seller may have a poor season and repay only part. A sensible lender asks the tea seller for a higher rate. The extra is the price of the added uncertainty, and its size depends on how much danger the lender sees.

To measure the extra, we need a starting point that has none. The **risk-free rate** is the return on an investment with no chance of loss, such as a short government bill. A *government bill* is a loan to the government, repaid within a year or less. No investment is perfectly free of risk, but a short loan to a stable government comes very close. The risk-free rate pays only for waiting, which is the time value of money. In the running case the natural choice would be a Bangladesh government bill; the 8% used below is assumed for illustration.

Now the extra has a name. A **risk premium** is the extra return investors demand for taking on risk above the risk-free rate. Risk, as Chapter 9 defined it, is the chance that results differ from those expected.¹ In words: risk premium equals required return less risk-free rate. If the risk-free rate is 8% and the tea seller must offer 12% before you will lend, the risk premium is $12\% - 8\% = 4\%$.

¹ Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', defines risk.

Notice the pattern. Moving from 800 to 1,000 notebooks adds Tk 1,600 to EBIT, and so does moving from 1,000 to 1,200. Each step of 200 notebooks adds $200 \times 8 = \text{Tk } 1,600$, because the fixed cost does not grow. Below 1,000 notebooks the EBIT is negative, which means a loss.

REALISTIC NUMBERS

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Break-even: $240,000 \div 40 = 6,000$ packs. EBIT at 7,000 packs: Tk 40,000. EBIT at 8,000 packs: Tk 80,000. EBIT at 9,000 packs: Tk 120,000.

We now follow the arithmetic behind these lines. *Given.* Price Tk 90, variable cost Tk 50, fixed cost Tk 240,000 a year. *Find.* The break-even point, and EBIT at 7,000, 8,000 and 9,000 packs. *Formula.* The same two rules as before: $Q = F \div CM$, and $EBIT = Q \times CM - F$. *Steps.* $CM = 90 - 50 = \text{Tk } 40$. Break-even $Q = 240,000 \div 40 = 6,000$ packs. At 7,000 packs, $EBIT = 7,000 \times 40 - 240,000 = 280,000 - 240,000 = \text{Tk } 40,000$. At 8,000 packs, $EBIT = 320,000 - 240,000 = \text{Tk } 80,000$. At 9,000 packs, $EBIT = 360,000 - 240,000 = \text{Tk } 120,000$. *Answer.* The unit breaks even at 6,000 packs, and each extra 1,000 packs adds Tk 40,000 to EBIT. *Check.* 1,000 packs $\times \text{Tk } 40 = \text{Tk } 40,000$, which matches each step in the list.

Figure 11.1 shows the same unit as a chart. The horizontal axis shows packs sold in a year, and the vertical axis shows taka a year. Each point on a line shows the money at that volume. The revenue line starts at zero and rises by Tk 90 per pack. The total cost line starts at the fixed cost of Tk 240,000 and rises by Tk 50 per pack. The lines cross at 6,000 packs, which is revenue of $6,000 \times 90 = \text{Tk } 540,000$. To the left of the crossing the cost line lies above revenue, and the gap is the loss. To the right, revenue lies above cost, and the gap is profit.

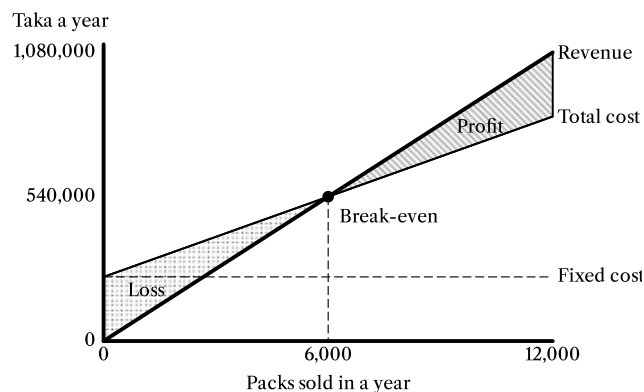


FIG. 11.1. Revenue and total cost of the printing unit, crossing at the break-even point

Three slips are common. The first is dividing the fixed cost by the price instead of the contribution margin; $240,000 \div 90$ gives about 2,667 packs, far too low, because it ignores the variable cost. The

Each extra 200 notebooks adds the same Tk 1,600 to EBIT, because the fixed cost does not grow.

The printing unit applies the same method to larger, realistic figures.

The steps behind the case file repeat the method: contribution first, then subtract the fixed cost.

The chart shows revenue and total cost as two lines, with loss and profit as the gaps between them.

Common slips are dividing by the price, mixing the periods, and forgetting that break-

Portfolio. a collection of investments held together by one investor

A portfolio is the whole set of investments that one investor holds.

Diversification. spreading money across different investments so that one loss hurts less

Diversification spreads money across different investments.

Diversifying removes some risk but not all, and the two kinds of risk need names.

Firm-specific risk. risk that affects one business only and can be reduced by diversifying

Firm-specific risk strikes one business and can be diversified away.

Market. Word in use. In daily life, a place where people buy and sell goods, such as a bazaar. In finance, all the buyers and sellers of one kind of investment, taken together.

Market risk. risk that affects all investments in a market and cannot be removed by diversifying

Market risk reaches every investment and remains however many are held.

The figure shows risk falling as different assets are added, until only market risk is left.

The trader's two stalls form a **portfolio**. A portfolio is a collection of investments held together by one investor. A person who owns common stock in five businesses holds a portfolio of five holdings. So does a business that owns a shop, a warehouse and a loan to a customer. What matters about a portfolio is the combined result, not the result of each part on its own.

Building such a portfolio has its own name. **Diversification** is spreading money across different investments so that one loss hurts less. The trader's move from one stall to two is diversification. The word *different* matters. Ten stalls that all sell umbrellas spread the money but not the danger, because one wet season hurts all ten.

Does diversifying remove all risk? It does not. Think of two kinds of bad news. A fire at one factory, a strike at one shop or a failed new product harms one business and leaves the others alone. A rise in the cost of borrowing, a fall in demand across the whole country or a war reaches almost every business at once.

The first kind is **firm-specific risk**: risk that affects one business only and can be reduced by diversifying. An investor who holds many businesses feels one failure as a small part of the portfolio, and good news at some businesses offsets bad news at others.

The second kind is **market risk**: risk that affects all investments in a market and cannot be removed by diversifying. Here *market* means all the buyers and sellers of a kind of investment, taken together. A fall in demand across the whole country lowers the profit of nearly every business, so no mixture of businesses escapes it.

To read the next figure, look first at the two axes. The horizontal axis counts how many different assets the portfolio holds. The vertical axis shows the total risk of the portfolio. The curve falls steeply at first, because each new asset removes much firm-specific risk, and then flattens at a floor. The shaded area above the floor is firm-specific risk; the floor itself is market risk.

An operating cost is money spent on the everyday work of the business, before any interest or tax is paid. A **fixed operating cost** means an operating cost that stays the same whatever the number of units sold. The market rent of Tk 8,000 a year is a fixed operating cost, because it is the same for ten notebooks or for a thousand. The depreciation of a machine and the salary of a shop manager are further examples.

A **variable operating cost** means an operating cost that rises and falls with the number of units sold. The Tk 12 paid for each notebook is variable. If the seller sells 100 notebooks, this cost is Tk 1,200; if the seller sells 200, it is Tk 2,400. Each unit adds the same amount. Materials, packing and sales commission are other examples.

We can now write a rule for the whole bill. In words: total operating cost equals fixed operating cost plus variable operating cost per unit multiplied by units sold. In symbols, $TC = F + v \times Q$. Here TC is total operating cost, F is fixed operating cost, v is variable operating cost per unit, and Q is the number of units sold. Suppose the seller sells 500 notebooks. Then $TC = 8,000 + 12 \times 500 = 8,000 + 6,000 = \text{Tk } 14,000$. Sales revenue is $500 \times 20 = \text{Tk } 10,000$, so EBIT is $10,000 - 14,000$, a loss of Tk 4,000.

The **contribution margin** is the selling price per unit minus the variable operating cost per unit. Each notebook gives $20 - 12 = \text{Tk } 8$. We call it a contribution because those Tk 8 go first towards the fixed cost and, once that is paid, towards profit.

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Selling price Tk 90 per pack. Variable operating cost Tk 50 per pack. Contribution margin Tk 40 per pack. Fixed operating cost Tk 240,000 a year.

Real costs seldom divide so neatly. An electricity bill may have a standing charge, which is fixed, and a charge for each unit of power used, which is variable; such a cost is part fixed and part variable. A cost is also fixed only within a range of volume. If the seller doubles sales, one table may no longer suffice, and a second table brings a second rent. In this chapter we assume that the selling

Cost. Word in use. In daily life, the price you pay for something. In business, any money given up to run the business, such as rent, wages or materials.

Fixed. Word in use. In daily life, fastened in place or repaired. Here, describing a cost that does not change with the number of units sold.

Fixed operating cost. an operating cost that stays the same whatever the number of units sold

The table rent is a fixed operating cost: it does not change with the notebooks sold.

Variable operating cost. an operating cost that rises and falls with the number of units sold

The Tk 12 paid for each notebook is a variable operating cost, because it grows with each unit sold.

Total operating cost equals the fixed cost plus the variable cost per unit multiplied by the units sold.

Contribution margin. selling price per unit minus variable operating cost per unit

Margin. Word in use. In daily life, the empty edge of a page or a small safety gap. In business, a difference between a price and a cost, as here.

Each unit sold contributes its price less its variable cost towards the fixed cost.

The printing unit's record of 6 April 2025 gives the price, costs and contribution margin used later.

Real costs are only roughly fixed or variable, and a fixed cost holds only within a range of volume.

Beta is found from past records, and it measures market risk only.

The section's ideas are the portfolio, diversification, the two kinds of risk and beta.

After Markowitz measured risk, one question remained: which risk should the market pay for?

GO SLOWLY HERE.

Capital asset pricing model. a formula setting required return as risk-free rate plus beta times the market risk premium

The model builds the required return from the risk-free rate and a premium scaled by beta.

The first example uses small, clean numbers so that only the method must be learned.

Two more betas show the formula at the ends of its range.

Two limits apply. First, beta is found by comparing the past returns of an investment with those of the market, and data services publish it; this book uses betas that are given and does not teach the estimation. Past movement may not repeat. Second, beta measures only market risk. The standard deviation of Chapter 9 measures total risk, which holds both kinds,³ so an investment can have a high standard deviation and a low beta when most of its risk is firm-specific.

Before moving on. A portfolio is the set of investments one investor holds, and diversification spreads money across different investments. It removes firm-specific risk but not market risk, and beta shows how strongly an investment moves with the market. *You should now be able to say why diversifying cannot remove market risk.*

* * *

10.3 THE CAPITAL ASSET PRICING MODEL

In 1952 Harry Markowitz showed how to measure the risk of an investment by the spread of its returns. A question stayed open. If investors diversify, which part of that spread should the market pay for? In 1964 William Sharpe, an American economist, gave the answer: only market risk, measured by beta. John Lintner in 1965 and Jan Mossin in 1966, working separately, reached similar results. Sharpe later shared the 1990 Nobel Memorial Prize in Economic Sciences for this line of work.

The **capital asset pricing model** is a formula setting required return as risk-free rate plus beta times the market risk premium. The *market risk premium* is the premium for the whole market: the return of the market less the risk-free rate. In words: required return equals the risk-free rate plus beta times (market return less risk-free rate). In symbols, with R_f for the risk-free rate, R_m for the market return and β for beta: required return $= R_f + \beta \times (R_m - R_f)$.

Given. Risk-free rate 6%; market return 10%; beta 1.0. *Find.* The required return. *Formula.* Required return $= R_f + \beta \times (R_m - R_f)$. *Steps.* First, the market risk premium: $10\% - 6\% = 4\%$. Second, beta times the premium: $1.0 \times 4\% = 4\%$. Third, add the risk-free rate: $6\% + 4\% = 10\%$. *Answer.* The required return is 10%. *Check.* A beta of 1 means the investment moves as the market does, so it must require the market return, 10%, and it does.

Try two other betas with the same rates. With a beta of 0 the required return is $6\% + 0 \times 4\% = 6\%$, the risk-free rate, because an investment that does not move with the market carries no market risk. With a beta of 2 it is $6\% + 2 \times 4\% = 14\%$, because the investment carries twice the market's risk and earns twice the market's premium.

PAUSE AND CHECK 10.P2 With a risk-free rate of 6% and a market

³ Chapter 9, 'Return and Risk', section 9.4 'Measuring the Spread of Outcomes', shows that standard deviation measures total risk.

The strength of the claim depends on which information is meant.

Weak-form efficiency. prices already reflect all past price information

The weak form covers past prices only.

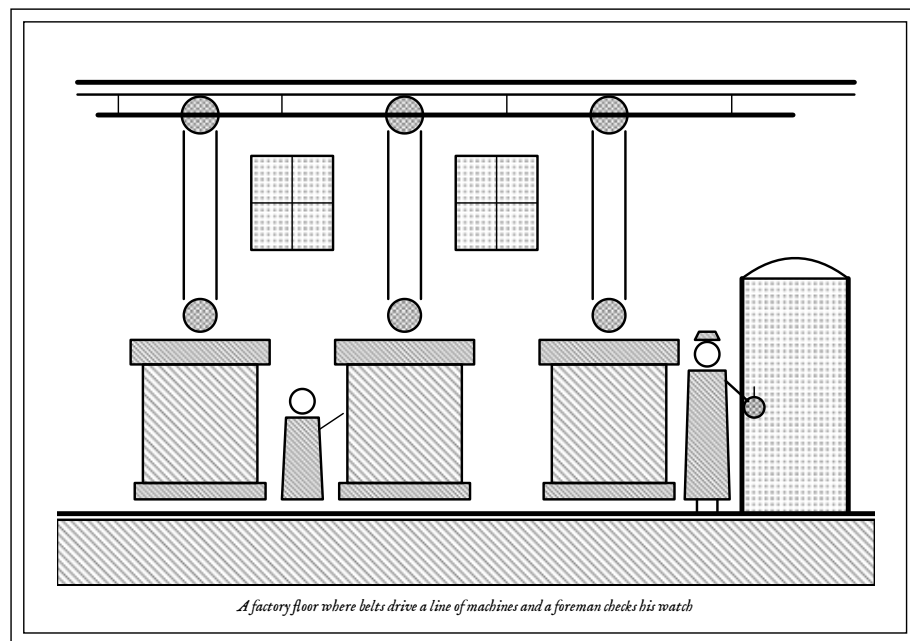
Semi-strong-form efficiency. prices reflect all public information, not only past prices

The semi-strong form covers everything that is public.

Strong-form efficiency. prices reflect all information, public and private

The strong form covers private information too.

The table sets the three forms side by side.



CHAPTER XI

Operating Leverage and Break-even Analysis

Why can a small rise in sales bring a large rise in profit?



Some costs stay the same however much a business sells, and others grow with every unit. This chapter separates the two kinds, finds the sales volume at which a business neither gains nor loses, and measures how strongly a change in sales moves profit. It closes by linking these ideas to risk.

Efficiency varies between markets and periods, so claims must be made with care.

The section's ideas are the efficient market and its three forms.

The case file gives the market rates and the beta and expected return of each line.

How much information do prices contain? There are three answers, each stronger than the one before. They differ in the kind of information that prices already reflect.

Weak-form efficiency means that prices already reflect all past price information. If it holds, studying old price charts cannot show which investment will rise next, because the pattern is already in the price.

Semi-strong-form efficiency means that prices reflect all public information, not only past prices. Annual reports, news and announcements are public. If this form holds, reading them after they appear brings no gain, because the price has already moved.

Strong-form efficiency means that prices reflect all information, public and private. Even a manager's secret knowledge would be in the price. Few people believe this form holds, and in many countries trading on secret information is a crime.

The table lists the three forms. Read each row from left to right: the form, the information already in the price, and an example of what would not help an investor if the form holds.

THE THREE FORMS OF MARKET EFFICIENCY

FORM	INFORMATION IN THE PRICE	WOULD NOT HELP
Weak	Past prices	Studying last year's price chart
Semi-strong	All public information	Reading a report published this morning
Strong	Public and private information	Using a secret known only to managers

Efficiency is a matter of degree. Studies of large, busy markets have found support for the weak form and mixed results for the others. Smaller markets, where fewer people trade and less information is published, may be less efficient. A student should say which market and which period a claim concerns. For a financial manager, the lesson is practical: where prices reflect information quickly, the market value of the business is a fair guide to the quality of the manager's decisions.

Before moving on. An efficient market is one whose prices already reflect the information available. The weak form covers past prices, the semi-strong form all public information and the strong form all information. *You should now be able to say* why new information, and not old information, moves prices.

* * *

WORKED CASE: TWO PRODUCT LINES ON THE LINE

Case file, illustrative. Meghna and Sons, 20 March 2025. The risk-free rate is 8%, a short government bill, assumed for illustration. The market return is 14%, so the market risk premium is 6%. Study-

since $9,000,000 \div 200,000 = 45$. Then $45 \times 329 = \text{Tk } 14,805$, close to the answer.

Before moving on. The method never changed: days, then taka, then add and subtract. A change in one day count moves the need by that many days of the right amount. Float adds more days of money that cannot be used.

* * *

SUMMARY

Working capital is money held in current assets such as inventory and accounts receivable. Net working capital is current assets minus current liabilities. The operating cycle is inventory days plus receivable days, and the cash conversion cycle subtracts the payable days. For Meghna and Sons the cycle is 50 days and the net investment is Tk 1,010,959. Float is money paid but not yet usable. For one cheque it lasted 6 days and cost Tk 329.

THINK IT THROUGH

15.T1 Meghna and Sons expects sales and cost of sales to rise by one fifth, with the days unchanged. Estimate the net investment in the cycle, and the extra money needed. Explain in plain words why the need rises.

WHERE PEOPLE GO WRONG

Wrong: a business that makes a profit must have money. *Right:* profit is counted when goods are sold, but cash arrives only when customers pay. Money can sit in inventory and accounts receivable while bills fall due.

Wrong: a long cash conversion cycle shows a badly run business. *Right:* some trades, such as selling books to schools, have long cycles. The danger is a long cycle with no plan to fund it.

Wrong: the later a supplier is paid, the better. *Right:* more payable days shorten the cycle, but a supplier who is paid late may raise prices or stop deliveries.

Wrong: float is the bank's problem. *Right:* the payer keeps the use of the money while the cheque travels, and the receiver loses it. The loss is measured in days and in interest.

IN HISTORY

W. T. Grant was a chain of variety stores in the United States. William Thomas Grant opened the first of them in Lynn, Massachusetts, in 1906. By the 1970s the chain was large and national, and its reports still showed profit. In October 1975 it filed for reorganisation under Chapter 11 of the United States bankruptcy law, which lets a business keep trading while it tries to settle its debts. It was liquidated soon afterwards.

In 1980 two authors. James Largav and Clvde Sticknev. published

Recap: the same method handles clean numbers, the case file, a changed case and a combined case.

Working capital is money held in short-term assets, and the cash conversion cycle measures how long the business must fund it.

A profit in the reports is not cash in the drawer.

A long cycle is not a failure; it is a need that must be funded.

Paying suppliers later helps only while the supplier agrees to wait.

Float helps the payer and costs the receiver.

A large American retailer reported profits yet failed, because its cash was held in credit sales and inventory.

Two authors later showed

Sales to earnings per share: $\times 7.27$ in all
Bar height = percentage change

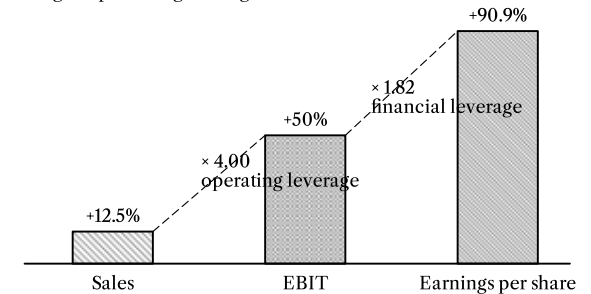


FIG. 12.1. Percentage changes in sales, EBIT and earnings per share, joined by the two degrees

Degree of total leverage. the percentage change in earnings per share caused by a one percent change in sales

The degree of total leverage measures the whole chain from sales to earnings per share.

Because the levers are joined, the two degrees multiply.

The **degree of total leverage** is the percentage change in earnings per share caused by a one percent change in sales. In words: degree of total leverage equals the percentage change in earnings per share divided by the percentage change in sales. From the table, $90.9\% \div 12.5\% = 7.27$. A one percent rise in sales therefore lifts earnings per share by about 7.27 percent.

The same number comes from the two degrees. The degree of operating leverage was 4.00, since the contribution margin of Tk 320,000 (8,000 packs at Tk 40) divided by EBIT of Tk 80,000 is 4.00. The degree of financial leverage was 1.8182. *Formula.* Degree of total leverage = degree of operating leverage $\times$ degree of financial leverage. *Steps.* $4.00 \times 1.8182 = 7.2727$. *Answer.* The degree of total leverage is 7.27. *Check.* A second route gives the same result: contribution margin $\div$ [EBIT - I - Dp $\div$ (1 - T)] = $320,000 \div 44,000 = 7.27$. Also $7.27 \times 12.5\% = 90.9\%$, which agrees with the table.

Now apply the lever to a fall. At 7,000 packs, sales fall by 12.5%. The contribution margin is $7,000 \times \text{Tk } 40 = \text{Tk } 280,000$, and EBIT is $280,000 - 240,000 = \text{Tk } 40,000$, a fall of 50%. Profit before tax is $40,000 - 20,000 = \text{Tk } 20,000$; tax is Tk 5,000; profit after tax is Tk 15,000; less the preferred dividend of Tk 12,000 leaves Tk 3,000. Earnings per share is $3,000 \div 5,000 = \text{Tk } 0.60$, a fall of 90.9%. The same 12.5% change in sales that nearly doubled earnings per share has almost wiped it out.

Before moving on. The degree of total leverage is the percentage change in earnings per share for a one percent change in sales, and it equals the degree of operating leverage times the degree of financial leverage. For the printing unit it is 7.27, and it works downward as strongly as upward.

* * *

12.4 LEVERAGE AND THE RISK OF A FIRM

Before moving on. Float is money paid by one person that the receiver cannot yet use. Its parts are mail, processing and clearing. Each day of float costs the receiver the use of the money.

* * *

15.4 ESTIMATING WORKING CAPITAL NEEDS

How much money does the cycle hold? Knowing that it lasts 50 days is not enough, because a manager must know how many taka to find. We turn each of the three items into taka. The idea is simple to state. One day of trading carries a certain amount, and the item holds that many days of it.

There are three rules, each using a 365-day year.¹ Inventory is held at cost, so inventory = cost of sales × inventory days ÷ 365. Accounts receivable is what customers owe at the selling price, so receivables = sales × receivable days ÷ 365. Accounts payable is owed at cost, so payables = cost of sales × payable days ÷ 365. The net investment in the cycle is inventory + receivables – payables.

Case file, illustrative. Meghna and Sons Limited, 1 September 2025. Using a 365-day year, inventory was Tk 1,294,521, receivables (on sales) were Tk 493,151 and payables were Tk 776,712. The net investment in the cycle was Tk 1,010,959.

So the shop must find about Tk 1,010,959 of its own money to keep the cycle turning. Its suppliers fund Tk 776,712 of the goods and the debts, and only for 45 days at a time. The full working is shown in section 15.5. How a business pays for this money is the work of Chapter 16, 'Short-Term Financing and Matching', section 16.4 'The Matching Principle'.

The need grows with trading. If sales and cost of sales rise by one fifth and the days stay the same, each of the three items rises by one fifth, and so does the net investment.

Three cautions apply. The estimate uses yearly averages, so a busy season can need more than the average. It leaves out cash and the other current assets. And the days can change: a customer who pays later raises the need at once.

Before moving on. Inventory and payables use cost of sales, and receivables use sales. Multiply by the days and divide by 365. Add inventory and receivables, then subtract payables. For Meghna and Sons the answer is Tk 1,010,959.

* * *

WORKED EXAMPLES: FROM DAYS TO TAKA

We work four examples. The first uses small, round numbers, so the method is the only new thing. The second uses the case file. The third changes the situation, and the fourth combines the ideas of this chapter. Each follows one order: given, find, formula, steps, answer, check.

¹ Some businesses use a 360-day year to make the arithmetic easier. This book uses 365 days throughout.

Recap: float is paid money that cannot yet be used, in three parts.

The days can be turned into taka to show how much money the cycle needs.

Each item equals one day's amount multiplied by the days it is held.

The case file gives the three amounts and the net investment for Meghna and Sons.

The shop must find about Tk 1,010,959 of its own money, because its suppliers fund only part of the cycle.

The need grows in step with trading: more sales means more money in the cycle.

The estimate uses averages and three items only, so it is a guide and not an exact budget.

Recap: turn each item into taka using days, then add inventory and receivables and subtract payables.

Four examples apply the method, from small numbers to a combined problem.

SUMMARY

A fixed financing charge is interest or preferred dividend owed at any level of EBIT. Financial leverage is the extra swing in earnings per share that such charges cause, and the degree of financial leverage measures it. The degree of total leverage equals the degree of operating leverage times the degree of financial leverage. For the printing unit the three degrees are 4.00, 1.82 and 7.27, and all three work downward as well as upward.

THINK IT THROUGH

12.T1 A business has a high degree of operating leverage. Explain in plain words why its owners might choose to borrow little.

WHERE PEOPLE GO WRONG

Error: "Borrowing always increases earnings per share." It does so only while EBIT is high enough to cover the charges. In a poor year the same borrowing lowers earnings per share more than EBIT falls.

Error: "Preferred dividend is deducted before tax." Interest is deducted before tax, but the preferred dividend is paid out of profit after tax. That is why the formula divides it by (1 – T).

Error: "Total leverage is operating leverage plus financial leverage." The degrees multiply: 4.00 × 1.82 = 7.27, and not 5.82.

Error: "The degree of financial leverage never changes." It is found at one level of EBIT and one financing plan, and it changes when either changes.

IN HISTORY

In 1988 Kohlberg Kravis Roberts, an American investment firm usually called KKR, won a bidding contest to buy RJR Nabisco, a large American maker of tobacco and food products. The price was about 25 billion dollars, or 109 dollars a share, the largest takeover of its time, and most of it was paid for with borrowed money.⁴ Such a purchase is called a leveraged buyout. The new owners hoped that profit from the business would exceed the interest, which would give them large gains on the small part they had paid themselves.

The lever turned out to be heavy. With so much interest to pay every year, the business sold parts of itself in the following years to reduce its debt. The episode is told in the book *Barbarians at the Gate*, by Bryan Burrough and John Helyar. It remains a plain lesson in the two edges of financial leverage: the owners gain when profit is high and carry a heavy burden when it is not.

REVIEW QUESTIONS

⁴ Sources describe the price as about 25 billion dollars; the amount differs slightly with the way debt is counted.

The gap has a name. The **cash conversion cycle** is the operating cycle minus the days a business takes to pay its suppliers. In symbols: $C = I + R - P$. Here C is the cash conversion cycle, I the inventory days, R the receivable days and P the payable days, which are the days the business takes to pay its suppliers. For the shop, $75 + 20 - 45 = 50$ days. Read it this way: for 50 days the shop has paid for goods and has not yet been paid by its customers.

IN SHORT

Cash conversion cycle = inventory days + receivable days - payable days. A shorter cycle means the business needs less of its own money.

The figure below is a timeline, and you read it from left to right. Each unit along the line is one day. Day 0 is the day the goods arrive, and day 95 is the day the cash arrives. The bracket at the top is the operating cycle. The three lower bars show inventory days, receivable days and payable days, and each bar starts and ends above the day marked beneath it. The outlined bar at the bottom is the gap, the cash conversion cycle.

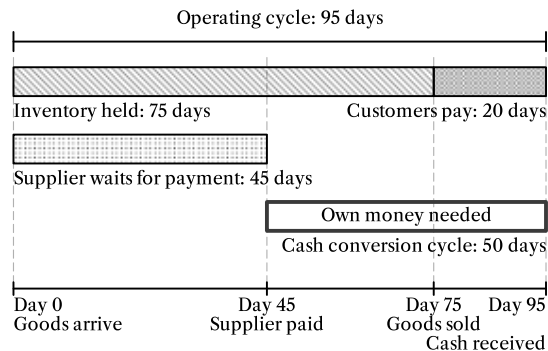


FIG. 15.1. Timeline of the cash conversion cycle for Meghna and Sons Limited

PAUSE AND CHECK 15.P2 A business has an operating cycle of 70 days and pays its suppliers after 30 days. What is its cash conversion cycle?

Three limits matter. First, the days are averages. A shop sells many products, and each has its own days. Second, a business that is paid at the till has no receivable days. A grocer who is paid in cash and pays suppliers after 30 days can have a cash conversion cycle below zero. This means customers' money arrives before the suppliers are paid. Third, a long wait for suppliers is not free. A supplier who is paid late may charge more, or may stop deliveries.

The financial manager therefore has three levers. Hold less inventory. Collect from customers sooner. Pay suppliers later, but fairly. A faster collection cuts the receivable days, so the operating

The cash conversion cycle counts only the days the business must supply its own money.

Cash conversion cycle. the operating cycle minus the days a business takes to pay its suppliers

The figure shows the three delays on one timeline and the gap that is the cash conversion cycle.

The cycle is an average, and it can be short, long, or even below zero.

The financial manager can shorten the cycle by holding less inventory or collecting sooner.

The cash, the rice and the customers' promises are what the woman uses to trade from day to day. Finance gives the money held in them a name. **Working capital** means the money a business has tied up in short-term assets such as stock and money owed by customers. "Tied up" means held in a form that cannot be spent at once. Her Tk 4,000 of rice cannot pay a bill until it is sold. The word *working* is used because this money keeps moving: from cash into goods, from goods into customers' promises, and back into cash.

Which assets belong in working capital? The short-term ones. A **current asset** is an asset expected to turn into cash within about a year. Her cash is already cash. Her rice will become cash when it is sold. The tea stall's promise will become cash when the tea stall pays. Her cart, which she will use for many years, is not a current asset.

The other side of the table is what she owes. A **current liability** is an amount owed that must be paid within about a year. The Tk 10,000 she owes the wholesaler is one, because it falls due on Friday. A loan that she may repay over five years is not.

The four sacks on her stall are her **inventory**. Inventory means goods held for sale or for use in making goods. The books on a bookshop's shelves are inventory. So are the cloth a tailor holds and the flour a baker holds before baking. Shopkeepers often say stock for the same thing. This book says inventory, and keeps the word stock for shares in a business, as in common stock.

The Tk 1,500 owed by the tea stall and the canteen is called **accounts receivable**. Accounts receivable is money owed to a business by customers who bought on credit. *Credit* here means permission to pay later. *Receivable* means "able to be received". A school that takes books in March and pays in April creates accounts receivable for a bookshop.

The mirror image is the Tk 10,000 owed to the wholesaler. **Accounts payable** is money a business owes to suppliers for goods bought on credit. *Payable* means "to be paid". The same debt appears on both sides of a trade. The wholesaler counts the Tk 10,000 as accounts receivable, and the stallholder counts it as

Working capital is the money a business keeps in the short-term things it uses to trade every day.

Working capital. the money a business has tied up in short-term assets such as stock and money owed by customers

Stock. Word in use. In daily life and in a shop, the goods on the shelves. In finance, also a share in a business, as in common stock. This book calls goods inventory.

A current asset is an owned item expected to become cash within about a year.

Recall: asset and liability, Chapter 1, section 1.3 'Where Funds Come From and Where They Go'.

Current. Word in use. In daily life, happening now, as in current news. In finance, due or expected within about a year, as in current asset.

Current asset. an asset expected to turn into cash within about a year

A current liability is an amount owed that must be paid within about a year.

Current liability. an amount owed that must be paid within about a year

Inventory is the goods a business holds, either to sell or to turn into other goods.

Inventory. goods held for sale or for use in making goods

Accounts receivable is what customers owe the business for goods taken on credit.

Account. Word in use. In daily life, a record of money kept at a bank. In accounting, a named record of one kind of item, such as the money owed by one customer.

Accounts receivable. money owed to a business by customers who bought on credit

Accounts payable is what the business owes its suppliers for goods taken on credit.

Accounts payable. money a business owes to suppliers for goods bought on credit

A certificate promises payments for five years, and the puzzle is how much a sensible person would pay for it.

A taka received later is worth less than a taka today, so the price depends on timing and on the return available elsewhere.

The chapter names the certificate, defines worth, calculates it, and then measures an investor's return; the pace is slow because two kinds of payment are valued together.

Many small certificates let a business borrow a large sum from many lenders.

GO SLOWLY HERE.

Bond. a loan sold to investors, repaid at a set date with fixed interest along the way

A bond is a loan sold to investors, repaid at a set date with fixed interest along the way.

Suppose a friend offers you a certificate. It promises Tk 90 at the end of each of the next five years, and Tk 1,000 at the end of the fifth year. The friend asks Tk 1,000 for it today. Should you pay? The promises total $Tk\ 90 \times 5 + 1,000 = Tk\ 1,450$, so the certificate seems a bargain. Many students find this step convincing, because totalling the promises feels natural. Yet the reasoning has a fault, and a second look will show it.

A taka received in five years is worth less than a taka held today, because the taka held today can earn a return in the meantime.¹ So the fair price depends on when each payment arrives and on what return you could earn elsewhere. A person who could earn 12% elsewhere would pay less than a person who could earn 7%. Both could be sensible. The task of this chapter is to turn that idea into a number.

We shall go in four steps. First we name the certificate and its parts. Second we ask what it means for any asset to have a value. Third we calculate the value, using tools from earlier chapters. Fourth we compare the value with the price and measure the return. The pace is slow, because two kinds of payment must be valued at once, and each idea is finished before the next begins.

YOU WILL NEED

Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'. Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'. Chapter 8, 'Capital Budgeting: Evaluation Techniques', sections 8.2 'Net Present Value' and 8.3 'Internal Rate of Return and Profitability Index'.

* * *

13.1 WHAT A BOND IS

Imagine a business that needs Tk 5,000,000 to buy equipment. Few lenders can lend so large a sum alone, and one lender would carry all the risk. The business can instead divide the loan into 5,000 pieces of Tk 1,000 and sell each piece to a different person. Each buyer lends Tk 1,000 and receives a certificate that states what the business has promised to pay. This arrangement is common, and the certificate has a name.

A **bond** is a loan sold to investors, repaid at a set date with fixed interest along the way. The business that borrows is the issuer, and the person who lends is the investor, also called the bondholder. A bond is therefore debt: the investor is a lender and not an owner of the business.² In the example above, the 5,000 certificates together form one loan of Tk 5,000,000.

¹ Recall: present value, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'.

² Preview: bonds and stock are compared in Chapter 14, 'Valuing Common and Preferred Stock', section 14.4 'Comparing Bonds and Stock'.



CHAPTER XV

Working Capital and the Cash Cycle

How can a business that sells well still have no money to pay its bills?



This chapter follows the money a business holds in goods and in customers' promises to pay. It measures the delay in days, shows what a slow cheque costs, and teaches you to estimate the cash a business must keep working.

A bond has a par value, a coupon interest and a maturity, and together these fix every payment.

To value a bond we first need the idea of value itself.

GO SLOWLY HERE.

Valuation. the work of finding what an asset is worth from the cash it is expected to bring

Valuation is the work of finding what an asset is worth from the cash it is expected to bring.

Intrinsic value. the worth of an asset worked out from the present value of its expected cash flows

Intrinsic value is the present value of the expected cash flows, discounted at the required return.

Intrinsic value is a calculation, while market value is a price, and a gap between the two may guide a decision.

Valuation turns expected cash and a required return into one number, the intrinsic value.

value than most assets. Two limits apply. An investor need not hold the bond to maturity, and may sell it to another investor at a price that differs from par. And the issuer may fail to pay, which is a risk the investor must judge. A comparison of bonds with stock follows later in the book.

PAUSE AND CHECK 13.P1 A bond has a par value of Tk 500 and a coupon rate of 8% a year. How much coupon interest does it pay each year?

Before moving on. A bond is a loan sold to investors. Its par value is repaid at maturity, and its coupon interest is paid each period. For the case bond these are Tk 1,000, Tk 90 a year and five years. *You should now be able to say* what a bond is and what its three facts are.

* * *

13.2 THE IDEA OF VALUE

A neighbour offers to sell you a mango tree. What is it worth? The cost of planting it does not decide the answer, and neither does its age. The tree is worth what it will bring: the mangoes of the coming seasons, counted by when they will arrive. Ten mangoes next year are worth more to you than ten mangoes in ten years. The same thinking applies to any asset, and to a bond in particular.

Valuation is the work of finding what an asset is worth from the cash it is expected to bring. The word asset here means anything that brings cash, such as a tree, a shop or a bond. Valuation needs two inputs: the cash the asset is expected to bring, and the required return, which is the return the investor needs from an asset of that risk.

The result of valuation has a name. **Intrinsic value** is the worth of an asset worked out from the present value of its expected cash flows. Take the smallest case. An asset will pay Tk 1,100 in one year, and the required return is 10%. Its intrinsic value is $1,100 \div 1.10 = \text{Tk } 1,000$. Today's Tk 1,000, invested at 10%, would grow to the same Tk 1,100.

Do not confuse intrinsic value with market value, the price at which the asset trades today. Intrinsic value is a calculation from expected cash and a required return. Market value is observed in the market. If the intrinsic value is above the market value, the asset looks cheap to the investor who made the calculation; if below, it looks costly. The two may differ because investors disagree about the cash to come or about the required return.

Before moving on. Valuation finds what an asset is worth from the cash it is expected to bring. The intrinsic value is the present value of that cash at the required return. *You should now be able to say* why Tk 1,100 due in one year has an intrinsic value of Tk 1,000 at a

³ Recall: the present value of an ordinary annuity, Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'. The par value is a single sum discounted as in Chapter 4, section 4.3.

grows 6% a year as a capital gain. Together they give 12%, equal to the required return.

Many students make the same slips here, so check for them. First, writing the percentage as a whole number: $8 \div 10$ gives 0.8, not 80.00, so convert 10% to 0.10 before dividing. Second, using the last dividend where the formula needs the next one: $4.00 \div 0.06$ gives Tk 66.67, which is wrong because D_1 is Tk 4.24. Third, adding the growth rate to the required return instead of subtracting it. Fourth, using the formula when the growth rate is not below the required return. Always write the unit, Tk, with the answer.

* * *

SUMMARY

Stock makes its holder a part-owner. Common stock carries a voting right and a residual claim, and its owner can also make a capital gain. Preferred stock pays a fixed dividend ahead of common stock. Stock is valued as the present value of its dividends: a level dividend gives $P = D \div r$, and a dividend growing at a steady rate g gives $P_0 = D_1 \div (r - g)$, provided g is below r . A bond holder lends and a stockholder owns, so the stockholder carries more risk for a chance of greater reward.

THINK IT THROUGH

14.T1 A cautious investor with Tk 10,000 wants steady yearly income and fears losing money. She can buy a bond or common stock in a growing business. Which would you advise, and why? Give two reasons from this chapter.

WHERE PEOPLE GO WRONG

Mistake: "All stock pays a fixed amount, as a bond does." Correction: only preferred stock has a fixed dividend. The directors decide each common dividend, and they may raise it, lower it or skip it.

Mistake: "The formula value is the price I must pay." Correction: the formula gives intrinsic value, which rests on estimates of growth and required return. The price in the market may be higher or lower.

Mistake: "The model counts dividends only, so it ignores capital gain." Correction: the model values the whole stream of dividends, and a rising price follows from rising dividends. The check in the worked case found a 6% dividend part and a 6% capital gain part.

Mistake: "A fast-growing business can be given any growth rate." Correction: g must stay below r . When it does not, the formula breaks down or gives a negative value, and the assumption of such growth for ever must be rejected.

IN HISTORY

Four slips are common when using these formulas.

The chapter's main ideas in a few lines.

Mistake: believing all stock pays a fixed amount.

Mistake: treating the formula value as the price.

Mistake: thinking the growth model ignores capital gain.

Mistake: letting growth reach or pass the required return.

The half-yearly value uses 6% for ten periods and gives Tk 889.60.

The case file records the half-yearly value of Tk 889.60.

The half-yearly value is slightly lower because 6% per half-year is the same as a higher return over a full year.

Common slips are using the coupon rate as the discount rate, counting years instead of periods, and forgetting to halve the rate.

The value of a bond is the present value of its coupons plus the present value of its par value, at the required return.

The same bond can be worth more than, equal to, or less than its par value, depending on the required return.

GO SLOWLY HERE.

At a required return of 9% the bond equals par, and at 7% it is worth more than par.

The case file records the three values, and the table sorts them into three kinds of bond.

The table names each case by comparing the value with par.

there are 10 periods, and the required return of 12% a year becomes 6% per half-year.

Steps. With $r = 6\%$ and $n = 10$, $f = 1 \div (1.06)^{10} = 0.558395$, and $a = (1 - 0.558395) \div 0.06 = 7.360087$. The present value of the coupons is $45 \times 7.360087 = \text{Tk } 331.204$. The present value of the par value is $1,000 \times 0.558395 = \text{Tk } 558.395$. The value is $331.204 + 558.395 = \text{Tk } 889.599$.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond with half-yearly interest (coupon Tk 45, 10 periods) at 12% a year (6% per half-year): Tk 889.60.

Answer. The value is Tk 889.60. *Check.* It is a little below Tk 891.86. Half-yearly coupons arrive earlier, which would raise the value, but 6% each half-year compounds to $1.06 \times 1.06 = 1.1236$, which is a return of 12.36% over a year. This higher yearly return lowers the value by more.

Four slips are common. The first is discounting at the coupon rate instead of the required return. The second is counting years when interest is paid twice a year, where periods are needed. The third is halving the coupon but not the rate, or the reverse. The fourth is adding the present values of the coupons and the par value without checking that both use the same rate and the same number of periods.

Before moving on. The value of a bond equals the present value of its coupons plus the present value of its par value, both at the required return. For half-yearly interest, halve the coupon and the rate, and double the periods. *You should now be able to say why the case bond is worth Tk 891.86 at a required return of 12%.*

* * *

13.4 DISCOUNT, PAR AND PREMIUM BONDS

The case bond has a par value of Tk 1,000. At a required return of 12% it is worth Tk 891.86. What would it be worth if the market required 9%, or 7%? Before calculating, guess the direction. The bond pays a fixed Tk 90 a year. If other investments offer more than 9%, this bond is less attractive and its price should fall. If they offer less, it is more attractive and its price should rise.

Steps at 9%. With $r = 9\%$ and $n = 5$, $a = 3.889651$ and $f = 0.649931$. The present value of the coupons is $90 \times 3.889651 = \text{Tk } 350.069$, and of the par value $1,000 \times 0.649931 = \text{Tk } 649.931$. The sum is Tk 1,000.000. *Steps at 7%.* Here $a = 4.100197$ and $f = 0.712986$. The coupons give $90 \times 4.100197 = \text{Tk } 369.018$, and the par value gives $1,000 \times 0.712986 = \text{Tk } 712.986$. The sum is Tk 1,082.004, which is Tk 1,082.00 to the nearest paisa.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond. Required return 12%: value Tk 891.86. Required return 9%: Tk 1,000.00. Required return 7%: Tk 1,082.00.

The table below collects the three results. Read each row from left to right: the required return, the value that follows from it, and the name of the kind of bond.

The pattern gives us the tool. The **dividend growth model** is a formula valuing stock as next dividend divided by required return minus growth rate. In words: value today equals the next dividend divided by the difference between the required return and the growth rate. In symbols, $P_0 = D_1 \div (r - g)$, where P_0 is the value today, D_1 is the next dividend, r is the required return and g is the growth rate, both written as decimals. The formula comes from the endless list of present values. Each dividend grows by the factor $1 + g$, yet is discounted by the factor $1 + r$. When r is larger than g , each present value is a little smaller than the one before, and the endless total settles at $D_1 \div (r - g)$.

IN SHORT

Value today = next dividend $\div$ (required return - growth rate). The next dividend is the last dividend $\times$ (1 + growth rate).

PAUSE AND CHECK 14.P1 A stock's next dividend is Tk 10, its required return is 15% and its growth rate is 5%. What is its value today?

The formula has firm limits. First, the growth rate must stay below the required return. If g equals r , the bottom of the fraction is zero. If g is larger than r , the answer is negative, which has no meaning. No business can keep raising its dividends faster than the required return for ever. Second, the formula assumes steady growth for ever, so it suits mature businesses better than young ones. Third, the answer is very sensitive when r and g are close. With $D_0 =$ Tk 4.00 and $r = 12\%$, a growth rate of 11% gives $4.44 \div 0.01 =$ Tk 444.00, though a growth rate of 6% gives only Tk 70.67. Fourth, a growth rate of zero returns us to the level formula of the last section.

Where does r come from? A stock is risky, so its required return is higher than that of a safe deposit. The capital asset pricing model, which you met earlier, builds r in steps: start with the return on a safe investment, then add the stock's beta multiplied by the extra return that the market pays above the safe rate. Take a safe return of 6%, a market return of 11% and a beta of 1.2. Then $r = 6\% + 1.2 \times (11\% - 6\%) = 6\% + 6\% = 12\%$. Every input is an estimate, so the value we find is an estimate too. It is the stock's intrinsic value, which can differ from the price at which the stock trades.

Before moving on. A growing dividend is valued by the dividend growth model: next dividend divided by the required return minus the growth rate. The growth rate must stay below the required return. The required return comes from the capital asset pricing model.

* * *

14.4 COMPARING BONDS AND STOCK

³ Chapter 13, 'Valuing Bonds', section 13.4 'Discount, Par and Premium Bonds': bond value is worked in the same way.

The dividend growth model values stock as next dividend over required return minus growth.

Dividend growth model. a formula valuing stock as next dividend divided by required return minus growth rate

The formula fails when growth equals or exceeds the required return.

The required return on stock reflects its risk and comes from the capital asset pricing model.

Recap of the section on growing dividends.

GO SLOWLY HERE.

Yield to maturity. the yearly return an investor earns by buying a bond at its price and holding it to maturity

Yield. Word in use. In daily life, to yield is to give way, or the amount a crop produces. In bonds, yield is a return stated as a yearly percentage.

Yield to maturity is the yearly return from buying at the market price and holding to maturity.

The case file gives a price of Tk 920 and a yield to maturity of about 11.2%.

The method of trial tries two rates, sees that the price lies between the values, and interpolates.

The answer is checked by calculating the value at the estimated rate.

The yield to maturity differs from the coupon rate whenever the price differs from par.

A discount bond has a higher yield than its coupon rate, a par bond equal, and a premium bond lower.

A new bond is valued with half-yearly interest, and then its yield is found from a

So far the required return was given, and we found the value. Now reverse the question. The price is given, and we want the return. The **yield to maturity** is the yearly return an investor earns by buying a bond at its price and holding it to maturity. It is found by the same idea as the internal rate of return: the rate that makes the present value of the payments equal the price paid.⁴

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. A bond priced at Tk 920 with the same coupon and 5 years left has yield to maturity about 11.2%.

Given. Price Tk 920, $C =$ Tk 90, par value Tk 1,000, five years. *Find.* The yield to maturity. *Formula.* There is no direct formula, so we use trial. Choose a rate, calculate the value, and compare it with the price. A value above the price means the rate is too low, because a higher rate lowers the value. *Steps.* At 11% the value is Tk 926.08, above the price. At 12% the value is Tk 891.86, below the price. The yield lies between 11% and 12%. The price is $(926.08 - 920) = 6.08$ below the value at 11%, out of a total gap of $926.08 - 891.86 = 34.22$. The fraction is $6.08 \div 34.22 = 0.178$, so the yield is about $11\% + 0.178 \times 1\% = 11.18\%$.

Answer. The yield to maturity is about 11.2%. *Check.* At 11.2% the value is about Tk 919.10, within Tk 1 of the price of Tk 920. The straight-line step is an estimate, and the exact figure is a little lower, at 11.17%.

Compare the three rates for this bond. The coupon rate is 9% and never changes. The price of Tk 920 is below par, so the yield to maturity, 11.2%, is above the coupon rate: the investor gains the coupon and also the rise from Tk 920 to Tk 1,000. A bond at par has a yield equal to its coupon rate. A premium bond has a yield below it.

Before moving on. A bond sells at a discount when the required return is above the coupon rate, at par when equal, and at a premium when below. The yield to maturity is the return from buying at the price and holding to maturity, found by trial. *You should now be able to say why the case bond at Tk 920 has a yield to maturity above 9%.*

* * *

WORKED EXAMPLE: HALF-YEARLY INTEREST AND YIELD BY TRIAL

Given. A bond has a par value of Tk 1,000, a coupon rate of 8% a year paid half-yearly, and three years to maturity. *Find.* First, its value at a required return of 10% a year. Second, the yield to

⁴ Recall: internal rate of return, Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.3 'Internal Rate of Return and Profitability Index'.

remains belongs to the common stockholders. The directors may pay all of it as dividends or keep some of it inside the business.

The residual claim has a main limit: it has no fixed size. A lender is promised a fixed amount, but an owner is promised nothing. When the business does well, the remainder can be large and the owner gains. When the business does badly, the remainder can be small or nothing. If the business ends, the common stockholders are paid last, after lenders and preferred stockholders, and they may receive nothing. This is why common stock carries more risk than a bond, and why owners expect a higher return for holding it.

Dividends are not the only reward. A **capital gain** is the profit made when an asset is sold for more than it cost. If you buy stock for Tk 50 and later sell it for Tk 56, your capital gain is Tk 6. Dividends and capital gain together form your return from holding the stock. If you sell for less than you paid, you have a loss instead.

Preferred stock is a different bargain. Its holders usually receive a fixed dividend, for example Tk 8 on each share every year. The business must pay the preferred dividend before it pays any dividend to common stockholders. If the business ends, preferred stockholders are paid before common stockholders, though after lenders. In return for this priority, preferred stockholders usually give up the voting right, and their dividend does not rise when the business prospers. Terms differ from one issue to another. Some preferred stock is cumulative, which means that a skipped dividend is still owed and must be paid before any common dividend. Unlike interest on a loan, a preferred dividend is not a legal debt, so the directors can skip it, although doing so harms the business's standing with investors.

Before moving on. Stock makes you an owner. Common stock carries a voting right and a residual claim, and its owner may also gain from selling it at a higher price. Preferred stock pays a fixed dividend ahead of common stock, usually without a vote.

* * *

14.2 VALUING PREFERRED STOCK AND LEVEL DIVIDENDS

How much should you pay for a stream of payments that never ends? The answer comes from an idea that you have used before: the value of any asset is the present value of the cash that it will bring. For stock, the cash is the dividends. A fixed dividend paid at the end of every year for ever is a perpetuity.² Because the dividend is level, the present values of all the payments add up to a short formula.

The rule is short. The value of stock with a level dividend equals the yearly dividend divided by the required return. In symbols, $P = D \div r$. Here P is the value of one share today, D is the dividend paid each year, and r is the required return written as a decimal, so 12% becomes 0.12. Preferred stock fits this rule directly, because its

² Chapter 5, 'Annuities', section 5.1 'What an Annuity Is': a fixed dividend for ever is a perpetuity.

The claim has no fixed size: it can be large, or it can be nothing.

A second reward is selling stock for more than you paid.

Capital gain. the profit made when an asset is sold for more than it cost

Preferred stock pays a fixed dividend ahead of common stock, usually without a vote.

Recap of the section on rights.

The value of stock is the present value of the dividends it will pay.

For a level dividend, divide the yearly dividend by the required return.

IN HISTORY

In the middle of the eighteenth century, the British government carried many separate debts. Henry Pelham, who was Prime Minister of Great Britain from 1743 until his death in 1754, took advantage of low market rates of interest to reduce the burden. In 1751 he merged several of these debts into one stock, later known as Consols, short for consolidated annuities. Its interest rate was brought down to 3 per cent.

Consols set no date at which the government had to repay. An investor received interest each year and, to turn the holding into cash, sold it to another investor at the market price. Because the interest was fixed, the price rose when market rates fell and fell when they rose, exactly as this chapter describes. The stock created a long-lived market in bonds, and later governments issued similar stocks. They have since been repaid.

REVIEW QUESTIONS

13.1 A bond has a par value of Tk 1,000, a coupon rate of 8% and four years to maturity. The required return is 10%. Find its value.

13.2 State how the required return compares with the coupon rate for a discount bond, a par bond and a premium bond, and give the reason for each.

13.3 A new bond has a par value of Tk 1,000 and a coupon rate of 6% paid half-yearly. It has two years to maturity, and the required return is 8% a year. Find its value.

13.4 The case bond (coupon 9%, par value Tk 1,000, five years) is priced at Tk 950. Using trial rates of 10% and 11%, estimate its yield to maturity.

13.5 Explain in plain words why the value of a bond falls when the required return rises.

adds interest every six months has a compounding period of six months. (*Chapter IV*)

Contribution margin. The selling price per unit minus the variable operating cost per unit. It is the amount each unit sold contributes towards the fixed operating cost and then towards profit. A pack sold at Tk 90 with a variable cost of Tk 50 has a contribution margin of Tk 40. (*Chapter XI*)

Cost of capital. The average return a business must earn on its funds to satisfy those who supplied them. It is a rate, not an amount, and it serves as the discount rate for judging projects. For Meghna and Sons Limited it is 12.85%. (*Chapter XVII*)

Coupon interest. The fixed interest a bond pays each period, set as a percentage of par value. The percentage is the coupon rate. A par value of Tk 1,000 and a coupon rate of 9% give coupon interest of Tk 90 a year. (*Chapter XIII*)

Covenant. A promise in a loan agreement that limits what the borrower may do. It may set a minimum level of owners' equity or limit new borrowing. If it is broken, the lender may ask for early repayment. (*Chapter XVII*)

Credit terms. The discount offered and the days allowed for payment, written like 2/10, net 40. This means 2% off if paid within 10 days, and otherwise the full amount due by day 40. The code lets a buyer compare the saving with the cost of waiting. (*Chapter XVI*)

Current asset. An asset expected to turn into cash within about a year. Cash, inventory and accounts receivable are examples. A delivery cart used for many years is not a current asset. (*Chapter XV*)

Current liability. An amount owed that must be paid within about a year. A bill owed to a supplier and due next week is an example. A loan repayable over five years is not. (*Chapter XV*)

Debenture. A bond backed only by the general standing of the borrower, with no particular asset pledged. Its holder trusts the borrower's record and profit, and so carries more risk than a lender with collateral. (*Chapter XVII*)

Debt. Money borrowed that must be repaid, with an extra charge for using it. The lender is promised repayment on agreed dates. The bank loan of Tk 600,000 at 10% a year is debt. (*Chapter I*)

Degree of financial leverage. The percentage change in earnings per share caused by a one percent change in EBIT. It equals EBIT divided by EBIT less interest less preferred dividend divided by one minus the tax rate. For the printing unit, $80,000 \div 44,000 = 1.82$. (*Chapter XII*)

Degree of operating leverage. The percentage change in EBIT caused by a one percent change in sales. It equals contribution divided by EBIT at the starting volume. A value of 4.00 means a 1% change in sales changes EBIT by 4%. (*Chapter XI*)

Degree of total leverage. The percentage change in earnings per share caused by a one percent change in sales. It equals the degree of operating leverage times the degree of financial leverage. For the

years before the usual ratios did. A profit in the reports does not pay a bill. Cash does.

REVIEW QUESTIONS

15.1 A business holds inventory for 60 days, its customers pay in 25 days and its suppliers are paid in 35 days. Find the operating cycle and the cash conversion cycle.

15.2 The business in question 15.1 has yearly sales of Tk 10,950,000 and cost of sales of Tk 7,300,000. Using a 365-day year, find the money it invests in the cycle.

15.3 A customer posts a cheque. Mail takes 3 days, processing takes 2 days and clearing takes 2 days. Name the three parts of float and find the days lost.

15.4 Explain in your own words why a faster collection from customers shortens the cash conversion cycle.

GLOSSARY

Accounts payable. Money a business owes to suppliers for goods bought on credit. One business's accounts payable is its supplier's accounts receivable. (*Chapter XV*)

Accounts receivable. Money owed to a business by customers who bought on credit. A school that took books in March and will pay in April creates accounts receivable for the bookshop. (*Chapter XV*)

After-tax cash outflow. Cash paid out in a year after taking away the tax saved by that payment. Example: a lease payment of Tk 100 at a tax rate of 25% saves Tk 25 of tax, so the after-tax cash outflow is Tk 75. (*Chapter XVIII*)

Agency conflict. A clash between what a principal wants and what an agent chooses to do. A manager paid on sales may choose discounts that raise sales, while the owners want profit. (*Chapter III*)

Agency cost. The loss in value, plus the cost of prevention, that comes from an agency conflict. If a bad choice costs the owners Tk 105,000 and checking the records costs more, both amounts count. (*Chapter III*)

Agent. A person who acts for a principal and takes decisions in the principal's name. A hired branch manager is an agent of the owners. (*Chapter III*)

Amortisation. Repaying a loan by equal payments that cover the interest and part of the amount borrowed. The instalment equals the loan divided by the present value factor, and the loan falls to zero at the last payment. (*Chapter V*)

Annuity. A series of equal payments made at equal time intervals for a set number of periods. Three deposits of Tk 1,000, one at the end of each of three years, form an annuity, as do monthly loan instalments over two years. (*Chapter V*)

Annuity due. An annuity whose payments fall at the start of each period. Rent paid in advance is an example. Each payment comes one period earlier, so the annuity due is worth $1 + i$ times an ordinary annuity. (*Chapter V*)

Asset. Something of value that a business owns and uses to earn money, such as books for sale, shelves or a bank balance. Each asset is the result of a use of funds, so on any one date the assets equal the sources. (*Chapter I*)

Bank overdraft. A bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used. Interest runs only for the days used. For example, using Tk 50,000 for 30 days at 12% costs Tk 493.15. (*Chapter XVI*)

Beta. A number showing how strongly an investment moves with the whole market. The whole market has a beta of 1. An investment with a beta of 1.5 tends to rise or fall about 15% when the market return changes by 10%. (*Chapter X*)

Bond. A loan sold to investors, repaid at a set date with fixed interest along the way. The business that borrows is the issuer. For

YOU WILL NEED

Chapter 15, section 15.1 'What Working Capital Is' and section 15.2 'The Cash Conversion Cycle'.

A loan that does not fit the time of the need can ruin a sound plan.

The opposite mistake is to borrow for many years to meet a need that lasts weeks.

Every business must choose how long to hold borrowed money, and each choice has a cost and a danger.

Financing is sorted first by how long the money is held.

Term. Word in use. In daily life, a period of time, such as a school term. In finance, the length of time of a loan, or a condition of a contract, as in credit terms.

There are three spans: up to one year, one to five years, and more than five years.

Short-term financing. money borrowed or raised for up to one year

Intermediate-term financing. money borrowed or raised for between one and five years

Long-term financing. money borrowed or raised for more than five years

Each span has its usual sources, shown in a table.

Suppose you need a motorcycle to start a delivery round. A friend lends you the money but says she must have it back in one month. The motorcycle will earn its cost only slowly, over five years. At the end of the month you have not earned enough to repay her. What happens now? You must find a new lender in a hurry, or sell the motorcycle at a poor price. Your delivery round was sound, but the loan did not fit it.

Now reverse the story. You borrow money for ten years to buy a stock of toys that you will sell in six weeks. After the sale you hold idle cash, yet you still pay interest every month for nine more years. Both mistakes come from one error: the time for which you hold the money does not match the time for which you need it.

Businesses face this choice every day. Money can be raised for a few days or for many years, and each choice brings its own cost and its own danger. This chapter shows how to sort financing by time, how to price the most common short-term source, which is credit from suppliers, and how to decide which kind of money suits which need. It ends with a bank that made the first mistake on a very large scale.

* * *

16.1 FINANCING BY TIME SPAN

Every business needs funds, and Chapter 15 showed that working capital is the money tied up in day-to-day operations.¹ The source of funds matters as much as the amount. The first way to sort sources is by time: for how long will the business hold the money before it must repay it? Three spans are in common use, and the dividing lines are one year and five years.

Each span has a name. **Short-term financing** means money borrowed or raised for up to one year. For example, a bank overdraft that is cleared in two months is short-term financing. **Intermediate-term financing** means money borrowed or raised for between one and five years, such as a loan for a delivery van repaid over three years. **Long-term financing** means money borrowed or raised for more than five years, such as the money that the owners put into the business as equity.

The usual sources follow the spans. The table below names them. Read each row from left to right: the first cell gives the span, the second gives its length, and the third lists the sources that busi-

¹ Chapter 15, 'Working Capital and the Cash Cycle', section 15.1 'What Working Capital Is': working capital is defined there.

771,608, so buying is lower before and after discounting; the sign of the answer is sensible. The gap is about 2.4% of the lease figure, which is small, so other facts could change the choice.

From the owners' viewpoint, the saving of Tk 15,017 is a gain in present value, because the same machine does the same work for less. The lender's viewpoint differs: buying with a loan adds Tk 780,000 of debt, while leasing adds no loan but does add a fixed promise to pay each year. A manager should weigh that fixed promise before signing either contract.

Common slips are these. Deducting tax on the whole instalment wrongly treats loan repayment as a tax saving. Leaving out depreciation overstates the cost of buying. Discounting at the loan rate of 10% instead of 7.5% mixes a before-tax rate with after-tax cash. Placing the year-5 payment in year 4 mixes up the periods.

* * *

SUMMARY

A lease is a contract in which a lessor lets a lessee use an asset in return for regular payments. An operating lease is short and leaves risk with the lessor; a financial lease is long and leaves risk with the lessee. To choose between leasing and buying, find each year's after-tax cash outflow, convert it to a present value at the after-tax cost of debt, and choose the lower total. For the second machine of Meghna and Sons Limited, buying costs Tk 15,017 less in present value.

THINK IT THROUGH

18.T1 A clinic needs a scanner for 3 years. Lease P asks Tk 60,000 at each year end for 3 years. Lease Q asks Tk 45,000 at each year end for 2 years, after which the clinic must hire a scanner for the third year at Tk 70,000. Tax is 25% and the discount rate is 7.5%. Which is cheaper in present value?

18.T2 Name the kind of arrangement in each case: a school hires a minibus for one week; a bakery signs a six-year lease of an oven that cannot be cancelled; a printer sells its press to a lessor and leases it back.

WHERE PEOPLE GO WRONG

Wrong: leasing is cheaper because no large sum is paid at the start. *Correct:* the lessor must earn a return, so total payments usually exceed the price. Only a present value comparison shows which road costs less.

Wrong: the tax saved on buying is the tax rate times the whole instalment. *Correct:* only interest and depreciation reduce taxable profit; the part of the instalment that repays the loan does not.

Wrong: the lessee owns the asset when the lease ends. *Correct:* the asset stays with the lessor unless the contract says otherwise,

The owners' viewpoint: a lower cost adds to the value of the business.

Common slips: wrong deduction, forgotten depreciation, wrong rate or period.

The chapter in a few lines.

Wrong idea: leasing is cheaper because no cash is paid at the start.

Wrong idea: tax applies to the whole loan instalment.

Wrong idea: the lessee owns the asset at the end of a lease.

A cash discount rewards early payment with a lower price.

Cash. Word in use. In daily life, coins and notes. In finance, money in hand or in a bank account, ready to be spent.

Cash discount. a reduction in price for paying a supplier early

If you refuse the discount, you borrow from the supplier; we first find the price of that loan.

The cost of giving up a discount has a standard formula.

A cost above the interest rate of other sources means the discount should be taken.

Two sets of terms can be compared by their yearly costs.

Recap of the section on trade credit.

The reduction itself has a name. A **cash discount** means a reduction in price for paying a supplier early. On a Tk 500,000 invoice at 2%, the cash discount is Tk 10,000. A supplier offers it because early money is useful: the supplier can pay its own bills and does not wait for slow customers.

Now the key question. If you do not take the discount, you in effect borrow money from the supplier for the extra days. What is the interest rate on that loan? Use a small example first. The invoice is Tk 10,000 on terms 2/10, net 30. If you pay on day 10, you pay Tk 9,800. If you wait until day 30, you pay Tk 10,000. By waiting 20 more days you keep Tk 9,800 in your hands, and the price of doing so is Tk 200. The cost for the period is $200 \div 9,800 = 2.04\%$. A year holds $365 \div 20 = 18.25$ such periods, so the cost for the year is about $2.04\% \times 18.25 = 37.24\%$.

The same steps give a formula. In words: the cost of giving up the discount equals the discount divided by what you pay if you take it, multiplied by the number of periods in a year. In symbols, $\text{cost} = d \div (1 - d) \times 365 \div (N - D)$. Here d is the discount as a decimal, so 2% is 0.02, D is the last day for the discount, and N is the day on which the full invoice is due. Check with the example: $0.02 \div 0.98 \times 365 \div 20 = 0.020408 \times 18.25 = 0.3724$, or 37.24%. The result is an approximation, because it does not allow for interest earning further interest.

PAUSE AND CHECK 16.P1 A supplier offers terms 1/10, net 30. What is the cost of giving up the discount, as a yearly rate?

What does the cost tell us? Compare it with the interest rate of the cheapest other source. If a bank charges 14% and giving up the discount costs 37.24%, the business should take the discount, borrowing from the bank if it lacks the cash. If giving up the discount costs less than the bank charges, the business should wait. Three cautions apply. Paying after the net date is not free: the supplier may add charges, refuse future credit or raise prices. Where no discount is offered, trade credit appears free, but the price may already include the cost of the delay. And the formula assumes that the full invoice is paid on the net date.

The formula also lets us compare suppliers. Take terms 2/10, net 30 and terms 1/10, net 40, both on an invoice of Tk 10,000. For the first, the cost of giving up the discount is $0.02 \div 0.98 \times 365 \div 20 = 37.24\%$. For the second, it is $0.01 \div 0.99 \times 365 \div 30 = 0.010101 \times 12.1667 = 12.29\%$. The first offer punishes late payment far more, because its discount is larger and its extra days are fewer. A business that must choose which invoice to pay early should pay the one with the higher cost of giving up the discount.

Before moving on. Trade credit is a supplier's permission to pay later. Credit terms such as 2/10, net 40 state the discount and the days. Giving up a cash discount is a costly loan, and its yearly cost is $d \div (1 - d) \times 365 \div (N - D)$.

The decision follows four steps. First, find the after-tax cash outflow of leasing in each year. Second, find the after-tax cash outflow of buying in each year, using the loan payments of Chapter 5. Third, convert each year's outflow into a present value at the after-tax cost of debt, and add. Fourth, compare the two totals: the lower total is the cheaper road. If the machine can be sold at the end, subtract the present value of the after-tax sale proceeds from the total for buying.

A resale value changes the comparison, so we note it now. Suppose, in a different case, that a machine can be sold for Tk 10,000 at the end of year 5, with no tax on the sale. That sum reaches the business only if it owns the machine. Its present value at 7.5% is $10,000 \times 0.696559 = \text{Tk } 6,966$, and this amount is subtracted from the present value of buying. The lessor, who keeps the machine, receives the resale value, which is one reason a lessor can offer a lease at all.

The test has limits. It counts cost only. It does not weigh the service that an operating lease may include, the risk of the asset becoming out of date, or the chance that the lessee earns no profit against which to deduct the payments. When the gap is small, such as two or three per cent, the manager should weigh these other facts before choosing.

Before moving on. The lease-or-buy decision compares present values of after-tax cash outflows at the after-tax cost of debt. Leasing deducts the whole payment; buying deducts interest and depreciation. The lower present value wins, if cost is all that matters.

* * *

WORKED EXAMPLE: LEASE OR BUY THE SECOND MACHINE

We first practise the method on small, clean numbers with no tax, and then repeat it on the case file.

Given. A delivery bicycle can be leased for Tk 4,000 at the end of each of 2 years, or bought now for Tk 7,000. There is no tax and no resale value. The discount rate is 10%. *Find.* The cheaper road. *Formula.* Present value is each payment divided by one plus the rate, once for every year of waiting. In symbols: $PV = P_1 \div (1 + r) + P_2 \div (1 + r)^2$, where P is a payment and r is the discount rate. *Steps.* $4,000 \div 1.10 = \text{Tk } 3,636.36$. $4,000 \div 1.21 = \text{Tk } 3,305.79$. The sum is Tk 6,942.15. Buying costs Tk 7,000.00 today. *Answer.* Leasing is cheaper by Tk 57.85. *Check.* The lease payments add to Tk 8,000, more than Tk 7,000, yet the second payment is worth less today, so a lower present value is sensible.

Case file, illustrative. 3 December 2025. Meghna and Sons Limited needs a second machine costing Tk 780,000. *Given.* The lease asks Tk 210,000 at each year end for 5 years. To buy, the business borrows Tk 780,000 at 10% a year and repays it in 5 equal year-end instalments of Tk 205,762.04. Depreciation is Tk 156,000 a year, and no resale value is assumed. Tax is 25%. assumed for

Recall: loan payments, Chapter 5, section 5.4 "Finding the Payment".

Four steps make the decision orderly.

A resale value would lower the cost of buying.

The test counts cost only, so a close result needs more thought.

Recap of section 18.3.

We learn the method on small numbers first, with no tax.

A bicycle example in the standard order.

Recall: loan payments, Chapter 5, section 5.4 "Finding the Payment".

The case file gives every figure for the second machine.

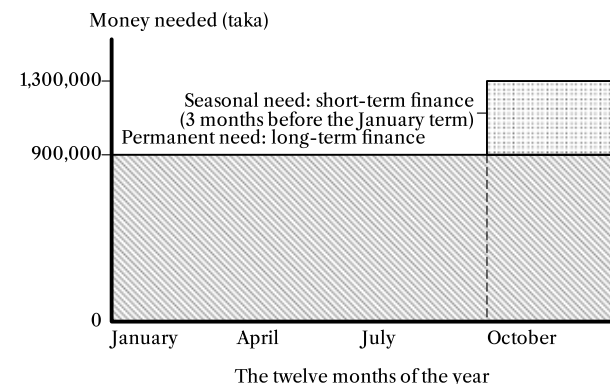


FIG. 16.1. Permanent and seasonal needs over twelve months, with the finance that matches each

The rule follows from the figure. The **matching principle** is the rule that short-term needs are financed with short-term funds and long-term needs with long-term funds. In the figure the lower band calls for long-term finance, and the upper band calls for short-term finance. Two reasons support the rule. The first is safety: money borrowed for three months to buy a machine that lasts ten years must be replaced three months later, and the lender may refuse. The second is cost: money borrowed for ten years to cover a three-month peak must be paid for through nine idle months.

IN SHORT

Match the time. Permanent needs are paid for with long-term funds, and seasonal needs with short-term funds.

A small example with round numbers shows the saving. A shop needs Tk 100,000 all year and a further Tk 50,000 for two months. Long-term money costs 10% a year and short-term money costs 12% a year. If the shop borrows Tk 50,000 short-term for two months, the interest is $50,000 \times 0.12 \times 2 \div 12 = \text{Tk } 1,000$. If it carries the same Tk 50,000 on a long-term loan for the whole year, the interest is $50,000 \times 0.10 = \text{Tk } 5,000$. The short-term rate is higher, yet the matched choice costs Tk 4,000 less, because the money is held for only two months.

The principle is a guide, not a law. A cautious business may use more long-term money than the rule needs, so that it is never forced to refinance in a hurry. The price is interest on idle cash. A bolder business may rely more on short-term money, because short-term interest rates are often lower. The price is the danger that lenders refuse to renew. Neither choice is wrong in itself, but each should be made knowingly. A lender prefers to know when it will be repaid, and an owner prefers a low cost and no sudden shortage. Matching serves both.

Now consider the lessor. The lessor owns many assets of one kind, can buy them at a good price and knows how to resell them. The lessor earns a return from the payments. Tax rules matter as well, because in many countries the lease payment reduces the lessee's taxable profit, as section 18.3 shows. Rules differ from country to country, so this chapter uses the tax rate of 25% assumed for illustration in the Meghna and Sons case.¹

Leasing has limits. The lessor must earn a return, so the payments over the whole lease usually add up to more than the price of the asset. The lessee must pay even if the asset is idle. When the lease ends, any resale value belongs to the lessor, not to the lessee. Leasing is therefore a different way of paying for use, not a cheaper way by nature. Section 18.3 shows how to test whether it is cheaper in a given case.

Before moving on. A lease lets a lessee use an asset owned by a lessor in return for regular payments. A business leases to save cash, to use an asset for a short time, or to share risk. Leasing is not free: the lessor must earn a return, and the lessee gives up any resale value.

18.2 KINDS OF LEASE

Leases range from a cab hired for an hour to a machine held for most of its working life. Two main kinds help us sort them, and both depend on one question: who carries the risks of owning the asset? These risks include repairs, damage and the chance that the asset becomes out of date.

An **operating lease** is a short or cancellable lease where the lessor keeps most of the risks of owning the asset. A clinic that hires a portable scanner by the month, with the right to return it, holds an operating lease. The lessor pays for repairs and bears the risk that the scanner becomes out of date.

A **financial lease** is a long lease that cannot be cancelled, in which the lessee in effect carries the cost and risk of owning. A bakery that signs for an oven for six years, with no right to leave early, holds a financial lease. The payments usually cover most of the oven's price plus the lessor's return, which is why the lessee in effect pays for the asset.

The two kinds are easier to remember side by side. The table below compares them on four features. Read each row from left to right: the feature, then how an operating lease treats it, then how a financial lease treats it.

OPERATING LEASE AND FINANCIAL LEASE COMPARED

FEATURE	OPERATING LEASE	FINANCIAL LEASE
Length	Short, often well below the asset's working life	Long, close to the asset's working life
Cancellation	Usually allowed	Not allowed

¹ Names and legal details of leasing contracts differ between countries. The ideas in this chapter are common to all of them.

The lessor has reasons too, and the tax rules matter.

Leasing is not free, and resale value passes to the lessor.

Recap of section 18.1.

Leases range from one day to many years; risk sorts them.

Operating lease. A short or cancellable lease where the lessor keeps most of the risks of owning the asset.

A short lease the user may cancel, with the owner holding the risk.

Financial lease. A long lease that cannot be cancelled, where the lessee in effect carries the cost and risk of owning.

A long, fixed lease where the user bears the burden of owning.

The table sets the two kinds side by side on four features.

÷ (N - D). A bank overdraft charges interest on the amount used. The matching principle pays for permanent working capital with long-term funds and for seasonal working capital with short-term funds.

THINK IT THROUGH

16.T1 A toy shop has a permanent need of Tk 300,000 all year and a seasonal need of Tk 150,000 for 4 months before December. A long-term loan costs 10% a year and short-term borrowing costs 13% a year. Choose how to fund each need, compute the interest for the year, and find the interest saved compared with funding everything with the long-term loan.

WHERE PEOPLE GO WRONG

Mistake: "Short-term money is always cheaper." Correction: the yearly rate of short-term money may be higher or lower than the long-term rate. What matters is the total interest, which depends on the rate and on how long the money is held.

Mistake: "Trade credit is free." Correction: it is free only when the buyer pays on the last day without a discount lost. If the buyer gives up a cash discount, the credit can cost more than a bank loan, as the worked case showed with 24.83% against 14%.

Mistake: "I can take the discount only if I have the cash." Correction: if giving up the discount costs more than the bank charges, the business should borrow to take it. The worked case gained Tk 4,362 by doing so.

Mistake: "Matching means every need is paid for by money of exactly the same length." Correction: matching is a guide. A business may hold some extra long-term money for safety, and it pays interest on idle cash as the price.

Mistake: "The cost of giving up the discount is exact." Correction: the formula is an approximation, because it does not allow for interest earning further interest. It is accurate enough to compare with a bank's interest rate, but it is not a precise measure.

IN HISTORY

Northern Rock was a bank in the United Kingdom that lent money to people buying homes. A loan for a home, called a mortgage, is repaid over many years. Yet the bank had funded much of this long lending with money borrowed for short periods from lenders in the money markets, and with sales of mortgages to investors, rather than with savings from ordinary customers.

Reports at the time said that only a small part of the money the bank lent came from its savers (Channel 4 News, 2007). For years this plan let the bank borrow cheaply and offer cheap mortgages. It worked while lenders were willing to renew, and it failed when they were not.

On 9 August 2007 the markets for short-term borrowing all but froze. and Northern Rock found that it could not renew its short-

Mistake: believing short-term money is always cheaper.

Mistake: believing trade credit costs nothing.

Mistake: taking the discount only when cash is in hand.

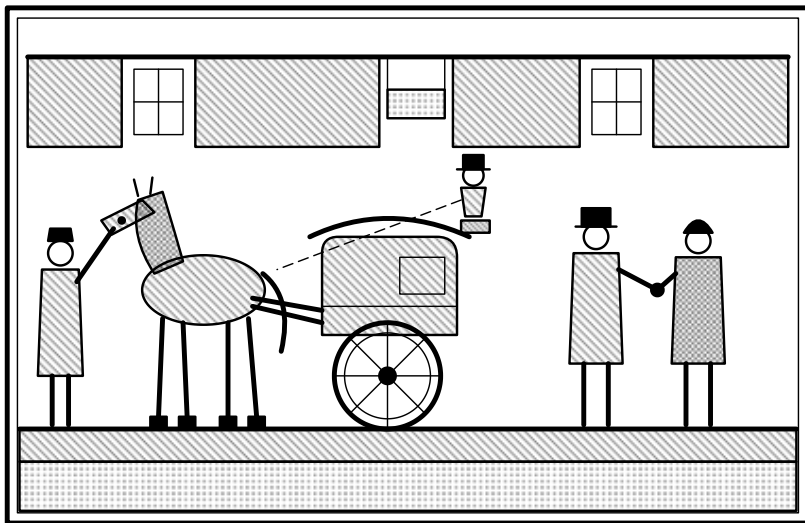
Mistake: thinking the matching principle must be exact.

Mistake: treating the formula result as an exact rate.

In 2007 a British bank that borrowed short to lend long met a queue of depositors.

Only a small part of the bank's lending came from its savers.

When short lenders stopped renewing, the bank had to ask



A carriage-hire yard, where the owner and the hirer agree their terms

CHAPTER XVIII

Leasing

Must a business own everything it uses?



This chapter explains what a lease is, who the two parties are and how the two main kinds differ. It then shows how to compare leasing with buying by turning each year's after-tax cash outflow into a present value.

Answer: The instalment is Tk 158,278.49 a year. The project has a net present value of about -Tk 620, so on these figures it should be rejected. *Check:* Five instalments repay $5 \times 158,278.49 = \text{Tk } 791,392.45$, which is Tk 191,392.45 more than the loan: this is the interest, and it is sensible for a five-year loan at 10%. At the loan's rate of 10% the project would have looked acceptable, since $225,000 \div 1.10 = \text{Tk } 204,545.45$. At the cost of capital it fails.

Before moving on. The method never changed: after-tax cost, weight, multiply, add. A change in the mix of funds changes the weights and so the answer. The result is the discount rate that a project must beat.

* * *

SUMMARY

A term loan is repaid in instalments over more than one year, and collateral and covenants protect the lender. A debenture is a bond with no particular asset pledged. Owners supply long-term money through retained earnings and new stock. The capital market issues and trades long-term loans and shares, and in Bangladesh banks, development institutions, the investment corporation, two exchanges and a regulator play their parts. The cost of capital is the weighted average of the after-tax cost of debt, the cost of preferred stock and the cost of common equity. For Meghna and Sons it is 12.85%.

THINK IT THROUGH

17.T1 A new business asks why long-term funds suit its first machines and its shop fit-out, and why short-term loans do not. Explain in plain words.

WHERE PEOPLE GO WRONG

Wrong: retained earnings cost nothing, because nobody is paid. *Right:* the owners could have taken the profit and used it elsewhere, so they expect a return on it. That is why the case file counts them inside common equity at 14%.

Wrong: the loan costs 10%, so use 10% in the calculation. *Right:* interest lowers taxed profit, so the cost after a 25% tax rate is 7.5%. Use the after-tax cost, and use it for debt only.

Wrong: debt is the cheapest source, so a business should borrow as much as it can. *Right:* debt must be repaid whatever the profit, and lenders and owners ask for more as debt grows.

Wrong: a project that beats the loan's 10% is acceptable. *Right:* a project must beat the cost of all the funds, which is 12.85% for Meghna and Sons.

IN HISTORY

Ford Motor Company was founded by Henry Ford in 1903. For

The instalment is Tk 158,278.49, and the project fails by about Tk 620.

Recap: the same steps give the cost of capital for clean, realistic and changed funds, and the rate tests a project.

Long-term funds come from loans, owners and the capital market, and the business must earn their combined cost.

Retained earnings are not free money.

The cost of debt is the interest rate less the tax it saves.

Debt is cheaper, but a business cannot use only debt.

The cost of capital is a rate for the whole mix, not the rate of one loan.

A famous offering of shares in

A half-built ship needs money that stays for years, not weeks.

You will need: Chapter 16, 'Short-Term Financing and Matching', section 16.1 'Financing by Time Span'; Chapter 13, 'Valuing Bonds', section 13.1 'What a Bond Is'; Chapter 14, 'Valuing Common and Preferred Stock', section 14.1 'Rights of Shareholders'.

The answer is money that stays as long as the asset takes to pay for itself, and it has a price.

The chapter surveys the sources of long-term funds and then computes their combined cost.

A lender will wait five years if it is paid along the way and holds rights that protect it.

A term loan is repaid in instalments over a fixed period longer than one year.

Recall: term loans and debt, assumed from earlier chapters; time spans of financing, Chapter 16, section 16.1 'Financing by Time Span'.

Term. Word in use. In daily life, a period of time such as a school term. In lending, the length of time a loan runs before it is fully repaid.

Term loan. a loan repaid in instalments over a fixed period of more than one year

The case file records the new term loan and the two protections the bank asks for

Collateral is an asset pledged to the lender as security for a loan

In a shipyard, an iron steamship rests on the slips with half its hull still open to the sky. The builder has paid for iron, rivets and wages for many months, and the ship will earn nothing until it is launched, fitted and sailing. Suppose the builder had borrowed the money for ninety days. On day ninety the lender would ask for it back, and the half-built ship could be neither sold nor repaid. What kind of money does such a builder need?

The answer is money that stays for as long as the ship takes to pay for itself: years, not weeks. This is long-term financing, and it is harder to find. The lender must trust the borrower for longer, and must be told what happens if things go wrong. The builder must also pay for it. Every lender and every owner who puts money in expects something back, and the sum of those expectations is a cost the business must earn.

This chapter has two parts. The first surveys where long-term money comes from: loans, owners' funds and shares sold to the public, with the institutions in Bangladesh that supply them. The second is one calculation, the cost of the whole mix of funds, which a business uses as the rate for judging its projects. Our case is Meghna and Sons Limited, which needs a machine for Project Press, the printing project you met in Chapter 8.

* * *

17.1 TERM LOANS AND DEBT INSTRUMENTS

Why would a bank wait five years for its money? Because the borrower pays interest and returns part of the money along the way, and because the bank holds rights that protect it if payments stop. Those rights are the heart of long-term lending.

The commonest long-term debt is the **term loan**. A term loan is a loan repaid in instalments over a fixed period of more than one year. An *instalment* is one of the regular payments that together repay the loan. The agreed length of time is the term. Each instalment usually holds some interest and some repayment of the amount borrowed. Meghna and Sons plans to buy a press machine and asks a bank for a term loan.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. A new term loan of Tk 600,000 over 5 years at 10% needs yearly instalments of Tk 158,278.49. Collateral: the machine. Covenant: owners' equity must stay above Tk 2,000,000.

A lender who waits five years wants a way out if the borrower

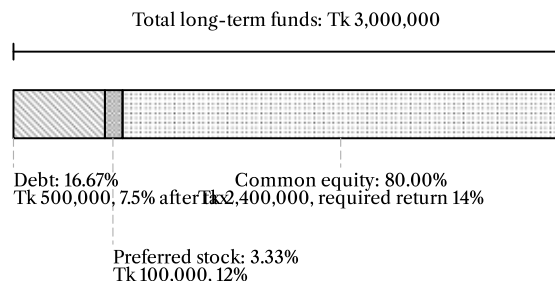


FIG. 17.1. The long-term funds of Meghna and Sons Limited, split by weight

PAUSE AND CHECK 17.P2 A business has Tk 200,000 of debt and Tk 800,000 of common equity. What is the weight of each source?

Now we combine them. The **weighted average cost of capital** is the cost of each source of funds multiplied by its share of the total and added together. In symbols: $K = (W_d \times K_d) + (W_p \times K_p) + (W_e \times K_e)$. Here K is the weighted average cost, W stands for a weight and K with a letter stands for a cost. The letters d, p and e mean debt after tax, preferred stock and common equity.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. Weighted average cost of capital = $0.1667 \times 7.5\% + 0.0333 \times 12\% + 0.80 \times 14\% = 1.25\% + 0.40\% + 11.20\% = 12.85\%$. Project Press (Chapter 8 flows) at 12.85%: net present value Tk 172,667, still positive.

Why compute it? A project must earn at least what its funds cost. So the cost of capital serves as the discount rate in net present value, as in Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value'. A positive net present value at 12.85% means Project Press earns more than all its suppliers of funds require.

Four cautions apply. The weights here use the amounts in the case file, and not current market prices. The cost of equity is an estimate, not a bill that arrives. The rate suits projects as risky as the business itself, and a riskier project needs a higher rate. And the cost changes when the mix changes: more debt lowers the average at first, but lenders and owners may ask for more as debt grows.

Before moving on. Find the after-tax cost of debt, and the costs of preferred stock and common equity. Find each weight. Multiply each cost by its weight and add. For Meghna and Sons the result is 12.85%, and Project Press still has a positive net present value.

* * *

The weighted average cost of capital multiplies each cost by its weight and adds the results.

Weighted average cost of capital. the cost of each source of funds multiplied by its share of the total and added together

The case file gives the result, 12.85%, and the test of Project Press against it.

The cost of capital is the discount rate for judging projects: a project must earn at least this much.

The result is an estimate that depends on the weights, the costs and the risk of the project.

Recap: find each cost, find each weight, multiply and add.

FOUR SOURCES OF LONG-TERM MONEY

SOURCE	REPAID ON A FIXED DATE?	RETURN PAID
Term loan	Yes, in instalments	Interest set by the agreement
Preferred stock	No	A fixed dividend (12% in the case file)
Common stock	No	Dividends that vary with profit
Retained earnings	No	None paid, but owners expect 14%

Owners' funds have limits. They cost more than debt, because the owners take more risk. A new issue costs fees and may reduce the control of the existing owners. Seen from the lender's side, a thick layer of owners' funds is a comfort, which is why covenants such as the one in 17.1 protect equity.

Before moving on. Retained earnings are profit kept in the business. New common and preferred stock bring new owner money. None of these has a fixed repayment date, and all of them expect a return.

* * *

17.3 RAISING FUNDS IN THE CAPITAL MARKET

A bank loan comes from one lender. What if a business needs more money than a single bank will lend, and its present owners cannot supply more? It may offer shares or bonds to many people. For that it needs a place where such offers are made and later traded.

That place is the **capital market**. The capital market is the market where long-term loans and shares are issued and traded. Securities are first sold there by the business that issues them. Later they may be traded between investors, who do not give any money to the business itself. A buyer who wishes to leave can sell to another buyer, and this makes people more willing to buy at the start.

Suppose a printing business owned by one family decides to sell 30% of its shares to people it does not know. This is an **initial public offering**: the first sale of a business shares to the general public. After it, the shares are usually listed on a stock exchange, so that they can be bought and sold. A later offer to the public by a business already listed is called a public issue. Both bring in new owners and new money, and both are costly and carefully regulated.

There is a quieter way to raise owners' money. A **rights issue** is an offer of new shares to existing shareholders in proportion to

Owner money carries no repayment date, but it asks for a higher return and may share control.

Recap: retained earnings and new stock are owner money, with no repayment date and a higher return.

A business that needs more than one bank or its owners can supply may sell securities to the public.

The capital market is where long-term loans and shares are issued and traded.

Recall: financial market, assumed from earlier chapters.

Capital. Word in use. In daily life, money or wealth, or the main city of a country. In finance, the long-term funds a business uses, as in cost of capital.

Capital market. the market where long-term loans and shares are issued and traded

An initial public offering is the first sale of a business's shares to the general public.

Initial public offering. the first sale of a business shares to the general public

A rights issue offers new shares to existing shareholders in proportion to what they

Discount bond. A bond that sells for less than its par value. This happens when the required return is above the coupon rate. The case bond at 12% sells for Tk 891.86. (*Chapter XIII*)

Discount rate. The rate used to turn a future amount into a present value. It is the same number as the interest rate when the money could earn that rate, used for working backwards. (*Chapter IV*)

Discounted cash flow. A way of valuing a series of future cash flows by turning each one into a present value and adding the results. For example, Tk 1,000 in year 1 and Tk 2,000 in year 2 at 10% are valued as Tk 909.09 plus Tk 1,652.89. (*Chapter VI*)

Discounting. Finding a present value by working backwards from a future amount at a rate, dividing by $(1 + i)$ once for each year. Tk 12,100 becomes Tk 11,000 and then Tk 10,000. (*Chapter IV*)

Diversification. Spreading money across different investments so that one loss hurts less. The investments must differ in what affects them. Ten umbrella stalls do not diversify the trader's risk from wet weather, but one umbrella stall and one sunshade stall do. (*Chapter X*)

Dividend. A share of profit paid out to the owners of a business. If a shop earns Tk 90,000 and pays Tk 30,000 to its owners, the dividend is Tk 30,000. A family may call its payout a withdrawal. (*Chapter II*)

Dividend decision. Choosing how much profit to pay out to owners and how much to keep in the business. A payout rewards owners now; keeping profit leaves more to fund growth later without asking a lender. (*Chapter II*)

Dividend growth model. A formula valuing stock as the next dividend divided by the required return minus the growth rate. It assumes the dividend grows at a steady rate for ever, and that this rate is below the required return. A next dividend of Tk 4.24, required return of 12% and growth of 6% give Tk 70.67. (*Chapter XIV*)

Earnings per share. The profit left for common stock owners divided by the number of common shares. It is found by taking interest, tax and preferred dividend from EBIT in turn. Tk 33,000 left for 5,000 shares gives earnings per share of Tk 6.60. (*Chapter XII*)

EBIT. Earnings before interest and tax: the profit from operations before paying lenders or the tax office. It equals revenue less cash costs less depreciation. The book uses EBIT in place of the term operating profit. (*Chapter VII*)

Effective annual rate. The single yearly rate that gives the same growth as compounding within the year. A rate of 12% compounded half-yearly has an effective annual rate of 12.36%. (*Chapter IV*)

Efficient market. A market in which prices already reflect the information available. New information moves prices, and since it cannot be predicted, price changes look random. Efficient does not mean that prices are always right or that investors never lose.

Trade credit, 153–154, 157–158

U

Use of funds, 5, 7, 9, 14

V

Valuation, 126, 138

Variable operating cost, 105–106, 110–111

Voting right, 134–135, 139, 163

W

Weak-form efficiency, 100

Wealth, 3, 6–8, 22–25, 28–30, 32–33, 64, 91, 97, 164, 170

Wealth maximisation, 23–25, 28–30, 33

Weighted average cost of capital, 167–168, 170

Working capital, 14, 67–72, 142–144, 147, 149, 152, 155, 158–159

Y

Yield to maturity, 130–132

Fixed operating cost. An operating cost that stays the same whatever the number of units sold. Shop rent of Tk 8,000 a year is the same for ten notebooks or a thousand, so it is a fixed operating cost. (*Chapter XI*)

Float. Money that has been paid by one person but cannot yet be used by the receiver. It has three parts: mail float, processing float and clearing float. A cheque with 2, 1 and 3 days of these parts has 6 days of float. (*Chapter XV*)

Funds. Money that a business has available to spend, save or invest at one moment. Funds may be notes in a purse or a balance held at a bank. Goods waiting to be sold are not funds until they are sold and paid for. (*Chapter I*)

Future value. What an amount today will grow to after a stated time at a stated rate, found by $FV = S \times (1 + i)^n$ to the power of n . Tk 10,000 at 10% for two years has a future value of Tk 12,100. (*Chapter IV*)

Independent projects. Projects where accepting one has no effect on whether another is accepted. A print-on-demand machine and a new set of shelves are independent: a business may accept both, either or neither. (*Chapter VII*)

Initial investment. Cash spent at the start to get a project going. It includes the price of the asset, delivery, installation and any working capital needed. It is paid at time zero, before year one begins. (*Chapter VII*)

Initial public offering. The first sale of a business shares to the general public. After it, the shares are usually listed on a stock exchange. A family printing business selling 30% of its shares to strangers for the first time is an example. (*Chapter XVII*)

Interest. The price paid for using money for a period, received by the lender. If a bank pays Tk 8,000 for the use of Tk 100,000 for a year, the Tk 8,000 is interest. (*Chapter IV*)

Interest rate. The interest for one period, shown as a percentage of the amount lent or borrowed. Tk 8,000 earned on Tk 100,000 in a year is an interest rate of 8% a year. (*Chapter IV*)

Intermediate-term financing. Money borrowed or raised for between one and five years. It suits assets that last a few years. For example, a loan for a delivery van, repaid in instalments over three years, is intermediate-term financing. (*Chapter XVI*)

Internal rate of return. The discount rate at which a project's net present value is exactly zero. It is the yearly return the project itself earns. Find it by trial between two rates and interpolation; accept when it is at least the required return. (*Chapter VIII*)

Intrinsic value. The worth of an asset worked out from the present value of its expected cash flows. An asset that pays Tk 1,100 in one year has an intrinsic value of Tk 1,000 at a required return of 10%. (*Chapter XIII*)

Inventory. Goods held for sale or for use in making goods. A bookshop's books, a tailor's cloth and a baker's flour are examples. Shopkeepers often call it stock. (*Chapter XV*)

I

Independent projects, 67, 71
Initial investment, 66–70, 74, 76–81
Initial public offering, 164–165, 170
Interest, 14, 26–28, 30, 33–36, 38–41, 44–52, 56, 58, 60, 68, ...
Interest rate, 33–34, 36, 39–41, 44, 46–52, 132, 146, 148, 154, 156–158, ...
Intermediate-term financing, 152
Internal rate of return, 77–81, 124, 130
Intrinsic value, 126, 131, 137, 139
Inventory, 143–145, 147–150, 163
Investment decision, 13–15, 17–18, 64, 174

L

Lease, 172–178
Lessee, 172–175, 177–178
Lessor, 172–175, 177–178
Liability, 6–8, 143
Liquidity, 16–18, 144
Long-term financing, 152, 162, 172

M

Market risk, 96–102
Market value, 23–26, 28, 33, 99–100, 126
Matching principle, 147, 155–156, 158–159
Maturity, 125–127, 129–132
Mixed stream, 54–56, 60–61, 76
Mutually exclusive projects, 67, 71, 74, 79–81

N

Net present value, 76–81, 84, 94, 99, 124, 167–169, 172, 174
Net working capital, 143–144, 149

O

Operating break-even point, 106, 108, 110–111
Operating cash flow, 68–71
Operating cycle, 144–145, 149–150
Operating efficiency, 16–17, 19
Operating lease, 173–175, 177–178
Operating leverage, 86, 108–111, 114, 117–121
Opportunity cost, 33, 42
Ordinary annuity, 45–52, 126, 176

P

Par value, 125–132

Operating break-even point. The sales volume at which EBIT is exactly zero. It equals the fixed operating cost divided by the contribution margin, and is stated in units. With fixed cost Tk 240,000 and contribution margin Tk 40, it is 6,000 packs. (*Chapter XI*)

Operating cash flow. Cash a project produces each year from running it, after tax. It equals EBIT, less tax on EBIT, plus depreciation, since depreciation reduced EBIT but no cash was paid. (*Chapter VII*)

Operating cycle. The days from buying stock to collecting cash from the customer. It equals the inventory days plus the receivable days. For Meghna and Sons Limited, $75 + 20 = 95$ days. (*Chapter XI*)

Operating efficiency. How little money and effort a business uses to produce each unit of sales. A tailor who spends Tk 70 for every Tk 100 of sales is more efficient than one who spends Tk 80 for the same Tk 100. (*Chapter II*)

Operating lease. A short or cancellable lease where the lessor keeps most of the risks of owning the asset, such as repairs and the chance that the asset becomes out of date. Example: a portable scanner hired by the month, with the right to return it. (*Chapter XVIII*)

Operating leverage. The effect of fixed operating costs in making EBIT change by more than sales change. The larger the fixed cost compared with the contribution, the stronger the effect. A business with no fixed operating cost has none. (*Chapter XI*)

Opportunity cost. The value of the best choice given up when you pick another. Cash kept idle in a drawer has an opportunity cost equal to the interest a bank would have paid on it. (*Chapter IV*)

Ordinary annuity. An annuity whose payments fall at the end of each period. Loan instalments and bank interest are usually of this kind. When no kind is stated, an annuity is taken to be ordinary. (*Chapter V*)

Par value. The amount printed on a bond that is repaid at maturity. It is also called face value. A bond with a par value of Tk 1,000 repays Tk 1,000 to the investor on the date of maturity. (*Chapter XIII*)

Payback period. The time a project takes to recover its initial investment from its cash flows. With equal inflows it is the initial investment divided by the yearly inflow; otherwise add the inflows into a running total. Tk 10,000 repaid by Tk 5,000 a year takes 2 years. (*Chapter VIII*)

Permanent working capital. The minimum working capital a business needs all year. A bookshop always needs some stock, some money in the till and some money owed by customers, whatever the month. This level of need is normally financed with long-term funds. (*Chapter XVI*)

Perpetuity. An annuity that never ends, such as a fund paying a scholarship of Tk 50,000 every year for ever. Its present value is

INDEX

A

Accounts payable, 143–144, 147
Accounts receivable, 143–144, 147, 149
After-tax cash outflow, 174–175, 177–178
Agency conflict, 25–30
Agency cost, 27–30
Agent, 9, 25–29
Amortisation, 50–51
Annuity, 44–52, 54–55, 57–58, 60–61, 70, 77, 124, 126–127, 134, 176
Annuity due, 45–52
Asset, 5–9, 13–15, 17, 64–65, 67–70, 90–92, 96–99, 101–102, 124, 126, ...

B

Bank overdraft, 152–153, 155, 157–159
Beta, 95, 97–102, 137
Bond, 6, 84–85, 92, 124–132, 134–135, 137–140, 153, 162–164, 169
Business function, 4–5, 8–9, 12–13
Business risk, 86, 92, 104, 109–111, 118–119

C

Capital asset pricing model, 98–99, 101, 137, 140
Capital budgeting, 14, 36, 64–66, 70–71, 81, 84, 94, 104, 124, 167, ...
Capital expenditure, 65
Capital gain, 135, 138–139
Capital market, 102, 164–166, 169–170
Cash conversion cycle, 144–145, 147, 149–150, 152, 155
Cash discount, 154, 157–158
Cash flow, 3, 7–9, 14, 32, 36, 54–55, 57, 60, 64, 66–72, ...
Coefficient of variation, 89–92
Collateral, 162–163, 165, 169
Common stock, 24–25, 96, 99, 115–118, 134–140, 143, 153, 163–164
Compound interest, 34–35, 40–41, 46, 56, 60
Compounding period, 38–39
Contribution margin, 105–108, 110–111, 118–119
Cost of capital, 164, 166–170, 172, 174
Coupon interest, 125–127, 131
Covenant, 162–165, 169
Credit terms, 152–154, 157
Current asset, 143–144, 147, 149, 157
Current liability, 143

pays Tk 6,000 and later sells for Tk 104,000 gives a return of Tk 10,000. (*Chapter IX*)

Rights issue. An offer of new shares to existing shareholders in proportion to what they already hold. An owner with 10% of the shares who buys the offered share of a 1-for-5 issue keeps 10% of the business. (*Chapter XVII*)

Risk. The chance that actual results differ from those expected, especially for the worse. The wider the range of possible results, the greater the risk. A certain loss carries no risk, because nothing is uncertain. A fixed-payment savings bond has little risk. (*Chapter IX*)

Risk premium. The extra return investors demand for taking on risk above the risk-free rate. It equals the required return less the risk-free rate. If the risk-free rate is 8% and a lender demands 12%, the risk premium is 4 percentage points. (*Chapter X*)

Risk-free rate. The return on an investment with no chance of loss, such as a short government bill. It pays for waiting alone. No investment is perfectly free of risk, but a short loan to a stable government comes close. The case file assumes 8%. (*Chapter X*)

Sale and leaseback. Selling an asset to a lessor and at once leasing it back. The business receives cash today and keeps using the asset, but makes lease payments from then on. Example: a baker sells a delivery van to a lessor and leases the same van back. (*Chapter XVIII*)

Seasonal working capital. Extra working capital needed only at the busiest times of the year. A bookshop buys extra books before the school term opens and needs the extra money for a few months. This peak is normally financed with short-term funds. (*Chapter XVI*)

Security market line. The straight line that shows required return rising with beta. It starts at the risk-free rate when beta is 0 and passes through the market return when beta is 1. An investment above the line promises more than its required return. (*Chapter X*)

Semi-strong-form efficiency. The condition in which prices reflect all public information, not only past prices. If it holds, reading a published annual report or news item after it appears brings no gain, because the price has already moved. (*Chapter X*)

Short-term financing. Money borrowed or raised for up to one year. It suits needs that last a short time. For example, a bank overdraft cleared in two months, or a supplier's trade credit of 30 days, is short-term financing. (*Chapter XVI*)

Simple interest. Interest worked out on the original amount only, so the same amount is added each period. Tk 10,000 at 10% earns Tk 1,000 every year. (*Chapter IV*)

Single sum. One payment or receipt made at one point in time. A deposit of Tk 10,000 made today is a single sum, unlike payments that repeat each year. (*Chapter IV*)

Sinking fund. Money put aside in equal deposits to reach a target sum by a set date. The deposit equals the target divided by the future value factor. A business may use one to replace a vehicle.

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Wealth maximisation. The aim of making the market value of the owners' stake as large as possible over time. It takes account of when profits arrive and how certain they are, because buyers pay for both. (*Chapter III*)

Weighted average cost of capital. The cost of each source of funds multiplied by its share of the total and added together. Debt is counted after tax. For Meghna and Sons Limited: $1.25\% + 0.40\% + 11.20\% = 12.85\%$. (*Chapter XVII*)

Working capital. The money a business has tied up in short-term assets such as stock and money owed by customers. It is the money that moves daily from cash to goods to customers' promises and back to cash. A stall's rice, cash and customer debts are an example. (*Chapter XV*)

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity. It is the rate that makes the present value of all payments equal the price. The case bond at Tk 920 has a yield to maturity of about 11.2%. (*Chapter XIII*)

15.1 Operating cycle: $60 + 25 = 85$ days. Cash conversion cycle: $85 - 35 = 50$ days.

15.2 Inventory: $7,300,000 \times 60 \div 365 = \text{Tk } 1,200,000$. Receivables: $10,950,000 \times 25 \div 365 = \text{Tk } 750,000$. Payables: $7,300,000 \times 35 \div 365 = \text{Tk } 700,000$. Net investment: $1,200,000 + 750,000 - 700,000 = \text{Tk } 1,250,000$.

15.3 The parts are mail float, processing float and clearing float. The days lost are $3 + 2 + 2 = 7$ days.

15.4 Customers then pay in fewer days, so the receivable days fall. The operating cycle and the cash conversion cycle fall by the same number of days, and less of the business's own money is tied up.

CHAPTER XVI · SHORT-TERM FINANCING AND MATCHING

16.p1 Cost = $0.01 \div 0.99 \times 365 \div 20 = 0.010101 \times 18.25 = 0.1843$, or 18.43% a year.

16.t1 Fund the permanent Tk 300,000 long term: $300,000 \times 0.10 = \text{Tk } 30,000$. Fund the seasonal Tk 150,000 short term: $150,000 \times 0.13 \times 4 \div 12 = \text{Tk } 6,500$. Total Tk 36,500. All long term: $450,000 \times 0.10 = \text{Tk } 45,000$. Interest saved: Tk 8,500.

16.1 Cost = $0.02 \div 0.98 \times 365 \div 20 = 0.020408 \times 18.25 = 37.24\%$ a year.

16.2 Cost = $0.03 \div 0.97 \times 365 \div 35 = 0.030928 \times 10.4286 = 32.25\%$ a year.

16.3 Discount = Tk 4,000, so the payment on day 10 is Tk 196,000. Interest for 30 days = $196,000 \times 0.13 \times 30 \div 365 = \text{Tk } 2,094$, which is less than Tk 4,000. Giving up the discount costs 24.83%, above 13%. Borrow and take the discount, gaining about Tk 1,906.

16.4 3/10, net 30: $0.03 \div 0.97 \times 365 \div 20 = 56.44\%$. 1/10, net 60: $0.01 \div 0.99 \times 365 \div 50 = 7.37\%$. The first is more costly to give up, by about 49.07 percentage points.

16.5 Short: up to one year, for example trade credit or a bank overdraft. Intermediate: one to five years, for example a loan for a van repaid in instalments. Long: more than five years, for example equity from the owners or a bond.

16.6 Fund the permanent need, the stock held all year, with long-term money. Fund the extra December stock with short-term money, such as an overdraft, and repay it in January when the stock is sold.

16.7 Permanent working capital is the minimum the shop needs all year, such as basic stock and money in the till. Seasonal working capital is the extra stock and money needed only before the school term opens.

16.8 An overdraft charges interest only on the amount used and only for the days used, while a short loan charges interest on the whole amount for the whole period. A business whose need rises and falls may prefer the overdraft.

CHAPTER XVII · LONG-TERM FINANCING AND THE COST OF CAPITAL

2.1 Planning: decide in advance what money the shop will need, such as stock for the new term. Raising: find that money from the family or a bank. Control: compare spending with the plan and act on any difference.

2.2 Buying the scanner is the investment decision; choosing a bank loan to pay for it is the financing decision. The dividend decision, how much profit to pay out and how much to keep, has not yet been made.

2.3 Planning decides in advance that Tk 30,000 will be needed for cloth and arranges it. Control compares the actual bill with Tk 30,000 afterwards, finds any difference, and acts on it.

2.4 Profit is not cash. If cash is short, suppliers may refuse deliveries, wages may be late and chances to buy cheaply may pass, even though the shop earns a profit.

2.5 Buying the fridge is an investment decision. Repaying the loan changes how the household is funded, so it is a financing decision.

2.6 The freezer is an investment decision, and paying herself is a dividend decision. The dividend is Tk 50,000, and the retained earnings are Tk 40,000. Using kept profit to pay for the freezer is also part of the financing decision.

2.7 The three decisions are investment (the machine), financing (the loan) and dividend (the payout). The more profit is paid out, the less is retained to pay for the machine, so the more the manufacturer may need to borrow.

2.8 Tailor A is more efficient, because A uses Tk 70 for each Tk 100 of sales against Tk 80. Efficiency looks at the cost per unit of sales; profitability compares profit with the money invested or the sales made.

CHAPTER III · THE GOAL OF THE FIRM AND AGENCY

3.p1 Wealth maximisation looks at the market value of the owners' stake over time. A buyer would pay less for a business that has lost its customers, so the cut lowers the owners' wealth even though this year's profit rises.

3.p2 The bonus is 5% of branch sales. It rises with sales whatever happens to profit, so Plan Y, with more sales and less profit, pays the manager more while the owners lose.

3.t1 Plan P: bonus 4% of Tk 1,000,000 = Tk 40,000; owners keep Tk 120,000 - Tk 40,000 = Tk 80,000. Plan Q: bonus 4% of Tk 1,400,000 = Tk 56,000; owners keep Tk 60,000 - Tk 56,000 = Tk 4,000. The manager prefers Q (Tk 16,000 more) and the owners prefer P. If the manager chooses Q, the owners lose Tk 80,000 - Tk 4,000 = Tk 76,000.

3.t2 The change in profit over the two years is Tk 80,000 - Tk 150,000 = minus Tk 70,000. Five per cent of that is minus Tk 3,500, a loss to him, so he would not cut restocking. The new rule lines his reward up with the owners' result.

3.1 Profit maximisation looks at one period only. It cannot see that money arriving earlier is more useful, or that one plan may be less

11.t1 Before: $CM = 90 - 50 = \text{Tk } 40$, so break-even is $240,000 \div 40 = 6,000$ packs. After: $CM = 90 - 30 = \text{Tk } 60$, so break-even is $480,000 \div 60 = 8,000$ packs.

11.t2 Before: contribution $9,000 \times 40 = \text{Tk } 360,000$, EBIT Tk 120,000, DOL 3.00. After: contribution $9,000 \times 60 = \text{Tk } 540,000$, EBIT $540,000 - 480,000 = \text{Tk } 60,000$, DOL $540,000 \div 60,000 = 9.00$. The new machine carries more business risk, because its break-even point and its leverage are both higher.

11.1A A fixed operating cost stays the same whatever the units sold, such as shop rent. A variable operating cost grows with each unit, such as the price paid for each book bought for resale.

11.2 $CM = 150 - 95 = \text{Tk } 55$. Break-even = $330,000 \div 55 = 6,000$ units.

11.3 At 5,000 units: $5,000 \times 55 - 330,000 = -\text{Tk } 55,000$, a loss. At 6,000 units: EBIT is 0. At 7,000 units: $385,000 - 330,000 = \text{Tk } 55,000$.

11.4 Contribution = $8,000 \times 55 = \text{Tk } 440,000$. EBIT = $440,000 - 330,000 = \text{Tk } 110,000$. DOL = $440,000 \div 110,000 = 4.00$.

11.5 Near break-even EBIT is small compared with contribution, so the quotient is large. For the printing unit it is 4.00 at 8,000 packs and 3.00 at 9,000 packs. At break-even EBIT is zero and the value cannot be found.

CHAPTER XII · FINANCIAL AND TOTAL LEVERAGE

12.p1 Degree of financial leverage = $50,000 \div (50,000 - 10,000) = 50,000 \div 40,000 = 1.25$.

12.t1 A high degree of operating leverage already makes EBIT swing widely when sales change. Borrowing adds a second lever, and the two multiply, so earnings per share would swing even more. Owners who dislike a wide swing may borrow little to keep total risk lower.

12.1 Profit before tax = $150,000 - 30,000 = 120,000$. Tax = 30,000. Profit after tax = 90,000. Less preferred dividend 18,000 = 72,000. Earnings per share = $72,000 \div 10,000 = \text{Tk } 7.20$.

12.2 Preferred dividend before tax = $18,000 \div 0.75 = 24,000$. Degree of financial leverage = $150,000 \div (150,000 - 30,000 - 24,000) = 150,000 \div 96,000 = 1.56$.

12.3 Degree of total leverage = $2.50 \times 1.60 = 4.00$.

12.4 Earnings per share rises by about $4.00 \times 10\% = 40\%$. From Tk 5.00 it becomes about $\text{Tk } 5.00 \times 1.40 = \text{Tk } 7.00$.

12.5 Plan X: at EBIT 100,000, $(100,000 - 40,000) \div 4,000 = \text{Tk } 15.00$; at 60,000, $20,000 \div 4,000 = \text{Tk } 5.00$, a fall of 66.7%. Plan Y: at 100,000, $100,000 \div 8,000 = \text{Tk } 12.50$; at 60,000, $60,000 \div 8,000 = \text{Tk } 7.50$, a fall of 40%. Plan X is riskier, because its earnings per share swings more for the same fall in EBIT.

CHAPTER XIII · VALUING BONDS

13.p1 $8\% \times 500 = \text{Tk } 40$ each year.

13.p2 The investor receives $50 + 1.000 = \text{Tk } 1.050$ in one year.

5.5 An ordinary annuity pays at the end of each period and an annuity due at the start. The annuity due has the higher future value, because each payment earns one more period of interest.

5.6 Present value = $\text{Tk } 20,000 \div 0.05 = \text{Tk } 400,000$. A fund of Tk 400,000 earns $5\% \times \text{Tk } 400,000 = \text{Tk } 20,000$ a year, so paying out only the interest leaves the fund whole.

CHAPTER VI · UNEVEN CASH FLOWS

6.p1 $\text{Tk } 3,000 \div 1.10 + \text{Tk } 2,000 \div 1.21 + \text{Tk } 1,000 \div 1.331 = \text{Tk } 2,727.27 + \text{Tk } 1,652.89 + \text{Tk } 751.31 = \text{about Tk } 5,131.48$. It is worth more, because the larger receipts come sooner.

6.t1 Plan X: $\text{Tk } 81,818.18 + \text{Tk } 49,586.78 + \text{Tk } 22,539.44 = \text{Tk } 153,944.40$. Plan Y: $\text{Tk } 27,272.73 + \text{Tk } 49,586.78 + \text{Tk } 67,618.33 = \text{Tk } 144,477.84$. Plan Y costs less in present value, because its larger payments fall later.

6.t2 Money has a time value, so a taka received sooner is worth more than a taka received later. The same total, paid in a different order, gives a different present value.

6.1 $\text{Tk } 18,181.82 + \text{Tk } 28,925.62 + \text{Tk } 37,565.74 = \text{Tk } 84,673.18$.

6.2 $\text{Tk } 20,000 \times 1.21 + \text{Tk } 35,000 \times 1.10 + \text{Tk } 50,000 = \text{Tk } 24,200 + \text{Tk } 38,500 + \text{Tk } 50,000 = \text{Tk } 112,700$.

6.3 Annuity part: $\text{Tk } 15,000$ for 5 years, $\text{Tk } 15,000 \times 3.790787 = \text{Tk } 56,861.80$. Odd part: $\text{Tk } 25,000$ extra in year 5, $\text{Tk } 25,000 \times 0.620921 = \text{Tk } 15,523.03$. Present value = $\text{Tk } 72,384.83$.

6.4 At 9%: rising Tk 86,416.60, falling Tk 90,774.03. At 15%: rising Tk 76,732.14, falling Tk 83,093.61. The falling stream is worth more at both rates, and the gap is larger at 15%.

6.5 In an annuity all payments are equal, in a mixed stream they differ. The annuity factor assumes equal payments, so it fits only a level part of a stream; the rest must be discounted as single sums.

CHAPTER VII · CAPITAL BUDGETING: DECISIONS AND CASH FLOWS

7.p1 No. The survey was paid in December 2024 and cannot be recovered whether or not the machine is bought, so it is a sunk cost and does not change the decision.

7.t1 Asset cost $560,000 + 40,000 = \text{Tk } 600,000$; initial investment $600,000 + 50,000 = \text{Tk } 650,000$. Depreciation $(600,000 - 100,000) \div 4 = 125,000$. EBIT $500,000 - 260,000 - 125,000 = 115,000$; tax 28,750; operating cash flow $115,000 - 28,750 + 125,000 = \text{Tk } 211,250$. Terminal cash flow $100,000 + 50,000 = 150,000$. Net cash flows: year 0 -650,000; years 1 to 3 211,250; year 4 361,250.

7.t2 They are mutually exclusive, because buying one means rejecting the other. The first step is to find and list the proposals, then estimate the cash flows of each scanner.

7.1 Capital budgeting is the process of planning, evaluating and choosing long-term investments. A business needs it because such decisions are long, large for the business and hard to reverse, so a mistake is costly and lasts for years.

7.2 Find proposals (which oven, which supplier); estimate cash